DLS Buddy — User Manual

Applies to v1.22.0 and later

Esam Alfalah

Contents

1. Introduction . . . 3
   - 1.1 Launching the Program . . . 4
2. The Workflow at a Glance . . . 4
3. Loading Data . . . 5
   - 3.1 Supported File Formats . . . 5
4. Generic (Plain-Text) File Formats . . . 6
   - 4.1 Generic DLS Format . . . 6
   - 4.2 Generic SLS Format . . . 7
   - 4.3 Generic Trace Format . . . 7
5. Instrument File Formats . . . 7
   - 5.1 Brookhaven DLS Correlation Function (`.CSV`) . . . 8
   - 5.2 Brookhaven Raw ASCII DLS Export (`.dat`) . . . 8
   - 5.3 Brookhaven Count Rate History (`.CSV`) . . . 9
   - 5.4 Brookhaven SLS Intensity Export (`.CSV`) . . . 9
   - 5.5 Malvern Zetasizer Export Template . . . 10
   - 5.6 NanoTemper Prometheus Panta Autocorrelation Export (`.xlsx`) . . . 11
   - 5.7 ALV Correlator Export (`.ASC`) . . . 12
   - 5.8 Malvern Zetasizer Clipboard Correlation Text . . . 12
6. The Data Tab . . . 12
   - 6.1 The Parameter Table . . . 13
   - 6.2 Update, Undo and Reset . . . 15
     - 6.2.1 Editing several measurements at once . . . 16
   - 6.3 Solvent Names and Proposed Values . . . 16
   - 6.4 The Solvent Library at a Glance . . . 17
   - 6.5 When You Have No dn/dc, or No Calibration . . . 18
   - 6.6 When You Have No Solvent Blank . . . 18
7. How the Program Organizes Data . . . 19
   - 7.1 Samples and the Sidebar . . . 19
     - 7.1.1 Average derived results… . . . 20
     - 7.1.2 The regroup / merge confirm . . . 21
     - 7.1.3 Repeat sets . . . 21
   - 7.2 Molecular-Weight Fractions . . . 22
   - 7.3 Traces in the Sidebar . . . 22
   - 7.4 Sessions . . . 23

8. The DLS Tab . . . 23
8.1 What Every Sub-Tab Shares . . . 24
8.1.1 The measurements checklist . . . 24
8.1.2 The analysis region . . . 24
8.1.3 Run, and results that are out of date . . . 25
8.1.4 What an analysis needs before Run . . . 25
8.1.5 A Run that is refused . . . 27
8.1.6 Fits a regroup or a removal takes away . . . 27
8.1.7 Notes and warnings . . . 27
8.1.8 Reading a plot . . . 29
8.2 Correlogram . . . 29
8.2.1 The plot and the fits shown . . . 29
8.2.2 Method, order and window . . . 30
8.2.3 Flags, notes and what the export records . . . 31
8.3 Distribution . . . 32
8.3.1 Methods and fits . . . 32
8.3.2 The bottom panel and the residual weighting . . . 33
8.3.3 Peaks and weighting . . . 34
8.4 Γ vs q^2 . . . 35
8.5 Mode Map . . . 37
8.5.1 Extracting the modes . . . 38
8.5.2 Families and their fits . . . 39
8.5.3 Recording, persistence and export . . . 40
8.6 D vs c . . . 40
8.7 DDLS . . . 41
8.8 Summary . . . 42
9. The SLS Tab . . . 43
9.1 Analysis . . . 45
9.1.1 Calibration panel . . . 45
9.1.2 Methods . . . 46
9.1.3 Masking, the manual M_w and the anisotropy check . . . 49
9.1.4 Results . . . 49
9.2 Derived Intensity . . . 51
9.3 Mode-Resolved . . . 51
10. The Cross-Sample Tab . . . 52
10.1 Architecture . . . 52
10.1.1 Membership, focus and the table . . . 52
10.1.2 Choosing the sources . . . 53
10.1.3 Form-factor and degenerate sources . . . 57
10.1.4 Branching factors g and h . . . 58
10.2 Scaling . . . 58
11. The Utilities Tab . . . 59
11.1 Traces . . . 60
11.1.1 Summary statistics and baseline . . . 61
11.1.2 Outliers . . . 61
11.1.3 Windowed Hampel . . . 61
11.1.4 Running average . . . 62
11.1.5 The Diagnostic panel . . . 62

11.2 I·sin θ . . . 63
11.3 Synthetic Generator . . . 64
11.3.1 Inputs . . . 66
11.3.2 Artifacts . . . 67
11.4 Solvent Explorer . . . 68
12. The Settings Tab . . . 69
12.1 DLS Analysis Defaults . . . 69
12.1.1 Cumulant method . . . 69
12.1.2 Cumulant order . . . 70
12.1.3 Cumulant residual weighting . . . 70
12.1.4 Skip initial lag channels . . . 70
12.1.5 Cumulant fit cutoff . . . 71
12.2 DLS Distribution Defaults . . . 71
12.3 File Loading . . . 71
12.4 SLS Defaults . . . 72
12.5 Solvent Library . . . 72
12.6 Uncertainty . . . 72
12.7 Display Units . . . 72
12.8 Result Formatting . . . 73
12.9 Appearance . . . 73
12.10 Update and Restore defaults . . . 73
13. Exporting Results . . . 74
13.1 Where the Buttons Are . . . 74
13.2 The File . . . 75
13.3 The Name . . . 75
14. Visualization . . . 76

1. Introduction

DLS Buddy is a general-purpose Python program for analyzing **dynamic (DLS)** and **static (SLS)** light scattering data. It is not tied to any specific instrument, solvent, or solute. The physical parameters — viscosity, refractive index, temperature, the refractive-index increment $\mathrm{d}n/\mathrm{d}c$, wavelength, and scattering angle — are yours to set. You enter them, or, for a common solvent, accept the refractive index and viscosity the solvent library proposes (§6.3) and commit them with **Update** (§6.2). The program never proposes a $\mathrm{d}n/\mathrm{d}c$: you enter it, or a file that records one fills it, and it may be left blank (§6.5).

The graphical interface is organized as a **Workspace** sidebar (your loaded data, auto-grouped into samples) plus tabbed modules, one chapter each in this guide:

- **Data** — confirm and edit the physical parameters of each measurement (§6).
- **DLS** — the hydrodynamic radius R_h from correlation-function fits (cumulants, exponentials, the stretched exponential KWW), CONTIN/NNLS distributions, and multi-angle / concentration-series analysis (§8).
- **SLS** — Zimm, Berry, and Debye analysis for the weight-average molecular weight M_w, the radius of gyration R_g and the second virial coefficient A_2, with a calibration panel, plus **Derived Intensity** and **Mode-Resolved** sub-tabs for a static analysis built from DLS measurements (§9).

- **Cross-Sample** — quantities that link samples or modules: the shape parameter $\rho = R_g/R_h$, the branching factors, and log–log scaling plots (§10).
- **Utilities** — data-quality diagnostics (intensity traces, I·sin θ), a synthetic data generator, and the Solvent Explorer (§11).
- **Settings** — global defaults and appearance (§12).

DLS and SLS measurements of the same sample — the same solute, solvent and temperature — group together, so the Cross-Sample tab sets the DLS R_h beside the SLS R_g in one row for $\rho = R_g/R_h$ (§10).

1.1 Launching the Program

Double-click the launcher for your operating system:

- **Windows** — `Launch DLS Buddy (Windows).bat`
- **macOS** — `Launch DLS Buddy (MacOS).command` (on first run macOS Gatekeeper may refuse an unsigned script — right-click it → **Open** → confirm once, and it opens normally thereafter)

Or run it from the repository root with:

```
python -m gui.main
```

---

2. The Workflow at a Glance

Most analyses follow the same short path. Here is the whole flow in one place; the rest of the guide fills in the detail.

Launch → Load → Confirm parameters + Update → open the DLS or SLS tab → pick a method and Run → read the results → Export → keep the workspace.

1. **Launch** the program (§1.1).
2. **Load** your data — **Load DLS correlogram...**, **Load SLS intensities...**, or **Load count-rate trace...** — the program recognizes the common instrument formats and groups the files into **samples** in the sidebar (§3).
3. **Confirm the parameters** on the **Data** tab — temperature, viscosity, refractive index, wavelength, scattering angle, and (for SLS) concentration and $\mathrm{d}n/\mathrm{d}c$ (blank is allowed — §6.5). Identity and optics are entered **once per sample**; press **Enter** to jump to the next field. Click **Update** to commit — analysis runs on the committed values (§6).
4. **Open the analysis tab** for what you need:
   - **DLS** (size, R_h): tick the measurement(s) in the sub-tab's list and choose a method — **Correlogram** (cumulant / exponential) or **Distribution** (CONTIN / NNLS / Lognormal) — then **Run**. The multi-angle Γ **vs q**^{2} and concentration-series **D vs c** sub-tabs tick the points to include in the fit (§8).
   - **SLS** (M_w, R_g, A_2): select the sample, set the **calibration**, choose **Zimm / Berry / Debye**, then **Run** (§9).
5. **Read the results** — each method fills a results **table** plus a plot. An *apparent* result is read at one angle or one concentration, so it still carries that angle's or concentration's dependence; a *thermodynamic* one is extrapolated to zero angle and zero concentration. Results are labeled with which of the two they are, and an *uncalibrated* result is flagged as such (§6.5).

6. **Export** with **Export CSV...** (Origin-compatible) (§13).
7. **Keep the workspace** for the next launch: tick **Reopen last session on startup**, the **Session** row at the end of the Settings tab (§7.4). It is off by default, and without it closing the program discards everything you loaded.
8. *(Optional)* Combine a sample's DLS R_h and SLS R_g on the **Cross-Sample** tab for $\rho = R_g/R_h$ and the scaling plots (§10).

---

3. Loading Data

The program reads data from several instruments plus a generic plain-text fallback (§3.1) into one common format, so measurements from different sources load into the same workspace and analyze the same way.

Every file load follows the same workflow:

1. Click **Load DLS correlogram...** (or **Load SLS intensities...**, or **Load count-rate trace...**) at the top of the sidebar and pick the file(s); you can select **several at once**.
2. The program reads each file and pre-fills any parameters it can find.
3. Confirm the parameters on the **Data** tab and press **Update** (§6).

Each file is recognized on its own. The program tries the instrument formats it knows (§3.1); if none matches, it falls back to a **plain two-column** file — a generic DLS file is asked for its delay-time unit and data form at load (§4.1), a generic trace for its units (§4.3), and a generic SLS file loads directly (§4.2). Only if that also fails does it report an unrecognized format. Files that cannot be read are listed at the end, and the rest still load. A file can also load with something to report: rows it could not read, channels it dropped, parameters it does not carry. The program then shows a **Loaded with notes** box. The box names the file and the problem, ends each note with the section of this manual that describes the format ("(User-Manual §5.2)"), and, where there is something to do, gives the remedy on the line below ("Re-export the file if it was not edited by hand.").

> **Note.** Pre-filled values from a file are suggestions, not locked inputs. You can always override what the file supplied — for example to correct an instrument software value or to apply a calibration you determined independently.

3.1 Supported File Formats

The table lists every format the program recognizes, the plain-text fallbacks last, and what each one loads as.

Format	Produces
Brookhaven DLS correlation function `.CSV` (§5.1)	One DLS measurement per file
Brookhaven raw ASCII DLS export `.dat` (§5.2)	One DLS measurement per file, with angle, wavelength, temperature, viscosity and refractive index pre-filled from the file
Brookhaven count rate history `.CSV` (§5.3)	One intensity trace per file
Brookhaven SLS intensity export `.CSV`, concentration series (§5.4)	One measurement per concentration (solvent blank included)

Format	Produces
Brookhaven SLS intensity export `.CSV`, single angular scan (§5.4)	The sample — plus the solvent blank when the acquisition recorded one beside it
Malvern Zetasizer export template `.txt`/`.csv` (§5.5)	One DLS measurement per row; pre-fills scattering angle, wavelength, refractive index, temperature, viscosity, mean count rate, and the sample/material/dispersant names
NanoTemper Prometheus Panta autocorrelation export `.xlsx` (§5.6)	One DLS measurement per file, nothing pre-filled
ALV correlator export `.ASC` (§5.7)	Every angle in the file as a separate DLS measurement (one sample), each with its count-rate trace
Malvern Zetasizer clipboard correlation text (§5.8)	Every record in the file as a separate DLS measurement, nothing pre-filled
Plain-text two-column DLS: delay, and $g_2(\tau)-1$, $g_2(\tau)$ or $g_1(\tau)$ (§4.1)	One DLS measurement per file; asks for the delay unit and data form
Plain-text two-column SLS: angle, intensity (§4.2)	One concentration per file; loads directly
Plain-text two-column trace: time, count rate (§4.3)	One intensity trace per file; asks for the time and count-rate units

---

4. Generic (Plain-Text) File Formats

The generic formats accept data from any source — home-built correlators, simulation output, collaborator data, spreadsheet exports — as long as it is a clean two-column numerical table. Because these files carry no metadata, every physical parameter is supplied in the Data tab.

4.1 Generic DLS Format

The program accepts a file that meets all of the following, and rejects one that does not:

- Exactly two columns of numbers; no header, comment, or blank lines.
- Comma- or tab-separated (auto-detected).
- Column 1: delay times (unit asked at load).
- Column 2: correlation values (interpretation asked at load).
- At least two data rows.

At load a **Generic DLS file** box asks about the file in four fields:

Field	Choices	Meaning
Delay-time unit	`s`, `ms`, `us` (µs), `ns`	The unit of column 1
Column 2 is	`g2(τ) − 1`, `g2(τ)`, `g1(τ)`	What column 2 holds
β **(only if g1)**	coherence factor $\beta \in (0, 1]$	Read only for `g1(τ)`
Baseline B (only if g2)	baseline $B > 0$	Read only for `g2(τ)`

Delays are shown in seconds afterwards. A `g2(τ) − 1` column is already baseline-subtracted, and from a `g2(τ)` column the program removes the baseline B you give, in that column's own units.

Every form is analyzed as $g_2(\tau)-1$, so the three load into the same workspace and fit the same way. A `g1(τ)` column, the field autocorrelation, is converted by the Siegert relation (the **Theory-and-Equations-Guide** §2.2.1), which needs the coherence factor β: an optical property of your instrument, determined independently of the file. The 0.8 the β field opens with is a placeholder, not a measurement.

> **Tip.** Unsure which form you have? Check the range of column 2. $g_2(\tau)-1$ starts near the intercept β (often 0.3–1.0) and decays to 0; unsubtracted $g_2(\tau)$ starts near $1+\beta$ and decays to 1; $g_1(\tau)$ starts near 1 and decays to 0.

Because g_1 is *normalized*, the program checks that a column declared `g1(τ)` peaks between 0.01 and 1.05 and refuses it otherwise, naming the peak it measured. A column of raw counts, a column in per cent, or an unsubtracted g_2 from an ordinary instrument all land outside that range. The one case the check cannot settle for you is an unsubtracted g_2 from a very low-coherence instrument (β below about 0.05), which peaks close enough to 1 to look like g_1 — so if the baseline has not been subtracted, choose `g2(τ)` and supply B.

4.2 Generic SLS Format

Each file is one concentration at multiple angles (one file per concentration).

- Exactly two columns; no header/comment/blank lines; comma- or tab-separated.
- Column 1: scattering angles in degrees, each strictly in (0, 180).
- Column 2: raw intensity values. A negative value, which some instruments report after background subtraction, is loaded as read, and the load notes count such values.
- At least two data rows.

> **Important.** Angle must be column 1, intensity column 2. If reversed, the program finds column-1 values outside (0, 180)° and refuses the file, suggesting the columns may be swapped.

For a Zimm/Berry/Debye dataset, load one file per concentration (including the solvent blank at $c = 0$) and set each concentration in the Data tab. The SLS tab assembles the dataset from the loaded measurements.

4.3 Generic Trace Format

A plain two-column `(time, count rate)` table — the counterpart of §4.1 for count-rate traces: exactly two columns; no header, comment or blank lines; comma- or tab-separated; at least two rows. At load the program asks for the **Time unit** (s, ms, min, h) and the **Count-rate unit** (cps, kcps, Mcps). The trace loads unattached; attach it to a DLS measurement in the sidebar (§7.3).

---

5. Instrument File Formats

Each section below describes one instrument's native export for reference: what the file holds, what it pre-fills, and what you enter by hand.

A DLS fit's residual weight reads three inputs a file may state, and each section below says which of them its file carries:

- the **per-channel sample time**, the interval each correlator channel accumulates over; with-

out one, the fit derives a sample time from the delay axis and records that it did (§8.2.3);
- the **run duration**, the length of the acquisition;
- the **mean count rate**, which can also be committed on the Data tab, from a trace or by hand (§6.1).

Without a run duration or a count rate the weight takes a form that does not need it; what the weight does with each input, and without it, is the **Theory-and-Equations-Guide** §3.2.9 (the weight's inputs).

A run duration a file states — the Brookhaven raw ASCII export (§5.2) and the ALV export (§5.7) — is carried onto the measurement only when it is longer than every lag that collected counts. A stated duration no longer than the last lag that collected counts is left off the measurement: the load notes give it beside that lag, and the weight runs without it.

5.1 Brookhaven DLS Correlation Function (`.CSV`)

- Row 1: sample label string (e.g. `PnVP (40k) in Water - 6/1/2026 4:46:59 PM`).
- Row 2: column headers (`t(µs)`, `C(t)`, `Fitted (CONTIN)`).
- Rows 3+: delay time (µs), $g_2(\tau)-1$, and the instrument CONTIN fit.
- Trailing padding rows with $t = 0$ and $C(t) = -1$ fill unused correlator channels; they are stripped automatically.
- The file states none of the three inputs of the residual weight (§5): no per-channel sample time (its delay axis meets at the block boundaries without a duplicated channel), no run duration and no mean count rate.

  Important. The DLS file contains no instrument metadata. Angle, wavelength, refractive index, temperature, viscosity, and sample name are all supplied manually in the Data tab; concentration is optional for DLS sizing (§6.1).

  Note. The program ignores the Brookhaven CONTIN fit column; its own CONTIN fit is the Distribution sub-tab's (§8.3).

The correlogram and count-rate history are separate files and are associated manually (§7.3). (Sample labels in the two files sometimes differ slightly in spacing — a known Brookhaven quirk — which is why automatic pairing is not attempted.)

5.2 Brookhaven Raw ASCII DLS Export (`.dat`)

The same software also exports a raw ASCII file, described in Appendix A ("Raw Data File Formats") of the Brookhaven TurboCorr correlator manual. It is a different format from the `.CSV` above, and a more complete one.

- Lines 1–37: one number per line, a fixed list of correlator and sample parameters. Among them are the angle, wavelength, temperature, viscosity and refractive index — the values that must be typed in by hand for the `.CSV`.
- Then one row per correlator channel: delay time (µs) and the **raw channel counts**, not $g_2(\tau)-1$.
- Then four closing lines: sample ID, operator, date, and time.
- Among the parameters are the correlator's sampling times — one for each middle- and low-speed section, and one for each fixed high-speed delay — the run's total counts, its sample count and run length, and the mean count rate. On this correlator the delay time and the

sampling time are de-coupled: the lags of a section run at constant-ratio spacing while every channel of the section accumulates over the same sampling time.

- So the file states all three inputs of the residual weight (§5): each channel's sample time, from its section; the run length, under §5's rule for a stated duration; and the mean count rate. The derived static intensity (§9.2) reads the same totals.
- The sample times are read only when the section layout holds together:
  - the three sections' channel counts are whole numbers that sum to the channels in the data;
  - each section that has channels states a positive sampling time;
  - every high-speed delay is one of the correlator's fixed high-speed delays;
  - every channel's delay is a whole multiple of its section's sampling time.

  When any of these fails, no sample time is read and the load notes name the condition that failed; the fit then derives one from the delay axis and records that it did (§8.2.3).

Because the channels hold raw counts, a baseline is needed before there is a correlogram at all, and the file carries two candidates, a **calculated** and a **measured** baseline (its parameter lines 22 and 23). The program uses the calculated one unless the file names one itself; **Settings ▶ File Loading ▶ Brookhaven .dat baseline** (§12.3) describes both, says which fits the choice reaches, and changes the default. The program does not display the two values; a large difference between them in the file is a sign of dust or a slow drifting mode, whichever you pick.

Sample identity — solute, solvent, concentration — is not in the file and is entered in the Data tab, as for every other format.

5.3 Brookhaven Count Rate History (`.CSV`)

- Row 1: sample label string.
- Row 2: column headers (`Time (seconds)`, `Count Rate (kcps)`).
- Rows 3+: elapsed timestamp (`HH:MM:SS.sssssss`) and count rate in kcps.

Timestamps are elapsed time from the start of the acquisition, not wall-clock time. Count rates are shown in counts per second (cps) after loading.

5.4 Brookhaven SLS Intensity Export (`.CSV`)

The instrument writes SLS intensities in two layouts. Both are flat lists of key–value pairs sharing a metadata header, and the program tells them apart on its own — the same **Load SLS intensities...** button reads either.

Layout	What the file holds	Loads as	Entered by hand
Concentration series (the Zimm or Debye plot export)	A metadata header, an angle table, a concentration table and an intensity grid	One measurement per concentration	Sample name; temperature
Single angular scan (one concentration, such as a Guinier measurement)	One acquisition's intensity per angle, and a blank's when one was measured with it	One measurement, or two with the blank	Sample name; temperature; concentration; $\mathrm{d}n/\mathrm{d}c$

The concentration series pre-fills every parameter but the two in the table; the SLS export has no temperature field. Its grid rows read `Intensity - Concentration N - Angle M, value`.

The single angular scan has no concentration table. Its rows read `Sample Intensity - Angle N`, and `Liquid Intensity - Angle N` when a blank was measured in the same acquisition; that blank loads as a second measurement at $c = 0$, marked **solvent ref** in the sidebar. For a solvent or calibrant scan, enter a concentration of **0**, which files it as the solvent blank of its sample. The $\mathrm{d}n/\mathrm{d}c$ may be left blank (§6.5).

Load several single angular scans into one sample and each series is referenced against the blank from **its own file**: a blank drifts with cell cleanliness, alignment and the temperature of the index-matching vat around the cell between acquisitions, which is why the instrument records one per acquisition. A series whose file carried no blank (the plain export, a separate-file blank, or a generic loader) uses the sample's solvent blank, the first one loaded. A blank you pick in the series' **Solvent blank** row on the Data tab takes the place of either (§6.1).

Every blank beyond the first is kept; the sidebar marks it **extra blank** because it is not the sample's solvent blank, not because no series uses it. When an SLS load leaves a sample with an extra blank whose intensity differs from the solvent blank's by a factor of 1.02 or more at an angle they share (at their mean intensities when they share none), the load opens **Multiple solvent blanks**: it gives the largest ratio between the blanks and the number of samples affected, since the excess Rayleigh ratio, and the M_w, A_2 and R_g computed from it, depend on which blank a series is referenced against. Blanks that agree, or whose intensities cannot be compared, open nothing.

Removing a blank (sidebar right-click ▶ **Remove measurement**, or the Delete key) re-pairs every series that used it by the same rules. The confirm names each series that will switch and the blank it will use instead, or says that it will have no measured blank left, so the SLS tab asks before fitting it (§6.6).

The single angular scan has no calibration block either, so M_w and the absolute A_2 need a calibrant point entered in the SLS tab, exactly as for an uncalibrated concentration series (§6.5 says what that reaches). A sample with **no solvent blank at all** can still be analyzed, once you confirm it at **Run** — see §6.6.

> **Note.** The Brookhaven software embeds its own toluene Rayleigh ratio. It is read and recorded for traceability but not used in analysis — the program computes the temperature- and geometry-corrected value from the toluene Rayleigh literature. See the **Theory-and-Equations-Guide** §1.6.2 (the toluene Rayleigh ratio).

5.5 Malvern Zetasizer Export Template

The Zetasizer software's **export** is configurable, so the program reads its columns by **name** (not by position) and accepts any column order. To produce a file the program recognizes, configure the exporter as follows:

- **Settings:** include the header row; use commas as the separator.
- **Parameters (any order):** Sample Name, Material Name, Dispersant Name, Dispersant RI, Temperature, Viscosity, Wavelength, Scattering Angle, Mean Count Rate, Correlation Delay Times, Correlation Data.

The two correlation blocks (Delay Times + Data) are required; each named parameter is optional and pre-filled when present. The dispersant name is taken exactly as the instrument recorded it,

as any solvent name is (§6.3). The program maps the columns as:

Export column	Pre-fills
Sample Name	sample label
Material Name	solute name
Dispersant Name	solvent name
Dispersant RI	solvent refractive index
Temperature	temperature
Viscosity	viscosity
Wavelength	wavelength
Scattering Angle	scattering angle
Mean Count Rate	mean count rate
Correlation Delay Times	lag-time axis
Correlation Data	correlogram ($g_2(\tau)-1$)

Each column's unit is read from its own header where the header declares one, so a template that exports temperature in kelvin or the count rate in cps is read correctly; a column with no unit in its header is read in the unit the template above uses for it. Columns the template carries that are not listed above are ignored, and a parameter whose unit or value the program cannot use is left blank for you to fill in rather than guessed at. The **Correlation Delay Times** column is the exception: its unit sets the lag axis itself, so a unit the program does not recognize there stops the file loading rather than leaving a blank. Concentration is entered at the confirmation step, as with any DLS file.

5.6 NanoTemper Prometheus Panta Autocorrelation Export (`.xlsx`)

The Panta writes its autocorrelation function to a spreadsheet with two worksheets: a `Chart` sheet, which the program ignores, and an `ACF` sheet holding the data under the headers `Tau [µs]` and `Autocorrelation`. The delay unit is read from that first header rather than assumed, so an export written in different units still loads correctly.

Two things about this format are worth knowing.

- **The file carries no measurement parameters.** There is no scattering angle, wavelength, refractive index, temperature, viscosity, concentration or sample name anywhere in it, so every parameter is entered at the confirmation step and nothing is pre-filled. The scattering angle deserves particular care: it enters the recovered size through the scattering vector q (the **Theory-and-Equations-Guide** §1.2), so an angle entered wrongly rescales every result from the file without anything looking out of place. Take it from your instrument's documentation rather than from memory. The file also states none of the three inputs of the residual weight (§5): no per-channel sample time, no run duration and no mean count rate.
- **Repeated delay times are expected, and are removed on load.** The file lists each multi-tau block boundary twice (the **Theory-and-Equations-Guide** §2.2.2); the program keeps the finer block's value, drops the duplicate, and a note at load says how many channels were removed. Which of the two values is kept has no bearing on any result you would report.

5.7 ALV Correlator Export (`.ASC`)

An ALV `.ASC` export (the ALV-5000/6000/7000 family) holds a header of `Key : value` lines, a `Correlation` block of lag time and $g_2(\tau)-1$ per detector channel, a `Count Rate` block of time and count rate per channel, and the ALV's own cumulant fits, which are ignored — vendor fit results are display-only, as for the Brookhaven CONTIN column. Two layouts are told apart on load:

- A **multi-angle** file (e.g. ALV-7012 CGS-12F, with per-detector `Angle(1)`…`Angle(12)` keys) loads as one DLS measurement per active angle, each with its own count-rate trace already attached (§7.3).
- A **single-angle** file (e.g. ALV-7004/USB, a bare `Angle [deg]` key) loads as one measurement with its trace.

The header's temperature, viscosity, refractive index, wavelength, angle(s), mean count rate per detector and run **Duration** pre-fill the Data tab and the fit's inputs; the solute name, the solvent name and the concentration are entered there. Of the three inputs of the residual weight (§5), the file states the mean count rate and, for a one-run acquisition, the run duration (its **Duration**, under §5's rule for a stated duration). It states no per-channel sample time, so the fit derives one from the delay axis, whose written lags carry rounding, and records that it did (§8.2.3).

A `Correlation` or `Count Rate` block that stops at a row the program cannot read, rather than at a blank line, the next header or the end of the file, is read up to that row and no further; the load notes give the line it stopped at and the rows read.

5.8 Malvern Zetasizer Clipboard Correlation Text

The tab-separated text produced by copying the correlation data out of the Zetasizer software: the first column is the shared lag-time axis in µs, and every column after it is one record's correlation coefficient ($g_2(\tau)-1$) under a `Record M: <label>` header. Every record loads as its own DLS measurement; a record column with a header but no data is skipped and listed at load. The file carries no measurement parameters, so every parameter is entered in the Data tab, as for the Panta export (§5.6); the structured export of §5.5 is the way to bring the instrument's metadata across.

---

6. The Data Tab

The Data tab is where every physical parameter is entered, checked and **committed**. The analysis tabs read only committed values; nothing you type reaches a fit until you press **Update**. Three of its fields — the solute name, the solvent name and the temperature — are a measurement's **identity**, which groups measurements into a **sample** in the sidebar (§7.1); until they are committed the sidebar shows the measurement as an **(unconfirmed sample)**.

To confirm a measurement's parameters:

1. Click the measurement in the sidebar. The table shows its fields, with whatever the file pre-filled and whatever the solvent library proposes (§6.3) already in place.
2. Fill in what your analysis needs (§6.1 says what each field feeds). Identity and optics are shared across the sample, so entering them once is enough; press **Enter** to jump to the next field.
3. Press **Update**. The parameters are committed, the sidebar regroups the sample if its identity changed, and every analysis tab reads the new values (§6.2).

Tip. To set a value on several measurements at once, Ctrl- or Shift-click them in the sidebar, edit the focused one, and press **Update** (§6.2.1).

6.1 The Parameter Table

You enter and read each quantity in whatever unit is convenient. Each parameter row has a **Unit** column: a dropdown for quantities with a choice of units (the value re-displays when you switch units, and nothing else changes), or a fixed label for quantities with a single natural unit. The identity and label fields carry no units; the physical quantities do.

The table below lists the rows in the order the tab shows them. A DLS measurement shows the rows marked DLS, an SLS measurement those marked SLS, and both show the rest.

Parameter	Shown for	Units (default first)	Purpose
Label	both	—	A short name for the measurement (below)
Solute name	both	—	Names the scatterer, which need not be a polymer; part of sample identity (§7.1)
Solvent name	both	—	Part of sample identity; picks the library's refractive index n and viscosity η (§6.3)
Concentration	both	**mg/mL**, g/L, µg/mL, g/mL	Required for SLS and D vs c; optional for every other DLS analysis
Temperature	both	**°C**, K, °F	Part of sample identity (§7.1); sets the Stokes–Einstein R_h and the library's properties
Scattering angle	DLS	°	The detector angle; sets the scattering vector q
Wavelength	both	nm	The laser wavelength; sets q
Solvent refractive index	both	—	Sets q; proposed by the library (§6.3)
Viscosity	DLS	**mPa·s**, Pa·s	The solvent viscosity η for Stokes–Einstein; proposed by the library (§6.3)
Mean count rate	DLS	**kcps**, cps, Mcps	The measurement's average count rate, for the residual weighting and Derived Intensity (below)
$\mathrm{d}n/\mathrm{d}c$	SLS	mL/g	The refractive-index increment, the most sensitive SLS input; may be left blank, and zero is refused (§6.5)
Solute refractive index	both	—	The particle's own index, for the **Mie sphere** form factor (§9.1) and the Distribution sub-tab's Rayleigh–Debye–Gans (RDG) and Mie weightings (§8.3.3); hand-entered
Shell refractive index	SLS	—	The shell's index, for the **Core–shell sphere** form factor with the solute index (§9.1); hand-entered
M_w fraction	both	—	An optional, case-sensitive label separating molecular-weight fractions of one solute (§7.2)
Polarization (DDLS)	DLS	—	The correlogram's polarization, VV, VH or VU, for DDLS (§8.7); "—" for ordinary DLS
Chain type	both	—	An optional tag for the branching factors (below)
Solvent blank	SLS	—	The $c = 0$ measurement subtracted from this series (below)

Parameter	Shown for	Units (default first)	Purpose
Solvent blank intensity at 90°	SLS	the file's own	Stands in for a blank that was never measured; hand-entered (§6.6)

The **Label** is shown wherever the measurement appears: the sidebar, the legends, the results tables and the warning lines. It adds to the sample and the angle rather than replacing them, so a label such as `3` is enough; left blank, the source file name is used, without its extension. It affects no grouping and no analysis.

The **Chain type** tag serves the Cross-Sample branching factors g and h (§10.1.4): **Linear** marks a reference that defines the linear R_g–M_w and R_h–M_w laws, and **Branched** a target that gets g and h. It is entered once and shared across the sample; the "—" entry leaves it untagged.

On an SLS series at $c > 0$, the **Solvent blank** row's first entry, **auto**, uses the blank recorded in the series' own file, else the sample's solvent blank; pick one of the sample's other blanks when the file carried none or the automatic pairing is wrong. The dropdown's tooltip says which blank is in use and why (§5.4). A blank is named by the last two parts of its path (`run1/SLS.csv`), extended one folder upward until no two of the sample's blanks read alike (`water/run1/SLS.csv`); two that only their ID tells apart end with it in brackets (`run1/SLS.csv [m0003]`). The tooltip, the removal confirm and the export use this name; the dropdown shows it with the blank's ID in brackets.

A count-rate trace has no row here: its units are the file's, or chosen when a plain-text trace loads (§4.3).

The delay window a DLS fit reads is not a Data-tab field: it is set on the DLS tab (§8.1), in µs by default.

The **Mean count rate** is read by the Nonlinear cumulant's and the distribution fits' default residual weight (§12.1.3, §8.3.2) and by Derived Intensity (§9.2). It is the one parameter more than one place can propose. A trace attached to the measurement (§7.3) proposes its mean; the instrument file's own value is offered as an alternative where the format records one. A replicate average (§7.1) arrives with the mean of its runs' committed rates, and has no file's own value to offer. The **From...** button beneath the table lists every offer, and a line beside it names the source in use; the value follows the rule for every proposed value (§6.3). If your value disagrees with the linked trace by more than 5 %, a note below the table says so; the value is still used exactly as you entered it.

Below the table, three notes appear when they have something to say: the solvent library's (a solvent it could not fill in, a condition outside its range — §6.3), a plausibility check on the values typed, and the count-rate disagreement above. A legend under the table explains the **teal dot** that marks every proposed value (§6.3).

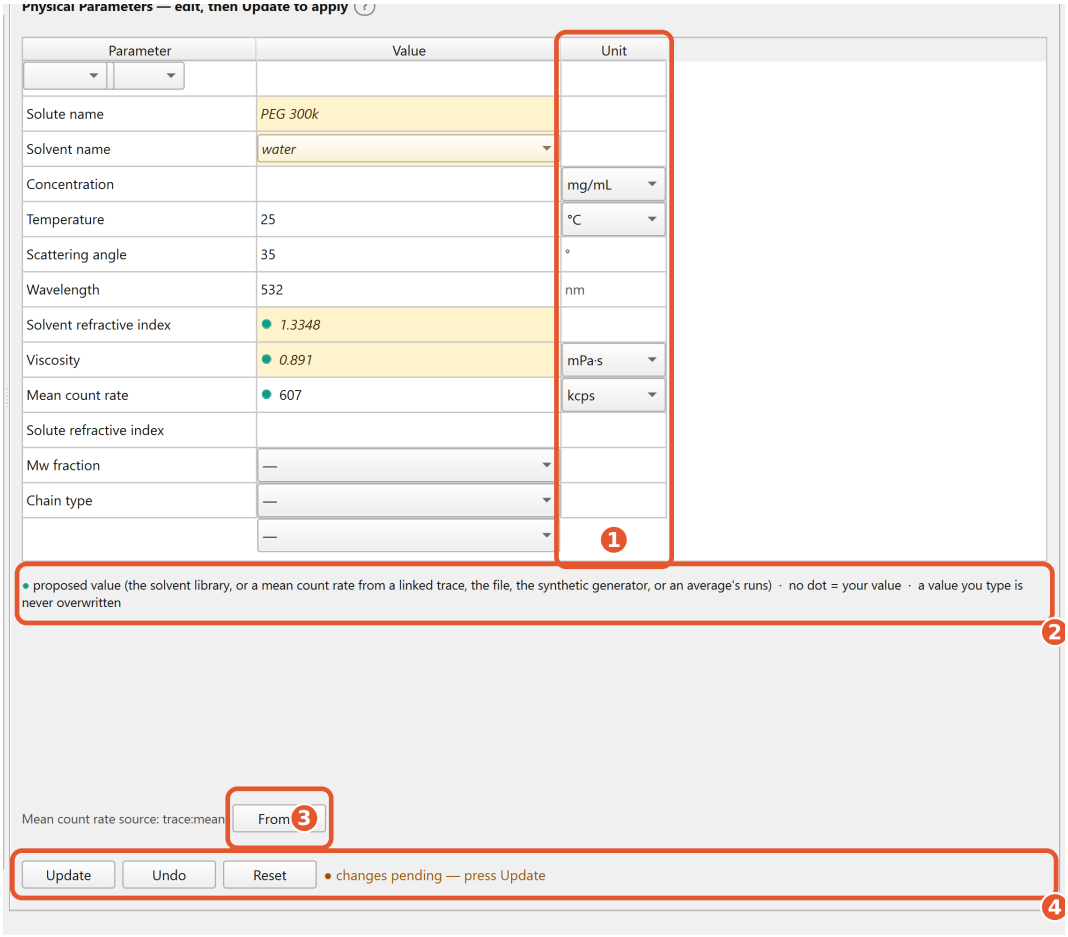


Figure 1: The Data tab with an edit pending. Each row has a **Unit** column (1): a dropdown where the quantity has several units, a fixed label otherwise. The legend under the table (2) explains the teal dot that marks a proposed value — here the solvent library's refractive index and viscosity, and the mean count rate from the linked trace. **From...** (3) lists the sources the mean count rate can be taken from. The **Update** / **Undo** / **Reset** row (4) shows the pending indicator and the highlighted cells while edits are not yet committed.

6.2 Update, Undo and Reset

The row beneath the table holds the three buttons that act on the whole edit, and a pending indicator.

- **Update** commits the edited parameters to the sample. If the change regroups samples — a corrected solute, solvent or temperature — the stored results follow the measurements (§7.1), and you are asked first if a merge would drop a hand-entered or picked value. Analysis tabs re-read the committed values; a fit already on screen that read a changed value is marked out of date rather than recomputed (§8.1).
- **Undo** steps back to the last committed parameters. With uncommitted edits showing, the first Undo discards those edits; the next steps back through earlier commits. Undo restores parameters, never results, so an Undo that splits a merged sample back into the samples it came from asks first, as the merge did, when a hand-entered or picked value would be dropped

(§7.1.2).

- **Reset** clears every entered parameter for this sample — the scattering angle and the raw data are kept — after a confirm that says how many measurements it reaches. Nothing is applied until you press **Update**, and **Undo** restores the previous values.
- The **changes pending — press Update** indicator, a dot and those words, appears beside the buttons while the table holds edits that are not yet committed, and the changed cells are highlighted until they are. Switching to an analysis tab with edits pending analyzes the *committed* values, not the ones on screen.

6.2.1 Editing several measurements at once

Ctrl- or Shift-click measurements in the sidebar, edit the focused one, and press **Update**: what you type, pick or clear reaches every highlighted measurement whose table has that row. Three kinds of value travel differently:

- a value shared across the sample, identity or optics (§6, step 2), reaches each highlighted measurement's whole sample;
- a typed **Label** lands on each highlighted measurement numbered in sidebar order ("PEG 1", "PEG 2" …, zero-padded from ten measurements);
- a proposed value (§6.3) stays where it was proposed: after a solvent, temperature or wavelength edit, each highlighted sample re-derives its own library refractive index and viscosity, and a **Mean count rate** taken with **From...** stays on its own measurement.

A refractive index or viscosity left blank on the focused measurement, cleared by you or by the library at the new condition, is blanked on each.

6.3 Solvent Names and Proposed Values

Solvent identity is part of sample identity: the same solute at the same temperature in two solvents is two samples, so the two can be compared.

A name is stored **exactly as you enter it** and matched **ignoring capitalization and surrounding spaces — nothing else is normalized**. `Water`, `water` and `WATER` are one solvent; `H2O` is a different one. The program does not decide that an abbreviation stands for a particular solvent, because it has no way to be right about that — a name you type is a name you own, and `My Phosphate Buffer` is as valid an entry as `toluene`.

> **Note.** The rule is the same for the solute name and the solvent name: both ignore capitalization and surrounding spaces, and neither is interpreted any further. `PVP` and `pvp` are one sample.

For common solvents the program can **propose** the two optical/hydrodynamic properties every scattering calculation needs — the **refractive index** n (which sets the scattering vector) and, for DLS, the **viscosity** η (which sets the Stokes–Einstein radius). The *Solvent name* row is a dropdown listing every library solvent in two sections — **Measured properties** and **Estimated properties**, the library's **primary** and **estimate** tiers as the Solvent Explorer's badge names them (§11.4) — followed by **Other...**.

Pick a solvent from **Measured properties** and n (and η) fill in automatically from the sample's temperature and wavelength, each marked with a small **teal dot**. This is the one rule for every proposed value in the program — the solvent properties here, the mean count rate above, and the

standard's refractive index in the SLS calibration panel (§9.1):

- A proposed value **re-derives** whenever the inputs it came from change (here, the temperature or wavelength).
- Each proposed value is **rounded at the last digit the library can stand behind**. The uncertainty itself only chooses that place; it is never used in any analysis.
- A value you **type by hand is never overwritten** — not by re-loading, not by re-deriving — and shows no dot. To return to the proposal, clear your entry (and, for a solvent property, re-pick the solvent).
- Where nothing can be proposed the program **does not guess**: it leaves the field blank and shows a note saying why. For the library that is a condition outside a solvent's validity range, or a solvent with only estimated properties.

Solvents under **Estimated properties** are listed but **not** auto-filled — their underlying data is too weak to put a number into an analysis unasked. Picking one sets the name, leaves n and η blank, and shows a note saying so; you enter both yourself. Use the Solvent Explorer (§11.4) to see what the library's estimate would be.

For a solvent the library does not carry, choose **Other...**. It opens a text box beside the dropdown; the name you type is matched by the rule above, and n and η are yours to enter.

The measured-properties solvents and their validity ranges are tabled in §6.4; the uncertainties and sources behind them are in the **Theory-and-Equations-Guide** §1.1 (the solvent inputs).

6.4 The Solvent Library at a Glance

The table below lists the **primary**-tier solvents — the higher-confidence set that auto-fills n and η in the Data tab (§6.3) — with the range of conditions each one covers. Ranges are **per-solvent**, not uniform: they follow the span of the underlying measurements, so refractive index and viscosity may be valid over different temperature windows for the same solvent. Outside a listed range the program stops proposing a value and asks you to enter it by hand (§6.3). The uncertainties behind these ranges, and the sources they come from, are in the **Theory-and-Equations-Guide** §1.1.

Solvent	RI λ-range (nm)	RI T-range (°C)	η T-range (°C)
acetone	476–633	15–40	−18 to 46
benzene	476–633	15–40	15–66
cyclohexane	476–633	15–40	15–76
ethanol	476–633	15–40	−20 to 70
ethylene glycol	400–800	1–45	0–71
glycerol	400–800	1–45	15–51
hexane	400–640	15–40	−3 to 66
methanol	450–700	15–50	0–70
toluene	405–830	15–40	−13 to 96
water	400–800	15–60	0–70

The library also carries 9 **lower-confidence (*estimate*-tier)** solvents, drawn from thinner or less consistent source data, named as the **Solvent name** dropdown lists them: `acetonitrile`, `ccl4`, `chloroform`, `cs2`, `dcm`, `dmf`, `dmso`, `ethyl acetate`, and `thf`. They appear in the Solvent Explorer

(labeled *estimate*) but are held back from the Data-tab auto-fill, so a value only ever reaches an analysis when you pick it there yourself.

6.5 When You Have No dn/dc, or No Calibration

The refractive-index increment is the one SLS parameter that is sometimes genuinely unavailable: no value has been published for your solute in your solvent, or you are measuring a particle dispersion, for which the quantity is not defined at all.

Leave the cell blank. Every analysis still runs, and the result divides into two kinds:

- **Unaffected** — every R_g, the form-factor sizes and axial ratio, and the calibration-free $2 \cdot A_2 \cdot M_w$. These come from *ratios*, and the unknown optical factor cancels out of a ratio exactly.
- **On an arbitrary scale, and flagged as such** — the absolute M_w and A_2. They are dashed out at the value, with a line under the results naming dn/dc as the reason.

This is the same treatment an **uncalibrated** run gets — a run with no calibration constant k_c (§9.1) multiplies every point by the same unknown factor, so the same quantities survive and the same ones are flagged. The two are reported separately: if a value is dashed, the label tells you whether to calibrate, to enter a dn/dc, or both. Calibrating will not repair a missing increment, so the program never implies that it would. Everywhere else in this guide, "the scale cancels, so R_g survives" points here.

> **Important.** Do not type 0 to get past the field. It is refused, and for a reason: zero is not "unknown", it is the claim that the sample does not scatter differently from its solvent, which makes every fitted quantity meaningless. Blank is the honest entry. A **negative** dn/dc is fine — the optical constant depends on the square, so the sign cancels, and a solute less refractive than its solvent really has one.

6.6 When You Have No Solvent Blank

The $c = 0$ solvent measurement is the other SLS input that is sometimes simply missing — the acquisition did not record one, or you want a quick look at a strongly scattering sample before committing to a full measurement. The **Solvent blank intensity at 90°** row lets the SLS tab run without one, once you confirm the **No solvent blank measured** prompt at **Run** (§9), in one of two ways:

- **You typed a value** — the pure solvent's reading at 90°, in the units of the loaded intensity column, before dark-count subtraction. The blank at each angle is then the dark count plus the rest of that reading divided by $\sin\theta$, so it is not a uniform subtraction: the number you type is the 90° value, and the solvent term at 30° is twice as large. The form, the assumptions behind it and where they are weakest are the **Theory-and-Equations-Guide** §1.7.2.
- **You left it empty** — or entered zero, which here means the same thing (unlike a zero dn/dc, which is refused, §6.5) — and no solvent is subtracted at all. The detector dark count is still removed.

A measured blank always wins where one exists; the field grays out then, and a value typed earlier is kept but inert. Removing a sample's last blank makes it the blank in use again at the next Run. What such a run produces, and how it is flagged, is §9's.

---

7. How the Program Organizes Data

Loaded measurements are grouped into samples by their identity and shown in the sidebar (§7.1), split by molecular-weight fraction when you tag one (§7.2), joined by their count-rate traces (§7.3), and kept between launches only when the autosave is on (§7.4). This chapter is the reference for each of those.

7.1 Samples and the Sidebar

A measurement's **identity** is its solute, its solvent and its temperature to the nearest 0.1 °C. Measurements that share one are grouped into a **sample**; concentration and angle are axes *within* it. The identity is also what pairs a DLS and an SLS measurement of the same sample for $\rho = R_g/R_h$.

The sidebar on the left is where you navigate and manage the data: click a measurement to load it into the active tab, or a sample or DLS/SLS heading to select everything under it; right-click a selection of measurements to move it to a new or an existing sample, average replicates, or remove the measurements or their whole samples, and right-click a sample or DLS/SLS heading to remove what it holds. The sidebar only *navigates* — each analysis tab decides what it actually analyzes (you **tick** measurements, or pick a sample from the tab's own **Sample** dropdown). The sidebar also **mirrors** the active tab's current selection — those measurements read **bold blue** — and shows a sample whose identity is not yet committed as an **(unconfirmed sample)**, in amber (set its identity in the Data tab and press **Update**).

Stored results **follow the sample, not its name**: correcting a typo in a sample's solute, solvent, or temperature (or moving all its measurements into another sample) carries the stored M_w, R_g, A_2 and R_h values along to the regrouped sample, but not the Γ vs q^2, D vs c, Mode Map and DDLS fits drawn for the sample it leaves (§8.1.6).

Right-click replicate DLS measurements and choose **Average correlation functions → new measurement** to average their correlograms channel by channel into one new measurement, named `avg of N` until you label it. It is a cleaner curve to fit, with no ± of its own; the replicates must pass the repeat-set check of §7.1.3 and share one lag grid. The average takes every parameter but its label and its mean count rate from the first measurement selected — the temperature, viscosity and refractive indices included — and joins that measurement's sample. Its mean count rate is the mean of the runs' committed rates when every run has one; when any run has none, the average has none either. Its tooltip, wherever its name appears, and the Comments note of an export of it record what went into it:

- the runs' temperatures and their span, a zero span included;
- the spread of the runs' amplitudes and the run whose amplitude lies furthest from the others', with its signed deviation;
- the runs' count rates and how far apart they lie, the highest against the lowest, or, when some runs had none, how many.

These are records, with no threshold and no flag; how a run's amplitude is measured, and why it is recorded, is the **Theory-and-Equations-Guide** §6.4 (the per-run amplitude). An average beside its own source runs is left unticked by default on the Γ vs q^2, D vs c and Mode Map sub-tabs (§8.4, §8.5), and DDLS fits it in their place (§8.7).

Two dialogs are raised from the sidebar rather than from a tab (§7.1.1, §7.1.2); the check every

average makes of its selection follows them (§7.1.3).

7.1.1 Average derived results...

1. Select replicate DLS measurements of one condition in the sidebar, right-click them and choose **Average derived results...**.
2. In the prompt, choose which fit each replicate gets, and tick what it takes from the DLS tab: **Use the Correlogram tab's delay window and baseline** (§8.1.2), and, for a cumulant average, **Use the Correlogram tab's cumulant order**, or for CONTIN, NNLS or lognormal, **Use the Distribution tab's settings**. Unticked, the average runs on the Settings values the prompt shows (§12.1–12.2).
3. Press **OK**. A ticked copy is taken then, once. The program fits each replicate and opens a **DLS replicate average** box with the result, the summary this section reads.

The summary reports the mean and its standard error: the replicates' standard deviation divided by the square root of their number (the **Theory-and-Equations-Guide** §6.4). It is the one $R_h \pm$ available at a single angle and concentration (§8.1.7). The summary also says which settings and which region the fits read and, when the runs' temperatures differ, their span; the row the average writes in the **Summary** table (§8.8) names the cumulant order it ran at and, when it is not zero, the temperature span of the fits it averaged. The summary repeats, once each, any remark the averaged fits carry (§8.1.7), with the measurements that carried it named in front; the mean, its ± and every flag are unchanged by it. The average replaces a picked R_h on the Cross-Sample tab (§10.1.2).

When any averaged fit was flagged, a ⚠ line in the summary says how many were and names each failed check once, with the range of its values and, when there are several, how many fits carry each; each run's own fit on the DLS tab shows its note. The mean and its ± are still reported and carry that caveat. A value the average writes carries the same verdict: its row in the **Summary** table is marked (§8.8) and its label names the causes, so the flag goes with the value to the Cross-Sample tab. The line appears for every method; a double exponential is never written to the sample, so for it the summary is the only place. In a cumulant average, when any averaged fit's μ_2 rests on its bound, the PDI line shows the mean without a ± and says how many fits sat on the bound, because their PDI is the bound rather than a measurement (§8.2.3); the other rows keep their ±.

A CONTIN, NNLS or lognormal average lists the peaks by size rank, and its summary carries these lines:

- the peak line, one per position, *Peak 1 (seen in k/N runs): geometric-mean Rh = …*: the mean of the runs' peak positions (§8.3.3), its ± and the mean weight;
- on the peak line, where every run puts that peak's weight on one grid point inside the R_h grid: the ± includes the grid's resolution, and the line says so;
- on the peak line, where a run puts it on one grid point at an end of the grid: the count of those fits, because the position may lie beyond the grid;
- a ⚠ on a position whose averaged peaks include flagged ones (a maximum on an end of the grid, a decay that never finished, a size outside the range, §8.3.3), with each cause and its runs on the lines under it; the position's mean and ± are unchanged and carry that caveat;
- a ⚠ line after the peaks: the count of flagged peaks at each position;
- an ⓘ line, when the fits ran under different residual weightings: which weightings, and how many fits ran under each;
- the closing line: each position is the peak's geometric mean, and the ± is a reproducibility

spread, with the grid resolution where stated.

What the ± holds is in the **Theory-and-Equations-Guide** §6.4 (the replicate ±).

Besides being a repeat set (§7.1.3), the selection must lie in one sample. The sidebar groups temperatures to the nearest 0.1 °C (§7.1), so a replicate logged just across that boundary lands in a sample of its own and the action refuses the selection as spanning more than one sample. Right-click that measurement, choose **Move to existing sample** and pick the others' sample, and the average runs.

The averaged R_h and its standard error are written into the sample when at least two replicates fitted completely and the fit gives one R_h — the cumulant, the single exponential or KWW; a double exponential's two modes and a distribution's peaks are reported only. Even for the cumulant, the single exponential or KWW, the averaged R_h and its standard error are not written, and the summary says why, when:

- fewer than two fits are complete;
- the mean is not finite;
- the sample already carries a hand-entered R_h;
- any fit returned a size outside the range DLS can establish.

The mean and its spread are still reported.

An average can also remove an earlier one. When you average the cumulant, the single exponential or KWW again within one sample and one M_w fraction and the new average fails, the earlier replicate average of that method is removed from the **Summary** table (§8.8) rather than left standing, and the sample R_h goes with it when it came from that average. The new average fails for the first, second or fourth reason in the list above, or when the selection is refused because its runs are not repeats of one condition or one of them is itself an average. The summary, or the refusal's message, names the value it cleared. A selection that spans samples or M_w fractions clears nothing, and neither does an average of a method that writes no R_h to the sample. A value you entered or picked yourself is never removed, and when you have picked one, re-averaging still refreshes the row.

7.1.2 The regroup / merge confirm

When a regroup would **merge** two samples that both hold results, each quantity keeps the higher-provenance value (hand-entered beats picked beats computed, and the surviving sample wins a tie) — and the program asks first whenever a hand-entered or picked value would be dropped; **Cancel** applies nothing. **Update** raises it, and so does **Undo** on the Data tab when it would split a merged sample back into the samples it came from.

7.1.3 Repeat sets

A selection is averaged only if it is a repeat set. Every measurement must agree on:

- solute and solvent;
- scattering angle and wavelength;
- concentration;
- M_w fraction and polarization.

A selection that does not is refused, naming the field. Temperature, viscosity and the refractive indices are not compared. Temperature is recorded instead, because repeats drift by a few hundredths

of a degree between runs; in **Average derived results...** each fit reads its own measurement's values, and they are yours to check. A measurement that was itself made by averaging is refused as well; select its source measurements instead. Average each concentration and each angle separately: a concentration series is deliberately one sample and an angular scan has its own analysis (§8.4). The check governs **Average derived results...** (§7.1.1), the correlogram average (§7.1) and the derived-intensity repeats (§9.2), whose derived point records the runs' temperature span in its export when it is not zero; the DDLS averaging (§8.7) compares the sample identity, temperature to the nearest 0.1 °C included.

7.2 Molecular-Weight Fractions

A scaling study measures several **molecular-weight fractions** of one solute (same chemistry, solvent, and temperature), each with its own concentration series. Because molecular weight is *not* part of the sample identity, all such fractions group into one sample. To keep them separate, give each measurement an optional free-text M_w fraction label (e.g. `250k`, `1M`) in the Data tab; editing it offers to apply the label to every concentration loaded from the same file. The label is *not* required — leave it blank for single-fraction samples or systems with no meaningful M_w (e.g. nanoparticles).

> **Important.** Tag before you fit. If you load more than one molecular weight of the same solute and leave them untagged, their concentration series share one fraction and an SLS fit pools them into a single meaningless mixed-M_w result that no flag catches and nothing on screen marks, because the program cannot tell two such series from replicates of one. Tag each series with its own M_w fraction to fit them separately; replicates of one molecular weight need no tag.

Within a sample, the fraction label **partitions the analyses**: the SLS tab shows a **Fraction** selector and runs an independent Zimm/Berry/Debye fit for the selected fraction, while the $c = 0$ solvent blank follows each series (its own file's blank, else the sample's solvent blank — the Data tab's **Solvent blank** row) regardless of fraction. The DLS tab partitions the same way, by the focused measurement's own fraction: the Γ vs q^2 and D vs c tables gray out rows of any *other* fraction (focusing an untagged measurement fits only the untagged series), so a single fit never mixes fractions. Each fraction therefore yields its own M_w, R_g, A_2 (and, from DLS, R_h). In the Cross-Sample tab each fraction contributes its own point to the scaling plots (R_g–M_w, A_2–M_w) and its own row in the Architecture sub-tab's table, so a molecular-weight series produces a proper scaling curve rather than a single merged point.

7.3 Traces in the Sidebar

Intensity traces (count-rate histories) load and analyze on their own, without a correlogram, and can be linked to a DLS measurement afterwards — so a concentration series' traces can be checked together.

A trace is loaded from **Load count-rate trace...** in the sidebar, beside the DLS and SLS load buttons. An ALV `.ASC` loaded as a correlogram brings its own trace with it, already attached to that measurement — attached traces appear nested under their measurement, unattached ones in a **Traces** group at the bottom. **Right-click a trace** to attach it to a DLS measurement, detach it, export it as `(time, count rate)` columns, or remove it. A measurement carries at most one trace, so attaching a second detaches the first (which becomes standalone — it is not deleted). Removing a measurement deletes the trace attached to it, and the confirm says how many before it happens.

An attached trace proposes the measurement's **Mean count rate** on the Data tab (§6.1). The **Utilities ▶ Traces** sub-tab (§11.1) chooses which traces are *plotted*; attaching and detaching happen here in the sidebar.

7.4 Sessions

The program's persistence is a single autosave, controlled by **Reopen last session on startup**, the **Session** row at the end of the Settings tab's Appearance group (§12.9; off by default) — there is no separate File ▶ Save or Open.

When it is on, closing the program keeps the whole workspace in a file the program manages for you, and the next launch reopens it automatically: every loaded measurement and trace, the sample results, the calibration (including any not-yet-committed edit), the SLS masks, the Summary table, and the DLS tab's Mode Map picks and recorded mode sets (§8.5) all come back exactly as you left them. A fit is recomputed on demand rather than kept, so a form-factor source on the Cross-Sample tab is offered again only after you re-run the fit (§10.1).

If that file is missing or damaged, the program starts empty. A session saved by an earlier version that stored its results' notes as sentences still opens, each note shown as it was written; a degenerate form-factor solution such a session picked keeps its stored value but matches none of the solutions offered until you pick one again (§10.1.3). A session this version cannot read — written by a version it does not support, or holding a note it does not know ("This session holds a note this build does not know (…): it was saved by a build whose notes differ. Start a new session here.") — is refused whole rather than half-read: you are asked at startup whether to keep the unreadable file (nothing is written over it) or let this version replace it the next time you quit.

---

8. The DLS Tab

The DLS tab extracts particle dynamics — decay rates, diffusion coefficients, and hydrodynamic radii — from a measured $g_2(\tau)-1$. It is organized into seven **sub-tabs**, each a persistent full-size view (running one never clears another), in this order: **Correlogram** (§8.2), **Distribution** (§8.3), Γ **vs** $\mathbf{q}^2$ (§8.4), **Mode Map** (§8.5), **D vs c** (§8.6), **DDLS** (§8.7) and **Summary** (§8.8). For the equations and the literature behind each method, see the **Theory-and-Equations-Guide** §3.

To fit a correlogram:

1. Tick one or more measurements in the **Measurements to Plot** checklist (§8.1).
2. Set the delay window and baseline region on the **Correlogram** sub-tab, or leave them unset to fit each measurement over its own full lag range (§8.1).
3. Choose a method in the **Method** box (or, on the **Distribution** sub-tab, tick the distribution methods you want).
4. Click **Run**, and read the result in the table and the plot.

Each sub-tab's own procedure is at the head of its section.

8.1 What Every Sub-Tab Shares

8.1.1 The measurements checklist

The **Correlogram** and **Distribution** sub-tabs carry a **Measurements to Plot** checklist of every loaded DLS measurement (any sample, angle, concentration, or M_w fraction); tick several to overlay them for comparison. This checklist — shared between the two sub-tabs and mirrored in the sidebar (ticked measurements read bold blue) — is the sole driver of what gets fit; selecting a measurement in the sidebar only navigates. Every measurement can be ticked, whatever its Data-tab parameters, since its raw correlogram is drawn either way; **Run** fits every ticked measurement that has the inputs its method needs (and, on the Distribution sub-tab, every ticked method), and the status line names the others (§8.1.4). Fitting many heavy distributions (especially CONTIN) at once is slow, so tick only what you want to compare. **Select all**, **Select none** and **Tick all at** a chosen angle or concentration are on every sub-tab that has a checklist.

8.1.2 The analysis region

A single shared region, set on the **Correlogram** sub-tab, defines the delay window and the baseline region. Set them by typing values into the **Delay window** min/max boxes or by dragging the markers on the correlogram: the green dotted lines are the window (points outside are ignored — useful to trim a noisy long-delay tail), and the purple dashed/shaded span is the **baseline region** — the parametric fits subtract the baseline measured over it before fitting (and show it in their **baseline** row), the nonlinear cumulant measures its floating baseline over it, and the distribution methods normalize by it. When a window and a baseline marker overlap, tell them apart by their offset caret handles: the window handles are green carets along the top of the plot, the baseline handles purple carets along the bottom — grab the top or bottom half of the plot to pick the one you want. The markers start at the full range of the ticked measurements and, for the baseline region, their last quarter of lag channels — counted in channels, not in delay time.

Nothing you have not set is applied, so a region proposed from one measurement never reaches one on another lag axis:

- Until you move a green handle, each measurement is fitted over its own full lag range.
- Until you move a purple one, each fit uses its own baseline estimate (the parametric fits none).
- **Reset window + baseline** returns to that state.
- Nothing outside the delay window counts: a baseline region you placed is read only over its part inside the window — the part outside is drawn gray and not used — and one with no channel inside the window counts as not placed.
- A ticked measurement whose lag axis does not overlap a window you set can have too few points inside; it is then skipped with a note.
- **Skip initial lag channels** (§12.1.4) composes with the window minimum, the later of the two winning.

Where the region reaches: the Correlogram and Distribution sub-tabs always read it. The paths that re-run a fit behind the scenes — the Γ **vs** $\mathbf{q}^2$, **D vs c**, **Mode Map** and **DDLS** sub-tabs and **Average derived results...** (§7.1) — read it only when you tick their **Use the Correlogram tab's delay window and baseline** box. The line under the box says what is currently set. The box is unticked for each new sample and not kept with the session, because one region serves every sample on the tab; ticking it, or moving the region while it is ticked, marks a shown result out of date, and the result's notes, its export and any Summary row it writes say it read the Correlogram tab's region.

8.1.3 Run, and results that are out of date

Only **Run** computes a fit, with one exception on this tab: on the DDLS sub-tab an angle tick refits a fit already on screen (§8.7). Switching sample, fraction or method keeps the last result on screen rather than blanking it. If you go back to the Data tab and commit a change to a parameter that fed a fit already on screen, the plot stays but a status line reads *"Inputs changed since this fit — press Run to refresh"* — so a result is never silently shown against parameters it was not computed with. The same hint appears when any other input of the shown fit moves:

- the delay window or baseline region, dragged or typed — on the Γ vs q^2, D vs c, Mode Map and DDLS sub-tabs only while their region box is ticked, and ticking or unticking it counts too;
- the cumulant order;
- the ticked set of a sub-tab that fits its ticked measurements together;
- a Settings change to any of the **DLS Analysis Defaults** (§12.1) or to the **Regression SE estimator** (§12.6) — each entry there says whether the change clears the result outright or marks it out of date.

The hint is specific to the results whose own inputs changed. The Correlogram and Distribution sub-tabs keep one fit per measurement and method, so on those two an out-of-date fit leaves the plot instead, and the status line names its measurement. A fit that finishes after you have clicked another sample or another M_w fraction is not painted over the new view; it waits in its own slot and shows when you focus that sample and fraction again. The SLS tab follows the same rule (§9), with one exception of its own.

8.1.4 What an analysis needs before Run

What the ticked measurements can run is judged before **Run**, from their committed Data-tab parameters and the ticks. An analysis they cannot run is grayed — a method box, a sphere model, a Mode Map **Source**, a table row — never hidden. A grayed box or entry keeps its tick or stays selected; a table row that stops being eligible loses its tick. Hover a grayed entry for what it lacks: its tooltip lists each missing input under *Needs:*, with the Data-tab row where it is entered or, for a count, how many are ticked against the minimum (• *at least two distinct angles (1 ticked, 2 needed)*). The sub-tabs judge in one of two ways:

- the Correlogram, Distribution and Mode Map sub-tabs fit or extract each ticked measurement on its own, so an entry grays only when no ticked measurement can use it; **Run** fits the measurements that can, and before Run the status line names each other ticked measurement, followed by *is left out:* and what to enter for it;
- the Γ vs q^2, D vs c and DDLS sub-tabs judge the ticked set as one: **Run** is off while the set cannot run, and the status line, headed *"Γ vs q^2 needs:"* (or *D vs c*, *DDLS*), lists what is missing.

A result already on screen stays while **Run** is off, and one whose inputs have changed is marked *"Inputs changed since this fit."*, without *press Run*; on the Correlogram sub-tab the same mark stands for an out-of-date fit of a measurement **Run** leaves out. A missing viscosity grays nothing on the Correlogram sub-tab, whose fits run without one and leave R_h reading *n/a* (§8.1.7). The table lists what each analysis needs.

Analysis	What it needs to run	Data-tab row
Every measurement analysed	a solute name	**Solute name**
	a solvent name	**Solvent name**
	a temperature above absolute zero	**Temperature**
	a scattering angle between 0° and 180°	**Scattering angle**
	a wavelength above zero	**Wavelength**
	a solvent refractive index above zero	**Solvent refractive index**
	a concentration, zero or above, or none	**Concentration**
	a viscosity above zero, or none	**Viscosity**
	a solute refractive index above zero, or none	**Solute refractive index**
Correlogram: Cumulant	—	—
Correlogram: Single exponential	—	—
Correlogram: Double exponential	—	—
Correlogram: Stretched Exponential (KWW)	—	—
Distribution: Lognormal	a viscosity	**Viscosity**
Distribution: NNLS	a viscosity	**Viscosity**
Distribution: CONTIN	a viscosity	**Viscosity**
Volume, surface or number view: Rayleigh point	—	—
Volume, surface or number view: RDG sphere	a solute refractive index above zero	**Solute refractive index**
Volume, surface or number view: Mie sphere	a solute refractive index above zero	**Solute refractive index**
Γ vs q^2	at least two distinct angles	—
	measurements of one polarization	**Polarization (DDLS)**
Mode Map: Double exponential	—	—
Mode Map: CONTIN	a viscosity	**Viscosity**
Mode Map: NNLS	a viscosity	**Viscosity**
D vs c	a concentration on every ticked measurement	**Concentration**
	at least two distinct concentrations	—
	measurements of one polarization	**Polarization (DDLS)**
DDLS	an angle with both a VV and a VH measurement	**Polarization (DDLS)**

8.1.5 A Run that is refused

A condition on the measured values is not judged before **Run**: the fit refuses it when it runs, and the refusal's message names its cause. The conditions refused this way are:

- a delay window that leaves too few channels;
- a placed baseline region at or above the correlogram's peak, refused by the Linear cumulant, by the Single, Double and KWW fits it seeds, and by the distribution methods; the Nonlinear cumulant falls back instead;
- a Γ vs q^2 or D vs c set whose per-point cumulant fits fail and leave fewer than two usable angles or concentrations.

What a sub-tab shows after **Run** comes from that Run, so a refused Run removes the result it would have replaced:

- on the Correlogram, Distribution, Γ vs q^2 and D vs c sub-tabs, the earlier fit leaves the sub-tab and its row leaves the **Summary** tables (§8.8);
- on the Mode Map, a refused mode-family fit removes its families' rows the same way;
- on the DDLS sub-tab, a refused Run empties the results table, hides the Shape Models table and turns **Export CSV...** off, with the warning under the column reading *DDLS could not run:* and the cause.

A sample R_h the program chose from a Γ vs q^2, D vs c or mode-family row goes with that row; a hand-entered or picked R_h is kept.

8.1.6 Fits a regroup or a removal takes away

A regroup that leaves a sample with no measurement — every run's identity edited on the Data tab and committed with **Update**, or every run moved to another sample — removes that sample's Γ vs q^2, D vs c, Mode Map and DDLS fits, and they do not come back when the sample re-forms under its old name.

Removing a DLS measurement (**Remove measurement**) removes the Γ vs q^2 and D vs c fits of its sample and M_w fraction, with every replicate-average, Γ vs q^2 and D vs c row of that fraction in the **Summary** tables (§8.8), whether or not the removed run was in the fit; a sample R_h the program took from one of those rows goes with it, and a hand-entered or picked one is kept. A fit that a **Regression SE estimator** change had already left on screen, marked out of date (§12.6), stays so. A sample that keeps a measurement keeps its DDLS fit, and the Mode Map drops the removed run's modes and re-fits its families. A Γ vs q^2 or D vs c sub-tab whose fit has gone reads *"Tick measurements, then Run."*, and the DDLS sub-tab *"Run DDLS to pair this sample's VV/VH correlograms."*, each with **Export CSV...** off until you run the fit again.

8.1.7 Notes and warnings

Three things can stand beside a result, and they mean different things.

A ⚠ **with a letter** is a failed precondition of the method. Each cause is one line under the results, its quantity against the limit and the section that explains it — "⚠[a] PDI 0.914 > 0.3, the cumulant limit (§3.2.1)", a **Theory-and-Equations-Guide** section — and the value it is about carries the same letter, "24.2 ⚠[a]". When the sub-tab shows several runs, one line holds every run with that

cause: it names up to three runs and counts them past that, giving the range of their values. A note whose own clause lists the runs it is about, such as a replicate average pooled beside its sources, is drawn the same way wherever the program shows it: two or three runs are named, more read as their count, *6 runs*; the export's Comments cell and the workspace kept between launches (§7.4) hold every name. Click a line or a lettered value for what to do about it; hovering shows the same while **Show tooltips on hover** is on (§12.9). A value whose flag is down with no line to state it keeps a bare ⚠ on its cell. The causes include:

- a size outside the range DLS can establish: above ~5 µm, or below 0.2 nm, a short-lag artifact;
- a cumulant PDI above 0.3;
- a decay rate that is not positive;
- a missing viscosity, which leaves the R_h cells reading *n/a*, since Stokes–Einstein needs it;
- a decay that never finished inside the delay window (§8.2.3);
- a distribution peak whose maximum sits on the first or last point of the R_h grid (§8.3.1).

A fit that ran without a usable result names why on its line. A Nonlinear cumulant fit that converges below 0.2 nm reports no values and names the size it reached, and one that reached a non-physical decay rate, β or moment reports none and names that quantity; an exponential, KWW or lognormal fit whose optimizer did not converge says so.

A **word on a label** says what the number is — *apparent*, *cumulant*, *mode-resolved*, *± statistical only*, *OLS* — on the row or column label of the value it qualifies, never in red; hover the label for the clause behind each word. The residual weighting a fit ran under is the one word that goes in the title of the results table instead, since it is about the whole fit: *sourced variance, classical* (the default weight's limit when no mean count rate was committed, §8.2.3), *unweighted* (§12.1.3) or $g_1{}^2$ *weight* (§8.3.2). A table of single fits — the Correlogram's results, the Distribution's peaks — follows a weighting not every fit ran under with *(2 of 3 fits)*; a pooled fit — Γ vs q^2, D vs c, a Mode Map family — names what its points' fits ran under, with *(2 of 3 points)* in the same way. Hover the title for the clause. The program's default, the Biganzoli–Ferri variance with a committed count rate and the Linear fit's weight, adds no word, as it adds none in the export; a DDLS result records no weighting and shows none.

An ⓘ marks *how* a number was obtained, and no precondition failed. What a fit records of this (a moment or a baseline that came to rest on its bound, channels that recorded nothing, a run a pooled fit left out) is found in these places:

- on the analysis sub-tabs, one record only: a residual weight the fit fell back to (§12.1.3), as one plain line after the ⚠ lines, in the ordinary text color, with no ⚠, no letter and no mark at any value, since it fails no check; it names the form the fit fell back to and why the default weight could not be formed, with the runs or angles in front when the sub-tab shows several, and a click on it opens its popup, as a ⚠ line's does;
- for a single measurement's fit, on the value's cell in the Summary's **Per-measurement results** table (§8.8) and in the export's Comments cell (§13.2);
- for a Γ vs q^2, D vs c or DDLS fit, in the export's Comments, since the **Sample-level R_h** table shows no ⓘ; a point the Γ vs q^2 or D vs c fit drops is also marked on its plot (§8.6).

The same causes appear in the Summary table, in the exported column's Comments cell and on the value's entry in the Cross-Sample source dropdown, which says when a flagged value can be chosen for you (§10.1.2). A view that pools several fits states its inputs' failed preconditions as lines of its own: Γ vs q^2 and D vs c count the flagged points ("2 of 8 angles carry a flagged fit") beside the first one's reason, naming the points, DDLS counts its flagged replicates the same way

(§8.7), and the replicate average names or counts the runs under each cause (§7.1.1). The SLS and Cross-Sample tabs mark their results the same way (§9.1.4, §10.1.1).

No value from a single correlogram carries a ± — its lag channels are correlated, so a regression standard error would under-report (the **Theory-and-Equations-Guide** §6, its single-correlogram rule). A dynamic ± comes from more than one correlogram: the replicate average (§7.1.1), the one ± at a single angle and concentration, and the fits that pool measurements — Γ vs q^2 (§8.4), the Mode Map's families (§8.5), D vs c (§8.6) and DDLS (§8.7) — whose ± is statistical only.

8.1.8 Reading a plot

Color and point shape are the measurement; line style is the method, and means the same thing on every sub-tab, so an NNLS curve and a CONTIN curve of one measurement are told apart at a glance. A residual panel below the plot shows the residuals of exactly the fits drawn above it, in the same colors and line styles, and names the methods on display. On screen the legend is a compact key in the corner — one entry per method drawn, plus the region markers — because the checklist and the results table beside the plot already name every ticked measurement against its color and shape; the figure written by the toolbar's **Save** button carries the full legend, every measurement by name grouped under its sample, in a figure enlarged to hold it. Every plot's axis-control bar is described in §14.

8.2 Correlogram

Fits the raw correlogram with a parametric model — cumulant, single exponential, double exponential or KWW (stretched exponential) — flagging anything the fit cannot honestly support. Tick the measurements, set the region and the **Method**, and click **Run**.

8.2.1 The plot and the fits shown

- **Scales.** The correlogram is shown in four scales at once — a large lin-log view plus lin-lin / log-lin / log-log thumbnails; double-click a thumbnail to promote it to the main view. The raw correlogram appears as soon as you tick a measurement, before any fit.
- **Fits to show.** Any fit the program has computed can be drawn over the data — not only the one in the **Method** box. The list above the plot offers all seven: the four parametric fits **Run** produces here, and **Lognormal**, **NNLS** and **CONTIN** from the Distribution sub-tab's own Run. An entry turns on once that fit exists for a ticked measurement, and a fresh Run ticks its own method, so a cumulant and a CONTIN reconstruction can be compared on one curve. A fit whose inputs have changed since it ran is marked *(stale)* and is not drawn until you re-run it on the sub-tab that owns it. The **results table** below follows the **Method** box, not this list: it shows one method's numbers at a time.

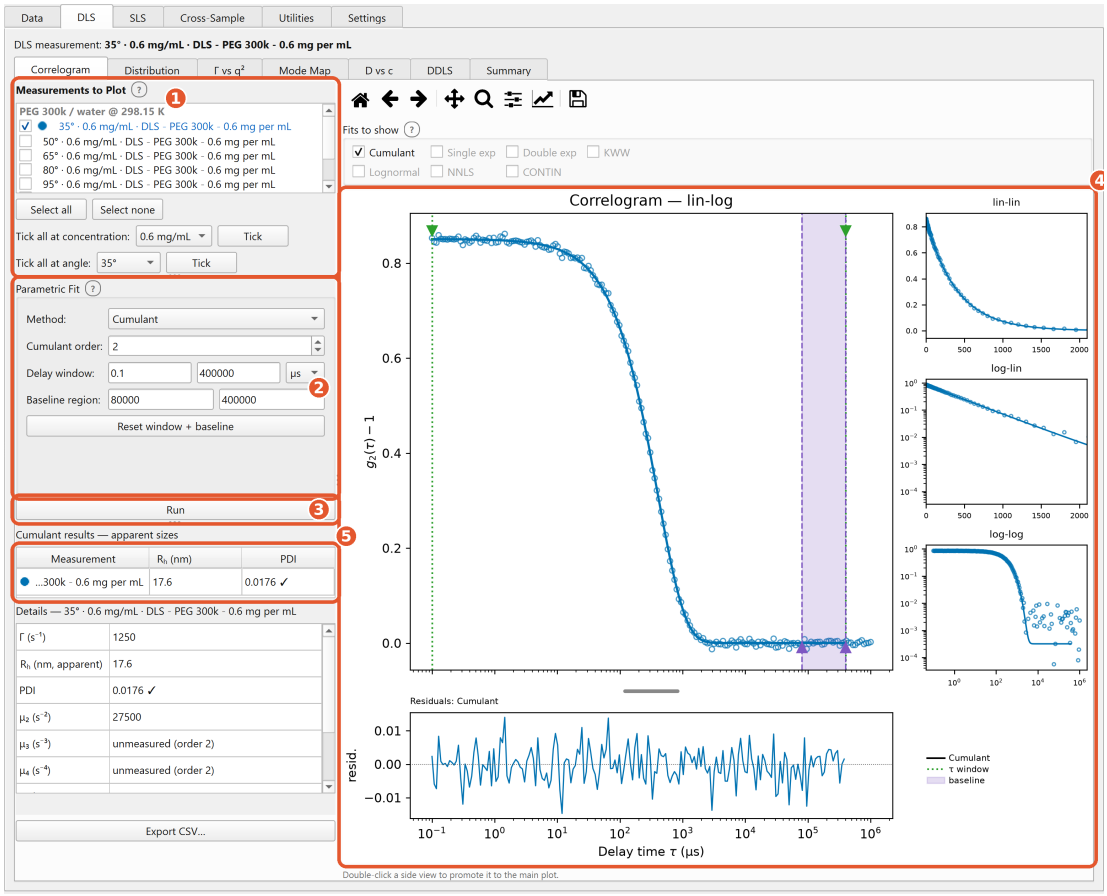


Figure 2: The **Correlogram** sub-tab after a cumulant fit. Tick the measurement (1), set the **Method**, the **Cumulant order** and the region in the **Parametric Fit** box (2), and press **Run** (3). The plot (4) shows the fit over the data in the main lin-log view with the three other scales as thumbnails, the residual panel below, and both region markers: the green dotted lines are the delay window, the purple span the baseline region, each with its caret handles. The results table (5) holds the apparent size and PDI; the details table under it every other fitted quantity.

8.2.2 Method, order and window

- **Method.** Cumulant, single exponential, double exponential or KWW. The cumulant's form — **Nonlinear** (Frisken's moment expansion, the default) or **Linear** (Koppel's log-polynomial fit) — and its residual weighting are global choices in **Settings ▶ DLS Analysis Defaults** (§12.1); the table shows which form produced each result (and its fitted baseline), and its title names a residual weighting other than the default, *Cumulant results — apparent sizes, unweighted* (§8.1.7).
- **Cumulant order.** 1 to 4, set here (seeded from **Settings ▶ Cumulant order**; the per-measurement value wins). Order 2 suits most samples. Orders 3 and 4 add the third and fourth moments, μ_3 and μ_4, to the results table and the export beside μ_2 — as bare values with no reliability mark of their own, because their uncertainty is at best comparable to their value; use them to compare fits, not to characterize the sample (hover the row label for the caution). Order 4 helps mainly on broad distributions, in the size rather than in the PDI; order 3 is a diagnostic choice, not a reporting one. The **Theory-and-Equations-Guide**

§3.2.6 explains what each order fits and where the higher moments stop being reliable.

- **Delay window.** Min/max, in µs, restricts which lags are fitted — the region of §8.1, typed rather than dragged.
- **The Settings that reach this fit.** Four global settings change the answer without a box on this sub-tab: **Cumulant method**, **Cumulant residual weighting**, **Skip initial lag channels** and **Cumulant fit cutoff**. Each is documented under §12.1; what a change does to a fit on screen is §8.1.3's rule. The distribution methods' seeds are §12.2's.

While this sub-tab's cumulant fit of a measurement is current, the Cross-Sample tab offers *that* fit — at this box's order and on your region — as the measurement's single-angle R_h, labeled *as fitted in the DLS tab*; once a committed change or a Settings change makes it out of date, it offers the measurement re-fitted at the cumulant order set in Settings instead (§10.1.2).

8.2.3 Flags, notes and what the export records

- **Undecayed modes.** A ⚠ marks a decay that never finished inside the delay window: the fit reports a rate, but the measurement did not observe the relaxation, so that rate — and the size from it — is an extrapolation. For the exponential fits the mark is per mode, so a healthy fast mode beside an undecayed slow one keeps a clean number, and the letter sits on the mode it is about; the cumulant and KWW fits judge the correlogram as a whole. Measure a longer delay range; a static or aggregated sample is the usual cause.
- **A moment at its bound.** When the nonlinear cumulant's μ_2 comes to rest on its lower bound of zero, the fit records it, and the Summary cell's ⓘ (§8.8) and the export name it: the PDI shown is that bound, not a measurement. At order 4 a μ_4 resting on its bound is recorded on its own, and the PDI is untouched by it. Neither record names a cause, because more than one thing can drive a moment onto its bound and the fit cannot tell them apart; the **Theory-and-Equations-Guide** §3.2.7 (*Two passive notes on a fitted moment*) explains them.
- **Empty channels.** A channel that collected no coincidence is dropped and counted in the fit's record (§8.1.7), on every method the sub-tab offers.
- **No committed count rate.** A fit that ran under the default residual weight, the Biganzoli–Ferri variance, without a committed mean count rate still runs, on the weight's classical form, its limit at an infinite count rate; the results title reads *sourced variance, classical* (§8.1.7) and its export's weighting column says so (below), and committing the rate on the Data tab (§6.1) adds the shot-noise terms the count rate enters. The two forms are the **Theory-and-Equations-Guide** §3.2.9 (the weight's count-rate input).
- **The other flags** — size range, PDI, decay rate, viscosity — are §8.1.7's.

The exported fit records:

- the cumulant order, method and fit cutoff, as values;
- the residual weighting the fit ran under, as a value in its own column — `biganzoli-full` when a committed count rate entered it, `biganzoli-classical` when none did, or the form you opted into;
- in the Comments of the scalar columns (the size, Γ, D, PDI, μ_2, μ_3, μ_4, β, baseline and RMS error): any non-default residual weighting, and where the per-channel sample time or the acquisition's duration had to be derived rather than read from the file (`per-channel sample time derived from the delay axis`).

The weightings themselves are explained in the **Theory-and-Equations-Guide** §3.2.8 and §3.2.11

(*The residual weight*; *The other two weightings, and the fallback*).

8.3 Distribution

Recovers a full distribution of decay rates without assuming a functional form. Tick the measurements and one or more **methods**, set the fields you want to change from their Settings seeds, and click **Run**; read the resolved peaks in the **Peak results** table.

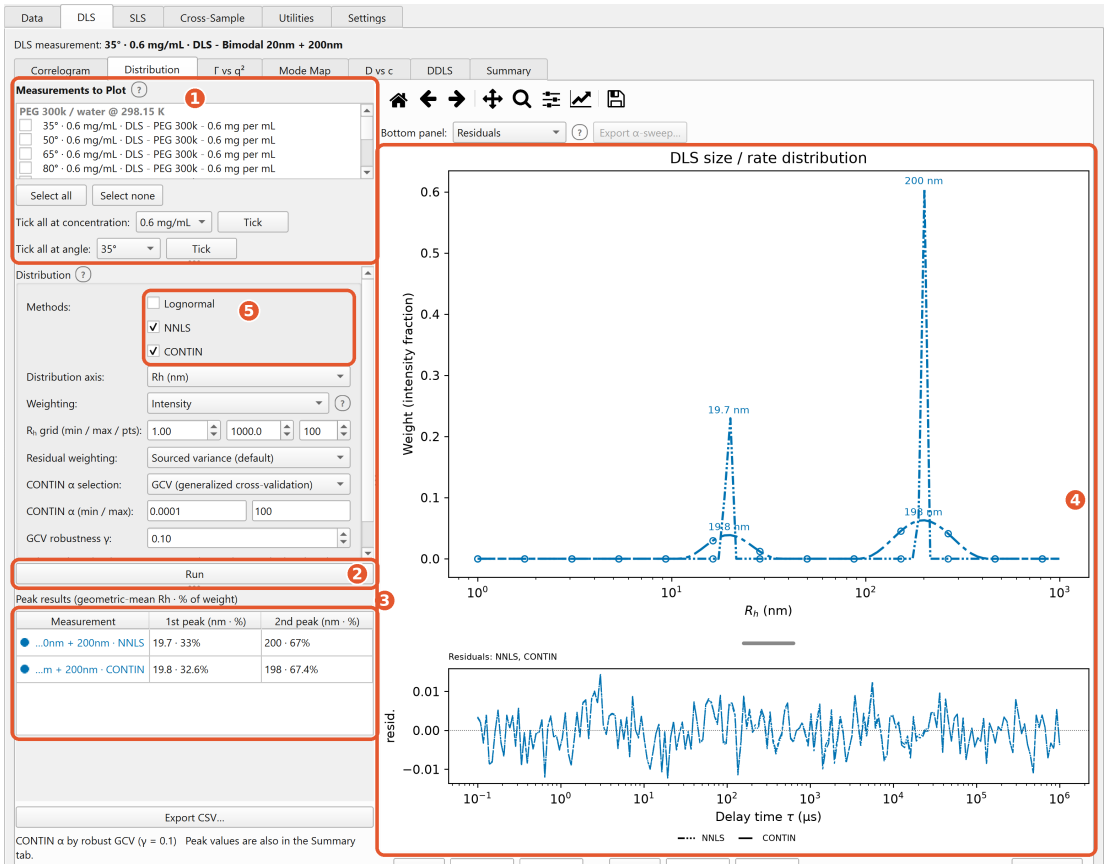


Figure 3: The DLS **Distribution** sub-tab on a bimodal sample. Tick the measurement(s) (1) and the methods in the **Distribution** box (5), then press **Run** (2); each method's size distribution is overlaid on the plot (4) with its peaks labeled, the residuals below (or, via the **Bottom panel** selector, the CONTIN α-sweep), and the resolved R_h and intensity fraction of every peak in the **Peak results** table (3). NNLS gives sharp peaks, CONTIN a smoother curve.

8.3.1 Methods and fits

- **The methods. NNLS** resolves well-separated peaks but is spiky; **CONTIN** is smoother and generally better for broad or real data; **Lognormal** is a robust single-mode form for a known-unimodal sample. The **Distribution axis** dropdown draws the distribution against R_h or against the decay rate Γ. All three are computed on an R_h grid, so each needs a viscosity: their boxes are disabled, keeping their ticks, while no ticked measurement has one, and a ticked measurement without one is left out of the Run (§8.1.4).
- **Each fit is kept**, one per measurement and method, as on the Correlogram sub-tab: unticking a measurement or a method hides its curves, peaks and export, and ticking it again brings

them back without a new Run while nothing it was fitted with has changed. A ticked measurement whose fit is missing or out of date is named in the status line with *press Run* when **Run** can fit it, and otherwise among those left out.

- **A Lognormal fit that fails to converge says so**: its legend entry names it as *failed*, the curve shows the starting estimate rather than a fit, a red flag names the measurement whose fit failed, and no peaks, Summary rows or ρ sources are offered from it (its export marks the affected columns' Comments the same way).
- **The fields.** Seeded from **Settings ▶ DLS Distribution Defaults** (§12.2); the per-measurement value wins:
  - the R_h grid (min / max / points) — a peak whose maximum sits on its first or last point is flagged; one whose weight only reaches an end is not, though its position (§8.3.3) may still not be its population's. Widen the grid and rerun;
  - the **CONTIN** α **selection** method — **robust GCV** by default, or the L-curve corner or F-test — and, for GCV, its robustness γ (0.1 by default; 1 = classical GCV);
  - the **CONTIN** α range and the **F-test** level;
  - the **Residual weighting** (§8.3.2).

  An α end that is not a number above zero is refused where you type it — the field puts its previous value back — and an α minimum above the maximum is refused at **Run**, with the reason.

8.3.2 The bottom panel and the residual weighting

- **Residual weighting.** This field sits two rows below the **Weighting** selector (§8.3.3) and does a different job: **Weighting** re-presents a finished distribution, while **Residual weighting** decides how the fit measures its own error, so it changes the answer. *Sourced variance (default)*, the Biganzoli–Ferri variance, weights each channel by a model of the noise a correlator has at that lag; ${g_1}^2$ *(older form)* is the constant-noise form to choose when you need to reproduce a result reported under it. The default takes about twice as long per fit, and changing the field marks a shown fit out of date rather than refitting live. The **Peak results** title reads ${g_1}^2$ *weight*, or *sourced variance, classical* for a fit run without a committed count rate (§8.1.7). What each form is, and why the Biganzoli–Ferri variance is the default, is in the **Theory-and-Equations-Guide** §3.4.2 (*The residual weight*).
- **Bottom panel.** A selector chooses what the lower panel shows: the fit **residuals** (data − reconstructed g_2-1 vs delay time) of the methods ticked here, through the same panel the Correlogram sub-tab uses — or, for CONTIN, the α**-sweep** diagnostic behind the automatic regularization: the L-curve corner, the GCV minimum, or the F-test curve, with the chosen α ringed, and a red halo when it landed on a sweep edge. The L-curve is drawn as the fit residual against the size of the distribution (its 2-norm), the chosen corner ringed. One sweep per CONTIN curve, colored to match the distribution; **Export** α**-sweep...** writes it to a CSV.

The red halo marks four cases. At the ceiling the selector wanted a larger α than the range allows — the fit shown is the smoothest the range holds, so raise the ceiling. At the floor it wanted a smaller α; at a low floor such as the default 1e-4 the fit is then essentially the unregularized NNLS one, whose extra peaks may be noise. Under the L-curve the halo also covers **no corner** — the sweep held no L-shaped bend, so the least-regularized α of the range was reported — and a one-point sweep, which has no interior to choose from.

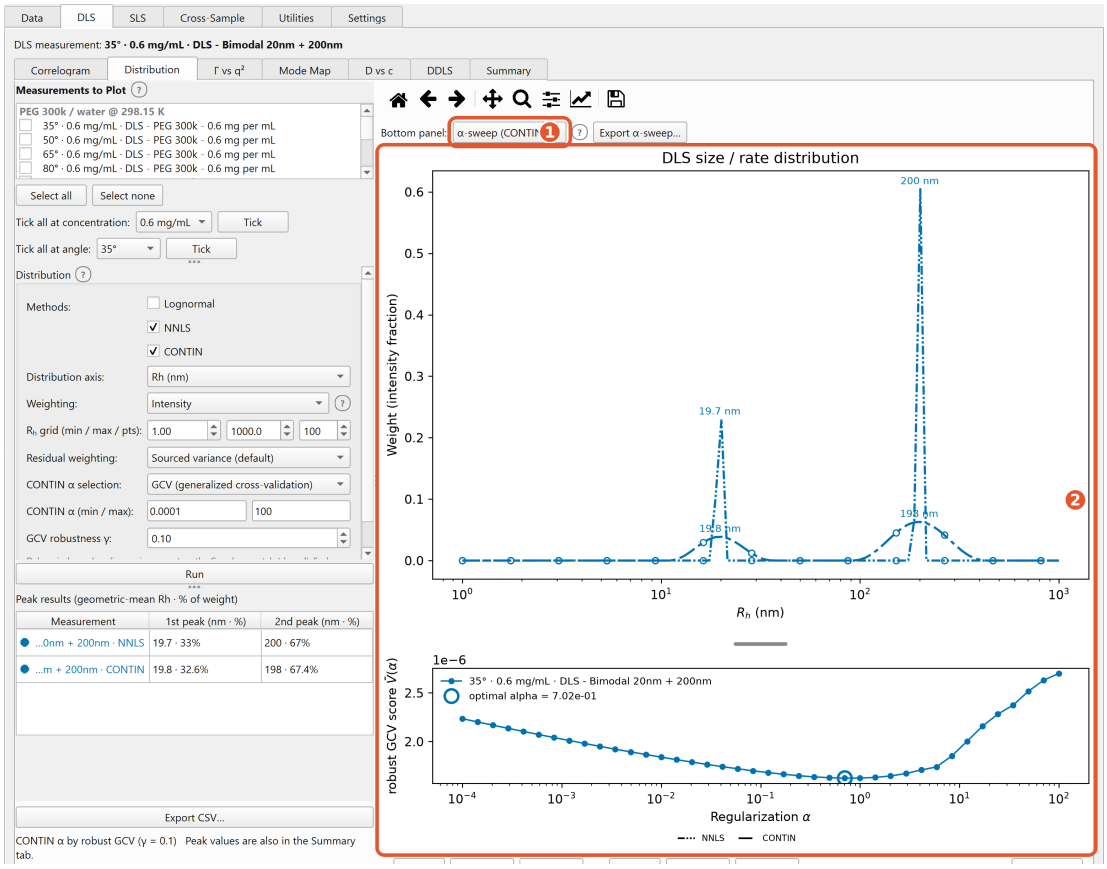


Figure 4: The **Bottom panel** selector (1) switched to the CONTIN α**-sweep** for the bimodal sample. The lower panel (2) draws the selector's curve against the regularization α — here the robust GCV score, whose minimum is ringed as the chosen α; under the L-curve corner or the F-test it draws that criterion instead. A red halo on the ring marks a choice that landed on a sweep edge.

8.3.3 Peaks and weighting

- **Peak results.** The table, titled *Peak results (geometric-mean Rh · % of weight)*, has one color-matched row per ticked measurement·method and one column per peak (*1st peak*, *2nd peak*, …), with the units in the column header and cells reading e.g. *"12.3 · 87%"*. The first is the peak's geometric-mean R_h, the average of ln R_h over the peak weighted by the distribution (for a lognormal fit, its fitted median). The second is the peak's share of the weight, in the **Intensity** view an intensity fraction, not a number or mass fraction, as is every peak percentage the program prints outside this table. The plot's peak labels print the same number and sit at it; the curve's top is the grid point with the most weight. The intensity view's values feed the Summary sub-tab (§8.8). Why the geometric mean is the position is in the **Theory-and-Equations-Guide** §3.4.1.
- **Flagged peaks.** A peak that fails a check — its maximum on an end of the grid, a decay that never finished inside the delay window, a position outside the size range — carries the letter of its line on its own cell, the line under the table stating the cause, on every method (§8.1.7); the flag follows it to the Summary and the Cross-Sample source list. The checks are the **Theory-and-Equations-Guide** §3.4.1's.

- **Small peaks.** A peak holding less than 5 % of the weight, in whichever weighting is shown, gets no column and no label; the Mode Map keeps weaker ones (§8.5).
- **Weighting.** Re-expresses the same intensity-weighted result as **volume**, **surface**, or **number** — a re-presentation, not a re-fit, so switching it never recomputes the distribution and carries no ±. The peaks are found again on the reweighted curve, each at its geometric mean over the reweighted peak, a lognormal fit's included, and each share is in the chosen weighting, which names the table rather than each column. Treat a large small-particle population in the number view as suspect — the **Theory-and-Equations-Guide** §3.7 says why, and why ISO 22412 deprecates number distributions from DLS.
- **Sphere model.** Reweighting is well-posed only for spheres; the **Sphere model** (Rayleigh / RDG / Mie) is chosen to match the sample, and RDG and Mie both need a **Solute refractive index** above zero, entered on the Data tab (§6.1). Each grays while no ticked measurement has one; under either, the view draws the measurements that have one, and the status line names the others as *left out of this view*.
- **Export.** The per-fit CSV gives each fit's *Peak Rh*, the grid point with the most weight, and *Peak mean Rh*, the geometric mean of the peak that holds that grid point (a lognormal fit's fitted median), its Comments cell saying which; both cells' Comments carry that peak's flag (**Flagged peaks**, above). A reweighted view exports the weighted-distribution CSV instead, whose *Peak Rh* and *Peak mean Rh* are read off the reweighted curve, the latter the reweighted peak's geometric mean for every method. The per-fit CSV and the Mode Map's carry three columns recording what the weight was and what it read — *Residual weighting*, *Sample time provenance* and *Run duration form* (§13).

8.4 Γ vs q^2

A sample-level analysis over the selected sample's DLS measurements (of the current M_w fraction) that tests whether motion is genuinely diffusive and returns D (and R_h). Tick the angles to include in the points table, tick **Use the Correlogram tab's delay window and baseline** if the fits should read the region set there, and click **Run**.

- **Points table. Measurements (tick to include in the fit)** — one row per measurement (Angle · c · Γ · D_{app}) — and the tick boxes choose which points enter the fit. On focus the table lists the sample's measurements as metadata only (Γ and D read "—"), and it refreshes as soon as you commit in the Data tab — no sidebar re-click needed. Every eligible row is ticked by default, a replicate average beside its sources excepted (below); changing the selection never refits or clears the plot, and a shown result reads *"Selection changed — press Run to refresh."* while the new set can run. **Select all**, **Select none**, and **Tick all at** a chosen concentration (in mg/mL) are the bulk helpers.
- **A replicate average beside its sources.** An average (§7.1) beside one or more of its own source runs is not independent of them, so it starts unticked, and **Select all** and **Tick all at** leave it so. Its row stays enabled, and a tick you give it is kept when you come back to the sample. A fit with it ticked runs, and a line under its results and the export's Comments name the average and the sources it was pooled beside: the fit counts their data twice, and its ± is computed as though the points were independent.
- **Ineligible rows** are grayed and unticked, and hovering one says why: its Data-tab parameters are not confirmed, listed under *Needs:* with what to enter, or it belongs to a different M_w fraction; a note also flags any same-solute/solvent measurements held in a separate sample because their temperature is not confirmed yet.

- **Run and results. Run** is off while the ticked set holds fewer than two distinct angles or mixes polarizations (below), the status line saying which (§8.1.4). After a Run the Γ/D_{app} columns fill in and unticked points gray out on the plot. The **Results** table holds D and R_{h}, each with its statistical ± (§8.1.7), R^2, *Diffusive?* and the number of angles fitted. A fit that fails the linearity criteria says so in a line under the results giving its R^2 and intercept, and on its Cross-Sample source label and in the export; it warns without flagging R_{h}, and the run can still be the sample's default R_{h} (**Theory-and-Equations-Guide** §3.5.1, the diffusive criteria).
- **Flagged points.** A point whose cumulant fit failed, whose PDI exceeds the validity limit or whose size is flagged for another cause (§8.1.7) is flagged in its row's tooltip, which names the cause. A converged point flagged this way still enters the line, which counts it.
- **How Γ is fitted here.** Each Γ comes from a cumulant fit of its correlogram — independent of any Correlogram or Distribution result — at the **Settings** cumulant order, with the Settings skipped channels, cumulant method, fit cutoff and residual weighting (§12.1). With the region box ticked, a measurement the region leaves too few channels is left out of the line and named in the export's Comments. The export closes on a **Cumulant order** row and a **Cumulant weighting (recorded)** row — the weighting each point's fit actually ran under, its Comments cell naming any point whose sampling time had to be derived from the delay axis. The **Results** title names that weighting when it is not the default, *Results — sourced variance, classical* when no point had a committed count rate (§8.1.7).
- **Mixed polarizations.** A ticked set that mixes polarizations turns **Run** off, the status line naming the polarizations ticked: a VV measurement beside a VH one, or beside an untagged or VU one, which count as one unpolarized class. A sample may hold both VV and VH measurements — that pairing is what DDLS is for — but one line cannot fit both (**Theory-and-Equations-Guide** §3.5.3, the rotational term in Γ_{VH}). Tick a single polarization, or run DDLS (§8.7).

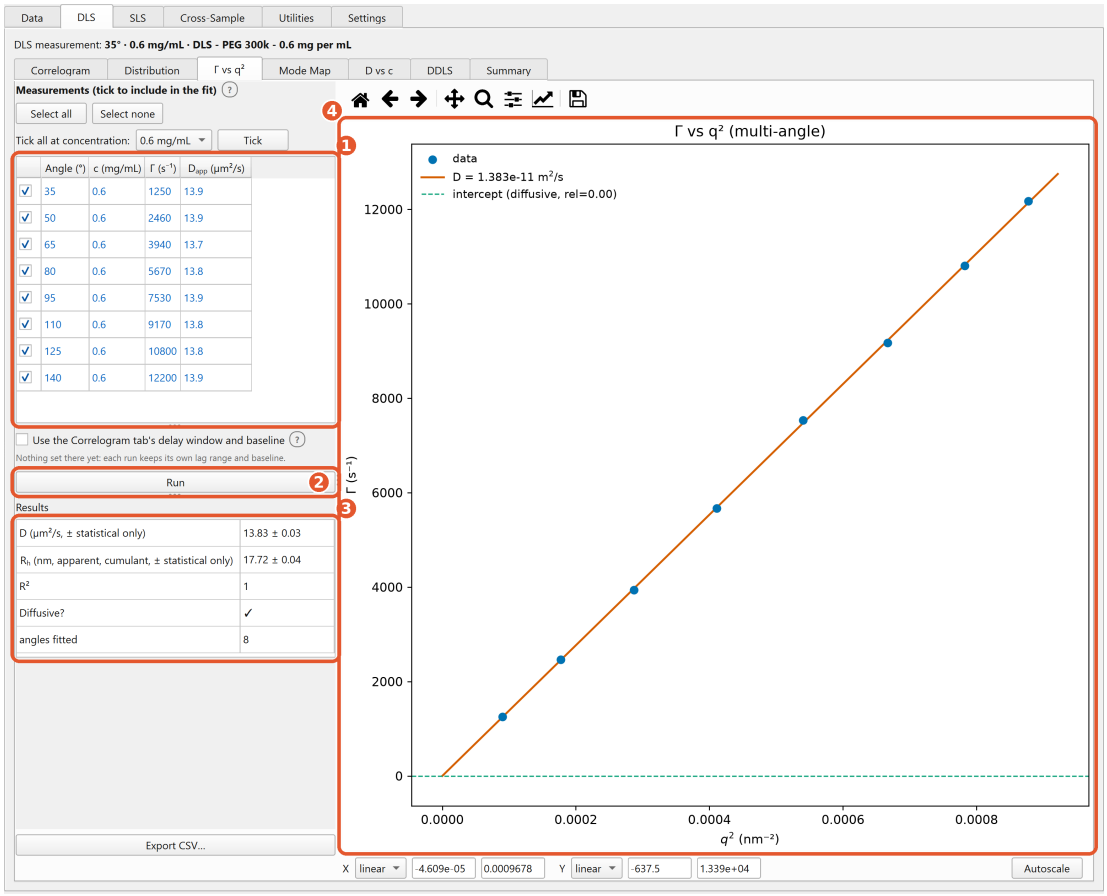


Figure 5: The DLS Γ **vs q**^{2} sub-tab. Tick which angles enter the fit (1) and press **Run** (2); the decay rate Γ is fitted against q^2 across the sample's angles (4). A straight line through the origin confirms diffusive motion; the results table (3) reports D and R_h with their statistical ±, R^2, and the diffusive check.

8.5 Mode Map

The mode-resolved companion to Γ vs q^2, for samples that relax on more than one time scale (a fast + slow mode).

1. Tick the measurements to include in the map.
2. Choose a **Source** and, for a distribution source, its **Inversion Settings**.
3. Click **Run**.
4. Add a **family** and click map points into it; each family is fitted on the spot.

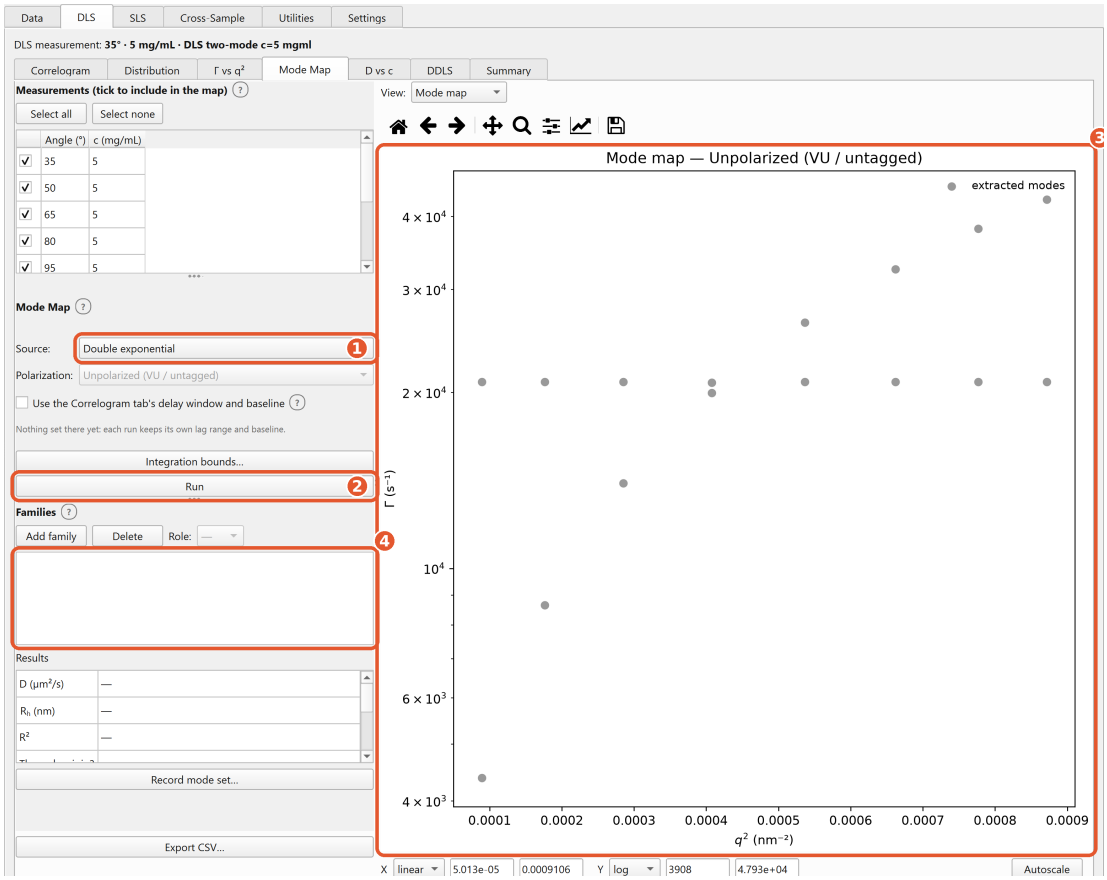


Figure 6: The **Mode Map** over the two-mode synthetic set. Choose the **Source** (1) and press **Run** (2); every extracted mode of every ticked measurement is plotted in (q^2, Γ) space (3) — here one mode rising with q^2 and the other flat across it, the two crossing near the middle of the range. Add a **family** (4) and click map points into it to fit one mode's line.

8.5.1 Extracting the modes

- **Ticks.** On first focus the table ticks every eligible measurement, and **Select all** ticks the same set, except a replicate average beside one or more of its own source runs, which you tick by hand if you want it (§8.4). A family whose picks hold the average's mode beside its sources' modes names them in a line under its fit, as the Γ vs q^2 fit does. A measurement whose Data-tab parameters are not confirmed is grayed, its tooltip listing what to enter under *Needs:*.
- **Run and Source.** **Run** extracts every mode of every ticked measurement — with the chosen **Source** (the two-mode double-exponential, or a CONTIN/NNLS distribution's resolved peaks) — and plots them all in (q^2, Γ) space. Both distribution sources invert on the map's own **Inversion Settings**, shown for those two sources only. A CONTIN or NNLS mode is plotted at its peak's geometric-mean Γ (§8.3.3), or the interval's under **Integration bounds...**, and a double-exponential mode at its fitted Γ; that one decay rate is the point's apparent D (Γ/q^2), decides its square marker (**Marker warnings**, below), enters the family fits and the mode set it is recorded into, and is the *Gamma* the export writes, each row's *D_app* its *Gamma* over q^2.
- **A source's viscosity.** **CONTIN** and **NNLS** need a viscosity, as on the Distribution sub-

tab: each grays while no ticked measurement has one, and the measurements without one are named as left out before **Run** (§8.1.4). The double-exponential source needs none.

- **Inversion Settings.** The R_h grid, the residual weighting and, for CONTIN, the α **selection**, its range and its γ or F-test level (NNLS is the unregularized case and has no α). They start from **Settings ▶ DLS Distribution Defaults** (§12.2), as the Distribution sub-tab's fields do, and can be changed here for this map; a denser grid resolves more closely-spaced modes and costs sharply more time per run. The α range and, under the F-test, its level are recorded with the map; each point records its own α, selector, γ and grid, and the export writes them all.
- **Copy from the Distribution tab** fills every field once with that sub-tab's current values, so a CONTIN setting tuned on one measurement there maps the whole sample in one click. It is a one-time copy, not a link: a later change on the Distribution sub-tab does not follow, and the map always runs on the values its own fields show.
- **Use the Correlogram tab's delay window and baseline.** Tick it to extract each measurement over the region set there; unticked, each keeps its own lag range and baseline estimate, so a measurement fitted on those sub-tabs under a region can resolve differently here.
- **Integration bounds...** sets, per measurement, the R_h intervals (in nm) that split a CONTIN/NNLS distribution into modes, in place of the automatic split where peaks overlap. The bounds apply at the next **Run**, travel with the results, and are ignored by the double-exponential source; a manual split of overlapping peaks is outside the decomposition's validity condition, and the export records that the bounds were set by hand.

8.5.2 Families and their fits

- **Families.** The program does not pair modes across angles; you do, by adding a **family** (Families panel), selecting it, and clicking map points into it — click again to remove, click a point held by another family to move it (the status line says which happened). Each family with points at ≥ 2 angles is fitted on the spot: through-origin D with an apparent, mode-resolved R_h (always computed, always labeled — whether the family passes through the origin is a label with stated tolerances, never a filter), plus the slow-mode deviation form ${\tau_s}^{-1} = aq^2 + bq^4$. The selected family's fit fills the **Results** table, whose title names the residual weighting the inversions behind its picked points ran under when it is not the default (§8.1.7); a double-exponential source runs no inversion and adds none.
- **Family verdicts.** A family's R_h is flagged when one of its points is undecayed, sits at the R_h grid's end or lies outside the size range, or when the sample has no viscosity, and a line under the fit counts each kind of point; a family that fails the linearity criteria says so in its line and on its Cross-Sample source label, as Γ vs q^2 does (§8.4).
- **Marker warnings.** Two independent warnings ride on the markers, one per channel so they compose: a **hollow** marker flags a low-amplitude mode (under 2 % of the measurement — imprecise), and a **square** marker flags a mode whose decay never finished inside the delay window (its rate is extrapolated) or whose peak's maximum sits on the R_h grid's first or last point; the legend names the causes present. A CONTIN or NNLS mode from a correlogram that never decayed is undecayed, whatever its own rate. A hollow square is both. Only a double-exponential mode or an **Integration bounds...** interval is ever hollow: an automatic CONTIN or NNLS peak under 2 % is left out of the map, a lower floor than the Distribution sub-tab's (§8.3.3), so the map can show a weak peak its table does not.
- **Polarization.** The map shows one polarization at a time (VV / VH / unpolarized); a family can never mix them.

- **Role and Amplitude view.** Designate one family **fast** and one **slow** (the Role dropdown) and switch the View to **Amplitude vs q** for the $A_\mathrm{f}/A_\mathrm{s}$ vs $\sin^2(\theta/2)$ report; the outcome of its angle-flatness check is written to the export.
- **Uncertainty.** Mode amplitudes and per-measurement Γ carry no $\pm$ (single-correlogram quantities, §8.1.7); the per-family D, R_h, a, b carry statistical SEs.

8.5.3 Recording, persistence and export

- **Record mode set** freezes the current decomposition — per-angle amplitudes with their provenance — for the SLS tab's Mode-Resolved sub-tab (§9.3).
- **Persistence.** Picks and recorded sets are kept with the session (§7.4) and matched to modes by value — re-running the extraction (or switching Source) never re-aims a pick; orphaned picks are counted in the Families panel and in the fit's export. A pick on a CONTIN or NNLS mode is keyed to its peak's grid point with the most weight, not to the plotted mean, so a click on a plotted point records that grid point's Γ, which can sit beside the plotted point rather than on it. A setting that moves the modes therefore orphans, at the next **Run**, a pick on a double-exponential mode, whose fitted Γ moves with it; a pick on a distribution mode still matches where its peak keeps that grid point.
- **Export CSV...** writes the per-point map (with each point's family, α, selector and γ, its Γ column's Comments counting the undecayed points and those at the R_h grid's end), every family's fit with its labels in the Comments cells, and the amplitude report.
- **Try it without data.** Utilities ▶ Synthetic Generator ▶ **Two-mode DLS (fast + slow, crossing)** (§11.3).

8.6 D vs c

The concentration-series companion of Γ vs q^2: it extrapolates the apparent diffusion coefficient of the selected sample (of the current M_w fraction) to zero concentration, returning the infinite-dilution $R_{\mathrm{h},0}$ and the interaction parameter k_D. Tick the concentrations to include and click **Run**.

- **What it shares with Γ vs q^2.** The sub-tab works exactly as §8.4 describes: the same points table ($c \cdot \mathrm{Angle} \cdot D_\mathrm{app}$), bulk helpers (**Tick all at** a chosen angle, in °), ineligible-row tooltips, the region box, the Settings the per-point cumulant fits read, the weighting in the **Results** title, the export rows, and **Run** off while the ticked set mixes polarizations or holds fewer than two distinct concentrations.
- **A measurement with no concentration** is grayed and unticked, its tooltip naming the **Concentration** row, so it is never counted toward the two.
- **Results.** The **Results** table holds D_0, $R_{\mathrm{h},0}$ and k_D, each with its statistical $\pm$ (§8.1.7), R^2 and the number of concentrations fitted.
- **A dropped concentration.** A concentration whose per-point fit returns no usable decay rate is dropped from the line: its position is marked with a dotted vertical line on the plot, its row's tooltip says why, and the drop is stated in the exported file's Comments cell, so a removed measurement is visible wherever the result travels. If fewer than two concentrations survive, nothing is extrapolated: D_0, $R_{\mathrm{h},0}$, k_D and R^2 read *n/a*.
- **Which value a flagged concentration flags.** A concentration whose own fit is flagged for a cause that bears on its decay rate (a decay that never finished, a PDI past the cumulant limit, a size outside the range, a failed or refused fit) flags k_D as well as $R_{\mathrm{h},0}$, and the line counting those concentrations stands at both cells; one lacking only a viscosity flags $R_{\mathrm{h},0}$ alone. The rule is the **Theory-and-Equations-Guide** §3.5.3 (the per-point fits).

- **A Settings change.** The extrapolation's per-point decay rates come from the cumulant fit, so a Settings change to a cumulant input or to the skipped channels clears it after you confirm (§12.1).

8.7 DDLS

Pairs a VV with a VH correlogram at each angle (tag each correlogram's **Polarization (DDLS)** on the Data tab, §6.1) to separate translational from rotational diffusion, and fits the sphere, rigid-rod and ellipsoid shape models for particle dimensions. Tick the paired angles to include and click **Run**; the shape models need the solvent viscosity.

- **Paired Angles.** A checklist ticks which paired angles enter the D_r/D_t fit (all ticked by default; untick an outlier), and **Include all angles** re-ticks them. A row whose angle lacks its VV or VH partner is grayed, and so is one whose Data-tab parameters are not confirmed, its tooltip listing what to enter under *Needs:*. With a fit on screen, a tick change refits at once while **Run** is on.
- **What Run needs** is §8.1.4's table. Its Data-tab rows apply to every measurement of the sample the fit uses, paired or not, since the fit builds them all before it pairs. While one is missing **Run** is off and the status line says what; unticking the last paired angle turns **Run** off rather than refitting.
- **Repeats.** Several VV (or VH) correlograms at one angle are treated as repeats and averaged, so DDLS first checks that every correlogram it is about to combine is from one sample and condition, and refuses the run if the concentration or M_w fraction differs — a check that matters because a concentration series is deliberately one sample in the sidebar, so a second concentration arrives here looking like a repeat.
- **A pre-averaged correlogram.** For a weak VH channel, pre-average the correlograms themselves with the sidebar's **Average correlation functions** action (§7.1) and tag the result; the averaged correlogram carries the better signal-to-noise. An average is fitted in place of those of its source runs that share its angle and polarization tag: they leave the fit and the angle's replicate count, and the export's Comments name the runs it replaced. A source tagged or re-angled after the average was made no longer shares its angle and tag and is fitted as itself.
- **Results.** The rotational-diffusion results fill the **results table**; the sphere, rod and ellipsoid models share the **Shape Models (assumed geometry — not measured)** table, a row per model and one per ellipsoid solution. The shape models are computed from the fit on screen, never from a fresh run, and are drawn with it: a Settings change that clears DLS results leaves the fit and its shape models on the sub-tab together, marked out of date. Without a viscosity the translational R_h reads *n/a*, marked with the letter of the line under the results that says there is no viscosity on the measurement. The D_r ±, and the sphere's rotational-radius ± derived from it, are standard errors over the angles, not regression ± values, so they never carry the *OLS* label of §12.6.
- **Shape models unavailable.** Without a viscosity, or with a D_t or D_r that is not positive, the Shape Models table is empty and the line under it, beginning *Shape models unavailable:*, gives the reason; the rotational-diffusion results are shown as usual.
- **An angle that cannot be fitted.** If no VV measurement at an angle gives a usable decay rate — its fit did not converge, converged to a size below the 0.2 nm size floor (§8.1.7) or gave a rate at or below zero, or, with the region box ticked, it refused a window its lag axis cannot fill — that angle is left out of the fit and named in the export's Comments, with the cause of

a refusal, and the other angles are fitted. A replicate whose fit gives no usable rate is left out of its angle's mean the same way and named there too, and the replicate count counts only the measurements averaged. An angle with no VH rate keeps its VV rate in D_t, and a line under the results counts the angles with no VH rate, since D_r then rests on fewer angles.

- **Flagged replicates.** When a VV measurement behind D_t carries a flag that bears on its decay rate — a decay that never finished, a PDI past the cumulant limit, a size outside the range, a failed or refused fit (§8.1.7) — D_t, $R_{h,t}$ and every shape value computed from D_t, which is all of them but the sphere's rotational radius, are marked, and a line under the results counts the flagged replicates, with their angles and the first one's reason. D_r, formed from both rates, is marked the same way for a flagged VV or VH measurement, on a line of its own. **Export CSV...** writes each verdict into the Comments cell of every column computed from that coefficient. The rules are the **Theory-and-Equations-Guide** §3.5.3 (the per-point fits) and §3.6.2 (validity and uncertainty).
- **Unphysical D_r.** If any angle comes back with $\Gamma_{VH} \leq \Gamma_{VV}$ the D_r is unphysical and a line under the results says so; **Export CSV...** repeats it in the Comments cell of every column whose value depends on D_r — including the rod, sphere and ellipsoid dimensions, which one bad angle can corrupt while leaving the average positive — so it reaches Origin with the number rather than staying on screen.

8.8 Summary

Two workspace-wide, exportable results tables, kept with the session (§7.4): **Per-measurement results** and **Sample-level R_h**. Tick the measurements to show, or untick **Ticked only** to see every row; **Export CSV...** writes one row per result or peak from both tables, in one set of columns (§13.1).

- **Per-measurement results.** One row per measurement shows cumulant R_h and PDI, single/KWW R_h, and NNLS, CONTIN and lognormal peaks, each as its geometric-mean R_h and its intensity fraction, which the column header names (§8.3.3).
- **Sample-level R_h.** Collects replicate averages, the Mode Map's family R_h values (§8.5) and the Γ vs q^2 ($q \to 0$, **apparent**) and D vs c ($c \to 0$, **thermodynamic**) extrapolations — the **Source** column names which, the R_h Type column marks the apparent/thermodynamic distinction, and **From** says where the value came from: its runs, angles or concentrations and the method, "replicate avg (3 runs, cumulant, order 2)" for a cumulant average.
- **Flagged sample-level rows.** A row whose fit failed a check — a flagged Γ vs q^2, D vs c or mode-family fit, a Γ vs q^2 line that fails the linearity criteria (§8.4), or a replicate average over flagged fits (§7.1.1) — carries ⚠ with a letter on its R_h cell, and on its ± cell when a cause is about the ±. Its lines follow the per-measurement ones under the first table (§8.1.7); a replicate average's first line counts the averaged fits that were flagged, and the lines after it name each cause once with the range of its values. The row keeps them in a saved session (§7.4), and the exported CSV's R_h Comments cell names the flagged rows.
- **Fit details.** Tick **Show fit details** to add every other quantity each parametric fit reports — the decay rate Γ, the moments, the fitted order, the baseline, the coherence factor β, the two amplitude fractions, the stretch and the stretched exponential's second size. Only the methods you actually ran add columns, so a cumulant-only workspace stays narrow; when the columns do not fit, the table scrolls sideways. Each detail column names its fit (`baseline · dbl exp`), because three of the methods report a quantity called *baseline*. A moment the fit did not reach reads **unmeasured (order N)** rather than a blank, so it is never mistaken

for a fitted zero. Every value comes from a single correlogram, so none carries a ± (§8.1.7).

- **Marks.** A flagged value in the per-measurement table carries ⚠ with the letter of its line under the table, which names or counts the runs it holds (§8.1.7). ⓘ on a cell marks how that number was obtained; hover to read it. Besides the lines §8.1.7 lists on the analysis sub-tabs, this cell is where the DLS tab shows a fit's records.
- **Filtering and export.** The left checklist is shared with the Correlogram and Distribution sub-tabs and doubles as the **Ticked only** filter. The CSV is unchanged by the detail columns — it carries the headline quantities either way — and ends with three columns recording what each per-measurement row's fit ran on: *Fit cutoff* (the cumulant's), *Skipped channels* and *Weighting*, the residual weighting that ran — each blank where the method has none, and on every sample-level row.
- **How rows fill and empty.** Rows fill in as you run fits and averages (**Average derived results...**, §7.1.1). A row goes when the Run that would replace it is refused (§8.1.5), when an average of the same method writes none (§7.1.1), when a DLS measurement of its sample and fraction is removed (a replicate-average, Γ vs q^2 or D vs c row, §8.1.6), or when a Settings change clears its result (§12.1). A hand-entered R_h is never removed.

---

9. The SLS Tab

The SLS tab extracts the weight-average molecular weight M_w, the radius of gyration R_g, and the second virial coefficient A_2. It first converts raw intensities to absolute excess Rayleigh ratios (via the calibration panel), then constructs the Zimm, Berry, or Debye plots. It has three sub-tabs: **Analysis** (§9.1), **Derived Intensity** (§9.2, a series built from DLS measurements when none was measured directly) and **Mode-Resolved** (§9.3, a series partitioned by DLS mode amplitudes). For the equations, see the **Theory-and-Equations-Guide** §4.

The plots' anatomy, the axis-control bar and the layout are described once in §14; the export buttons and files in §13.

Before you run, the sample needs what the analysis reads: a solvent blank, or your confirmation at **Run** without one (below), a dn/dc or a blank cell (§6.5), a concentration on every series (§6.1), and — if the sample holds more than one molecular weight — a fraction tag on each (§7.2). Then, on the **Analysis** sub-tab:

1. Pick the sample from the **SLS sample** dropdown at the top of the tab (only samples with SLS data appear); the whole set — solvent blank plus the concentration series — is analyzed together. Choose the **Fraction** if the sample has more than one.
2. Set the calibration in the **Calibration** panel and press its **Update**, or accept the uncalibrated fallback (§9.1.1).
3. Select a **Method** (§9.1.2), and mask any bad points or angles (§9.1.3).
4. Click **Run**, and read M_w / R_g / A_2 in the **Results** table, each label saying what its value is and each failed precondition a line under the table (§9.1.4).

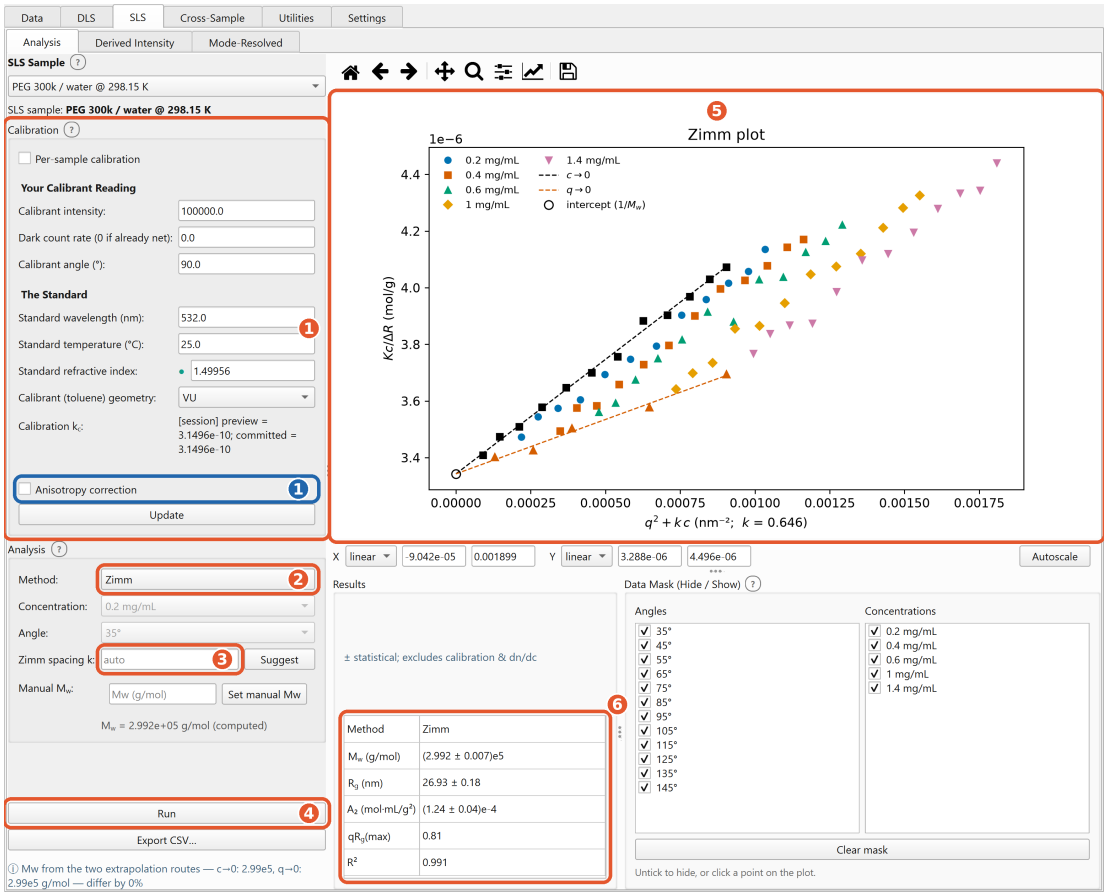


Figure 7: The SLS **Analysis** sub-tab after a Zimm run. The **Calibration** panel (1) holds the calibrant values and the proposed standard refractive index with its teal dot, and reports k_c; the **Method** (2) and, for Zimm/Berry, the **Zimm spacing k** box (3) sit in the Analysis group; **Run** (4) draws the Zimm grid (5) with its two extrapolation lines and fills the **Results** table (6). In blue: the **Anisotropy correction** checkbox that opens the Cabannes check.

Two rules hold across the sub-tabs:

1. Only **Run** computes a fit — on the Analysis and Mode-Resolved sub-tabs; Derived Intensity's **Derive** builds measurements rather than fitting — exactly as on the DLS tab (§8.1.3): switching sample, fraction or method keeps the last result on screen, and a committed change to an input the shown fit read flags it *"Inputs changed since this fit — press Run to refresh"*. The one exception is masking on the Analysis sub-tab, which refits a fit already on screen on its own, unless the new mask leaves that fit unable to run (§9.1.2).
2. A sample with no solvent blank at all can still be run: press **Run** and the **No solvent blank measured** prompt asks first, naming what it is about to do — a blank reconstructed from the typed 90° value, or no solvent subtraction at all (§6.6). **Cancel** is the default button, so Return or Escape declines. Everything such a run produces is flagged unreliable, R_g included — stricter than the uncalibrated case of §6.5, because a missing blank is *additive*, not a scale factor: it does not cancel out of a ratio, and bends the very slope R_g is read from. Nothing is withheld — the fit is shown in full and exports normally, carrying its reason in the affected columns' Comments cells — but it is not offered to the Cross-Sample tab; if you want its

value on a scaling plot, enter it there by hand.

Important. Running without a measured blank is a convenience, not a published method: no source states that a solvent blank may be reconstructed from one 90° reading; it is an inference from two separate facts about scattering geometry and simple liquids. The **Theory-and-Equations-Guide** §1.7.2 gives the equation, the assumptions behind it, where they are weakest (low angles, and solvents that are not simple liquids), and what they are inferred from. Treat the result as a preliminary estimate to decide whether a full measurement is worth making, not as a substitute for measuring a blank.

9.1 Analysis

9.1.1 Calibration panel

The program computes its own calibration constant k_c from a single calibrant measurement — the intensity of a standard liquid (usually toluene), the angle it was measured at, and that standard's geometry-aware Rayleigh ratio. The panel holds, top to bottom:

- **Per-sample calibration** — ticked, this sample gets its own calibration; unticked, it uses the session-wide one, which is the usual case.
- **Your Calibrant Reading**, the measurement you made:
  - **Calibrant intensity** — left blank, the run is uncalibrated (below);
  - **Dark count rate (0 if already net)** — the detector's dark count, taken off the sample, solvent-blank and calibrant intensities alike; a cleared field is 0;
  - **Calibrant angle (°)**.
- **The Standard**, the liquid whose Rayleigh ratio the program looks up:
  - **Standard wavelength (nm)**;
  - **Standard temperature (°C)**;
  - **Standard refractive index** — proposed by the solvent library (below);
  - **Calibrant (toluene) geometry** — VU, VV or VH, seeded from **Settings ▶ SLS Defaults ▶ Standard geometry** (§12.4).
- Below the two blocks, the resulting k_c with its note, the **Anisotropy correction** check (§9.1.3), and **Update**, which commits the calibration.

The calibrant intensity and the dark count rate are in the same units as your SLS intensity file, whatever the instrument exported: the program does not convert them, because only the *ratio* of calibrant to sample intensities enters the Rayleigh ratio. Any calibration constant stored in a data file is kept only for reference and not used in the calculation.

Where the dark count matters depends on the solvent blank. With a measured blank it cancels in the subtraction, so it changes only k_c, and with it M_w and the absolute A_2; without one (a typed 90° value, or none, §6.6) it also changes the excess Rayleigh ratio by a different amount at each angle, and R_g with it (the **Theory-and-Equations-Guide** §1.7 and §1.7.2 say why). Enter 0 when your readings are already net of it: the program cannot tell whether a reading was net, so a dark count taken off a net reading changes k_c with nothing to flag it.

A value the calibration cannot use is refused where you type it: the field puts back the value in use and the status line says why. That covers text that is not a number in any field; a cleared **Calibrant angle (°)**, **Standard wavelength (nm)** or **Standard temperature (°C)**; a standard refractive index that is not a positive finite number; and a dark count rate that is not finite.

If no k_c can be supplied, the analysis still runs on an arbitrary scale, flagged uncalibrated: M_w and the absolute A_2 are marked unreliable, while R_g and the calibration-free product $2A_2M_w$ remain valid (§6.5 says why).

The standard's refractive index is proposed from the solvent library for toluene at the standard's wavelength and temperature, marked with the teal dot and following the one rule for proposed values (§6.3). The field is blank only where the library has no value for the standard's condition, and a blank is not the same as leaving the correction out. With the field blank the $(n_{solvent}/n_{standard})^2$ correction is unapplied, so the run sits on an arbitrary scale as it does without a k_c, its M_w and A_2 flagged. The cause is named on the fits, the plots and the export, and wherever a value the run stored on the sample appears: the stored M_w line under **Manual M_w** and the Cross-Sample sources and plots (§10.1.2, §10.2). The two causes read differently:

- a blank standard refractive index reads *no standard refractive index, arbitrary scale*;
- a missing k_c reads *uncalibrated*.

A note beside k_c says, before you run, when:

- the standard temperature (outside 10–50 °C) or wavelength is off the toluene Rayleigh-ratio table, so the value is extrapolated;
- the standard wavelength is not the wavelength your SLS data were measured at;
- the standard refractive index differs from the library or is blank; for a blank index the note names the cause the run will be flagged with.

9.1.2 Methods

The **Method** dropdown lists these, in this order.

- **Zimm** and **Berry** — the double extrapolation: they combine several concentrations and angles and extrapolate to both zero angle and zero concentration, giving M_w, R_g, and A_2 at once. The result is marked **thermodynamic**. Zimm fits $Kc/\Delta R_\theta$ itself; Berry fits its square root with a q^2c cross term added, and drops that term where the grid cannot determine it; the program reports which construction it used. Past the **Guinier qR_g validity limit** (§12.4) the Zimm R_g and M_w are flagged, the Berry ones not (§9.1.4). How the two constructions compare is the **Theory-and-Equations-Guide** §4.4, and how conservative the Berry ± is on a two-concentration series its §6.1. The **Zimm spacing k** box sets the purely aesthetic spacing constant of the Zimm/Berry grid (**Suggest** picks an automatic value); it only changes how spread-out the curves look — M_w, R_g and A_2 come from the real fit and are unaffected.
- **Form factor P(q)** — fits the full angular form factor instead of the small-qR_g slope, so it stays valid to larger qR_g where the Zimm/Guinier plots curve. The **P(q) model** chooser offers five models: a **Sphere** (which also reports the geometric radius R), a **Gaussian coil**, an **Ellipsoid** (which returns a prolate and an oblate solution), a **Mie sphere** (single-valued in R given the contrast, and needing the **Solute refractive index** from the Data tab, §6.1) and a **Core–shell sphere** (needing both the solute and the **Shell refractive index**, and returning more than one core/shell pair). One concentration gives an apparent result; several give a thermodynamic M_w, R_g, and A_2. R_g (and R) survive an uncalibrated run (§6.5); how the Cross-Sample tab treats a model-dependent size and a degenerate solution is §10.1.3. The models, and what each assumes of the sample, are the **Theory-and-Equations-Guide** §4.8–§4.11.
- **Debye (single c, apparent)** — at one concentration, gives an apparent M_w and apparent R_g, both flagged past the **Guinier qR_g validity limit** (§12.4). It needs at least two angles;

a concentration series read at one angle is the ladder route below. Apparent values are never presented as final.

- **Guinier (single c, apparent)** — a single-concentration route to R_g from the log intensity, scale-independent, valid up to the **Guinier qR_g validity limit** set in Settings (§12.4); past it both its R_g and its M_w are flagged.
- **Single-angle M_w + A_2 (ladder, apparent)** — the Debye plot at one angle over the concentration series: apparent M_w ($= M_w \cdot P(q)$) from the intercept and apparent A_2 from the slope, with the calibration-free $2A_2M_w$ product beside them. The route for a single-angle ladder, such as a 90° series built on the Derived Intensity sub-tab; pick the angle in the **Angle** box. Not thermodynamic — a multi-angle Zimm/Berry set is.
- **Calibration-free A_2** — obtains the product $2A_2M_w$ from ratios of excess intensities across concentration at the angle chosen in the **Angle** box, without any absolute calibration. Valuable when absolute calibration is uncertain. Like the ladder it is apparent, read at one finite angle, and its export says so in an *Is apparent* column. Each ratio is taken against a reference concentration, the lowest one whose point at that angle is shown and has a positive excess; which concentration is the reference does not change the product (the **Theory-and-Equations-Guide** §4.5 says why).
- **Excess Rayleigh ratio** — the excess Rayleigh ratios themselves.

What the shown sample can run is judged before **Run**, from its committed Data-tab parameters, the committed dark count rate (§9.1.1), the mask (§9.1.3) and the **Concentration** and **Angle** boxes. A fit it cannot run is grayed in the **Method** or the **P(q) model** dropdown, never hidden, and stays selected if it was. Hover a grayed entry for what it lacks: its tooltip lists each missing input under *Needs:*, with the Data-tab row where it is entered or, for a count, how many are shown against the minimum (• *at least two concentrations above zero that are not hidden (1 shown, 2 needed)*). A concentration or an angle counts while one of its points is shown. While a grayed entry is selected, **Run** is off and the status line lists the same lines, headed with the fit's name (*"Mie sphere needs:"*). A fit already on screen stays; one whose inputs have changed is marked *"Inputs changed since this fit."*, without *press Run*, and the mark asks for **Run** again once a fit the sample can run is selected. The table lists what each fit needs.

Fit	What it needs to run	Data-tab row
Every fit	a solute name	**Solute name**
	a solvent name	**Solvent name**
	a concentration, zero or above, on every series	**Concentration**
	a temperature above absolute zero	**Temperature**
	a wavelength above zero	**Wavelength**
	a solvent refractive index above zero	**Solvent refractive index**
	a solute refractive index above zero, or none	**Solute refractive index**
	a shell refractive index above zero, or none	**Shell refractive index**

Fit	What it needs to run	Data-tab row
	when no solvent blank was measured, a solvent blank intensity at 90° at or above the dark count rate, or none	**Solvent blank intensity at 90°**
	a solvent blank measured at the same angles as the series it is paired with	**Solvent blank**
Zimm	at least two concentrations above zero that are not hidden	—
	at least two angles that are not hidden	—
Berry	at least two concentrations above zero that are not hidden	—
	at least two angles that are not hidden	—
Form factor P(q): Sphere	at least one concentration above zero that is not hidden	—
	at least two angles that are not hidden	—
Form factor P(q): Gaussian coil	at least one concentration above zero that is not hidden	—
	at least two angles that are not hidden	—
Form factor P(q): Ellipsoid	at least one concentration above zero that is not hidden	—
	at least three angles that are not hidden	—
Form factor P(q): Mie sphere	at least one concentration above zero that is not hidden	—
	at least two angles that are not hidden	—
	a solute refractive index above zero	**Solute refractive index**
Form factor P(q): Core–shell sphere	at least one concentration above zero that is not hidden	—
	at least three angles that are not hidden	—
	a solute refractive index above zero, for the core	**Solute refractive index**
	a shell refractive index above zero	**Shell refractive index**
Debye (single c, apparent)	at least two angles not hidden at the chosen concentration	—
Guinier (single c, apparent)	at least two angles not hidden at the chosen concentration	—

Fit	What it needs to run	Data-tab row
Single-angle Mw + A_2 (ladder, apparent)	at least two concentrations above zero not hidden at the chosen angle	—
Calibration-free A_2	at least two concentrations above zero not hidden at the chosen angle	—
Excess Rayleigh ratio	—	—

Grayed means an input is missing or, as entered, unusable, so three things never gray an entry. A missing calibrant intensity, standard refractive index or $\mathrm{d}n/\mathrm{d}c$: the fit runs and flags its scale (§9.1.1, §6.5). A sample with no measured solvent blank: **Run** asks first (the head of this chapter). And the measured values themselves: with every input present, a fit the data cannot support, such as an excess Rayleigh ratio at or below zero at every angle, still fails at **Run**: its warning line names the cause, and the earlier fit's numbers leave the **Results** table with it.

9.1.3 Masking, the manual M_w and the anisotropy check

- **Data masking.** The **Data Mask (Hide / Show)** panel lists the angles and the concentrations: untick one to hide all of its points. On every plot but the ladder's and **Calibration-free A_2**'s, a click on a drawn point hides just it, leaving a hollow gray marker, and a click on the marker shows it again; a point hidden with its angle or concentration comes back from the list. The Debye, Guinier and form-factor plots do not draw a point whose excess Rayleigh ratio is zero or negative, because their fits leave it out (the **Theory-and-Equations-Guide** §4 says why). The mask is kept with the session, and **Clear mask** shows every point again. A mask edit refits the fit on screen at once (the head of this chapter), unless it leaves the selected fit too few concentrations or angles, which grays it (§9.1.2).
- **Manual M_w.** An override for the case where the trustworthy M_w was characterized independently (say, in a different solvent or by GPC/MALS) and you want the analysis to use that value; it is never overwritten by a re-analysis.
- **Anisotropy correction.** For an optically anisotropic scatterer the depolarized scattering inflates the apparent M_w. Tick **Anisotropy correction** in the **Calibration** panel and enter the depolarization ratio ρ_v — directly, or as I_{VV} and I_{VH} intensities with an optional **Dark count** in the same units, taken off both before the ratio (blank is 0). It reads out ρ_v, the optical anisotropy δ^2, the Cabannes factor f (the **Theory-and-Equations-Guide** §4.7.1), and the isotropic-corrected weight $f{\cdot}M_w$ for the sample's analyzed M_w. The first three are ratios and always show; $f{\cdot}M_w$ is a mass, so an uncalibrated run dashes it out like every other absolute M_w — unless the M_w is one you entered by hand, which is trusted and still corrected.

The anisotropy check is display-only — nothing is written to the sample or exported, so a corrected M_w can never silently propagate into Cross-Sample or an export. If you trust the corrected value and want it used downstream, enter it by hand as the **Manual M_w**. When the correction matters, and what it does to R_g, is in the **Theory-and-Equations-Guide** §4.7.

9.1.4 Results

Each method fills a two-column **Results** table:

- **Rows.** Zimm/Berry give M_w, R_g, A_2, qR_g(max) and R^2; Debye the apparent M_w and R_g,

and Guinier the apparent R_g, each with its qR_g(max) and R^2; the single-angle ladder an apparent M_w and A_2; calibration-free A_2 the $2A_2M_w$ product.

- **The qR_g(max) row.** The largest fitted q times the reported R_g, the number the **Guinier qR_g validity limit** (§12.4) is compared with; the export writes it as a *qRg (max)* column whose Comments cell names the limit.
- **The ±.** A fitted value carries its statistical ± where an honest one exists (the **Theory-and-Equations-Guide** §6.5.2 lists where none does), and the ± excludes the calibration and dn/dc uncertainties, as the caption above the table says (*± statistical; excludes calibration & dn/dc*). The calibration-free $2A_2M_w$ product's row reads *± statistical only* instead, because both scales cancel in it. Which estimator produced a ± is §12.6's setting.
- **Labels and lines.** A row label carries the words that say what its value is (*apparent*, *OLS*), and each failed precondition is a line under the table with its letter at the value, as on the DLS tab (§8.1.7).
- **Dashes.** A value dashed for a missing scale names the scale in the dash, "— (uncalibrated)", "— (no standard refractive index)" or "— (no usable dn/dc)", beside the line that states it.
- **The excess Rayleigh ratio.** It has no scalar table: it shows a one-line status (*"Excess Rayleigh ratio for N concentrations."*) alongside its plot. Under the Results box an uncalibrated series reads "⚠[a] uncalibrated, arbitrary scale" (or "no standard refractive index, arbitrary scale" when the standard's index is blank) and a series with no measured solvent blank "no solvent blank measured", each line pointing at the **Theory-and-Equations-Guide** section on its absence (§1.7.1, §1.7.2); a missing dn/dc gives no line there, since it does not enter ΔR.

Past the **Guinier qR_g validity limit**, each method marks its result in its own way:

- **Zimm** and **Debye** — the R_g and M_w cells show ⚠ and one line under the table states the qR_g(max) against the limit; its popup gives the remedy. Where the M_w is dashed for a missing scale, the dash carries no letter and the line still names the limit.
- **Guinier** — the R_g cell shows ⚠ with the same line. The Guinier M_w, which the table does not show, carries the same clause in the export's Comments.
- **Berry** — not flagged: its R_g and M_w labels read *past the Zimm limit*, on the Analysis and Mode-Resolved sub-tabs alike, and the label's tooltip and the export's Comments give the qR_g(max) and the limit.

Under a Berry result an ⓘ line at the foot of the control column gives the fitted q^2c cross term as a multiple of the value Berry's equation predicts to first order — a readout, not a flag. A Berry fit run without its cross term says so in its export's Comments cells, and carries no flag for it.

Under a Zimm or Berry result a line at the same foot gives the M_w read off each of the two extrapolation routes, $c \to 0$ and $q \to 0$, and the percentage by which they differ: a check on the grid, not a ± (the **Theory-and-Equations-Guide** §6.3 says what it can show). On an arbitrary scale the two values are withheld and the percentage still shows. Above 10 % the line turns to a ⚠ warning that the angle × concentration grid is incomplete (a level with one usable angle, or a missing concentration), since on a complete grid the two routes agree by construction; it flags no value. The line is on screen only, and the export does not carry it.

The explicit-Run rule at the head of this chapter applies, with masking as its one exception.

9.2 Derived Intensity

Build an SLS intensity series directly from this sample's DLS measurements, so a goniometer that recorded only correlograms still yields a full static analysis. Each DLS measurement contributes its mean count rate as the intensity at its angle, and measurements at the same angle and concentration are averaged into one point per concentration.

Tick the DLS measurements to use (**Select all** / **Select none** help with a long list), optionally enter a $\mathrm{d}n/\mathrm{d}c$ for the derived points, and click **Derive**; the created measurements appear on the **Analysis** sub-tab like any other SLS data.

- The table lists each measurement's **Angle**, **Count rate**, **Source**, concentration, and a **Status** of *ready* or *not usable* (a not-usable row's reason is its tooltip).
- A count-rate trace that is not attached to any measurement — a solvent blank recorded without a correlogram, typically — is listed too, un-ticked. Tick it to derive a point from its mean: it takes the sample's identity from the first ticked DLS measurement, its concentration defaults to 0 (a blank), and its angle is filled in when the measurements share one; otherwise double-click the **Angle** cell (or the **c** cell) and enter the value. **Select all** leaves traces as they are.
- Measurements at one angle are compared before they are averaged: when their mean count rates lie more than 5 % apart, the highest against the lowest, a ⚠ line under **Derive** names the runs, or counts them past three (§8.1.7), with their angle and the spread. Click the line for what to check. The averaged value is used as is, and no flag is set. The derived measurement records the same spread, so it reaches the export's Comments (§13.2), and the averaged point records the standard error of the repeats' mean.
- Deriving again replaces only the points of the same fraction and concentration; another fraction's blank at $c = 0$ is left alone.
- The mean count rate and its source are set on the **Data** tab (§6.1) and shown here read-only — there is exactly one place that number is chosen.
- Derived points are ordinary SLS measurements: calibrate and analyze them on the **Analysis** sub-tab like any loaded series. Their intensities are count rates in cps, so the calibrant intensity and any solvent blank analyzed with them must be in cps as well (§9.1.1).
- A derived point is a snapshot: re-running a fit or editing a source measurement never changes it. Click **Derive** again to refresh it.

See the **Theory-and-Equations-Guide** §4.1 for how a derived intensity series is built and analyzed, and for what can make repeats at one angle disagree.

9.3 Mode-Resolved

Split an SLS intensity series by the amplitudes of this sample's recorded DLS modes (the Sedlák partition) and analyze each mode's share of the scattering on its own.

1. Record a mode set on the DLS tab's **Mode Map** (**Record mode set**, §8.5).
2. Select it here and click **Partition**.
3. Pick a **Mode** and an **Analysis** method, and click **Run**.

- A mode set appears here as a read-only snapshot with its method, regularization, and recording date — re-fitting on the DLS side never changes it; record again for a fresh one. **Open in Mode Map** jumps to where it was defined.

- **Partition** splits every SLS series of this sample (loaded or derived) that shares a concentration with the set: each mode's share is $\mathrm{d}R_{\mathrm{mode}}$ = amplitude fraction × $\mathrm{d}R$.
- **Amplitude Guinier** fits one mode's series alone and gives that mode's apparent size, flagged past the **Guinier qR_g validity limit** (§12.4).
- Guinier, Debye, Zimm, and Berry run on a mode's series like any SLS analysis; every result's labels read *apparent* and *mode-resolved*, with the partition recorded in its export Comments. A per-mode series is never added to the workspace as its own measurement, and a mode's M_w is never written to the sample — it describes that mode's share alone.
- A fit on a mode's share flags its M_w, R_g and A_2 when, at some of its angles, that mode:
  - was undecayed;
  - sat at the R_h grid's end;
  - fell below the Mode Map's amplitude floor, drawn there as a hollow marker (§8.5);
  - shared one run's decomposition with a mode that was undecayed or at the grid's end.

  A line under the results counts the points. The Amplitude Guinier R_g's Cross-Sample source label names the same cause. The rule is the **Theory-and-Equations-Guide** §4.2 (the Sedlák partition).

See the **Theory-and-Equations-Guide** §4.2 for the partition and what each per-mode analysis assumes.

---

10. The Cross-Sample Tab

The Cross-Sample tab computes quantities that link samples or modules: the shape parameter $\rho = R_g/R_h$ and the branching factors on the **Architecture** sub-tab (§10.1), and the log–log scaling laws on the **Scaling** sub-tab (§10.2). It is *aggregate-scoped*: it works over every included sample at once.

The plots' anatomy, the axis-control bar and the layout are described once in §14; the export buttons and files in §13.

1. Tick the samples to include in the tab's **Samples (Include / Exclude)** list (every sample with SLS data starts included).
2. Pick the focused sample and M_w fraction with the **Sample** and **Fraction** selectors in the **Source Selection** panel, or click the sample in the sidebar.
3. Choose the R_g / R_h / M_w / A_2 source for that sample (§10.1.2), and read ρ and its Shape label in the Architecture sub-tab's table.
4. Switch to the **Scaling** sub-tab for the log–log plots and fitted exponents.

For the equations, see the **Theory-and-Equations-Guide** §5.

10.1 Architecture

10.1.1 Membership, focus and the table

The tab makes two selections. **Membership** is the tab's **Samples (Include / Exclude)** list: untick to drop a sample from the table and the scaling plots; clicking a row there changes nothing else. **Focus** is the sample and M_w fraction the **Source Selection** panel edits, named by its own **Sample** and **Fraction** selectors. The sample focus is the one every tab shares: clicking a sample, or one of its measurements, in the sidebar focuses it here, and picking one in the **Sample** selector

focuses it in the sidebar; a sample with no SLS data is not in this tab, so clicking it leaves the focus where it was. The Fraction list holds every fraction the table shows for that sample — a label carried by the SLS series and one carried only by DLS measurements alike — so every row can be pointed at.

The Architecture sub-tab's table is a read-only results view with one row per included sample (or fraction): R_g, R_h, ρ, a **Shape** label naming the likely architecture, whether ρ is **apparent** or **thermodynamic**, and — for tagged samples — the branching factors of §10.1.4. The program checks that the two measurements share solute, solvent, and temperature. ρ is flagged apparent if *either* input is apparent; for a physically meaningful shape parameter, prefer the thermodynamic (infinite-dilution) source on both axes. An R_g or R_h cell whose value failed a check carries ⚠ with the letter of the line under the table that names the check, as its **Source Selection** row does (§10.1.2); ρ and the Shape label built from it do not repeat that mark. The R_g and R_h cells print no ±, only the value at the place its ± supports; a value that place would round to zero is shown beside its ± instead (§12.8), and that ± carries its words — *± statistical only*, and *OLS* when it comes from the classical OLS estimator (§12.6) — with each word's clause in the cell's tooltip. Under the table, a line names the samples whose arbitrary-scale M_w the linear reference laws and the branching factors rest on. The size columns follow **Settings ▶ Display Units** (§12.7).

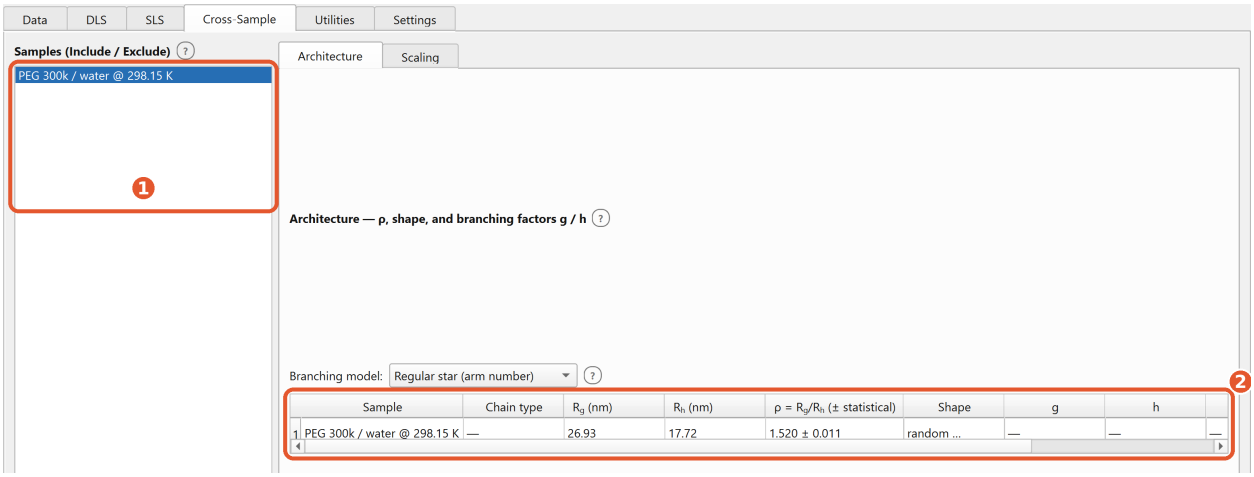


Figure 8: The Cross-Sample **Architecture** sub-tab's table. Tick which samples to include (1); each included sample gets one row (2) with its chain type, R_g, R_h, ρ, Shape label and branching factors g and h ("—" for an untagged sample); the **n (branch)** and **Type** (apparent or thermodynamic) columns lie past the scroll bar. The source dropdowns of the **Source Selection** panel, below the table, are not shown.

10.1.2 Choosing the sources

Because a sample can have several R_g and R_h values, you choose the source for each quantity with its **source dropdown** in the **Source Selection** panel, and the program never picks silently. For an SLS-only sample the R_h (DLS) dropdown is shown but disabled. A value you type in the box beside a dropdown and **Set** is read in the unit the box shows.

Each row's name carries the words of the value the sample holds for that quantity, read off the value itself — A_2 (SLS, apparent), R_g (SLS, model-dependent) — so what the chosen value is stays in view however far the dropdown cuts its text; hover the name for what each word means.

Each of the four rows has a badge right after its dropdown. When the value the sample holds failed a check, the badge reads ⚠ with the letter of the line under the rows that names the check, each line with its **Theory-and-Equations-Guide** section: a Zimm R_g past the qR_g limit, for instance, or an M_w or A_2 on an arbitrary scale (§6.5, §9.1.1) — "⚠[a] uncalibrated, arbitrary scale", "no standard refractive index, arbitrary scale", or "no usable dn/dc, arbitrary optical scale". R_g is never marked for the scale. Hover the badge or click the line for what to do (§8.1.7).

Quantity	Source	Kind	Hand-entry?
R_g (SLS)	Zimm/Berry double extrapolation	*thermodynamic*	Yes
	Debye or Guinier, one concentration	*apparent*	
	a **Form factor P(q)** fit: sphere, coil, ellipsoid, Mie or core–shell (§10.1.3)	*model-dependent*	
R_h (DLS)	D vs c extrapolation, $c \to 0$	*thermodynamic*	Yes
	Γ vs q^2 at one concentration, $q \to 0$	*apparent*	
	replicate average	*apparent*	
	a Mode Map family	*apparent, mode-resolved*	
	cumulant, one condition	*apparent*	
	a CONTIN, NNLS or lognormal distribution peak	*apparent*	
M_w (SLS)	Zimm/Berry double extrapolation	*thermodynamic*	Yes
	Debye or Guinier, one concentration	*apparent*	
	a **single-angle ladder**, labeled with its angle	*apparent*	
	a **calibrated Form factor P(q)** fit	*model-dependent*	
A_2 (SLS)	Zimm/Berry double extrapolation	*thermodynamic*	No (dropdown only)
	a **single-angle ladder** (no single-*concentration* A_2 exists)	*apparent*	

Each quantity's sources run from the best tier down. Every M_w and A_2 source is **calibration-dependent**.

The replicate average is the one R_h ± available at a single angle and concentration (§8.1.7), which is why it ranks above a single cumulant, and the program's re-selection of defaults, which runs whenever results change, never overwrites it.

A_2 has no hand-entry because it is a solvent/temperature-specific interaction coefficient with no external "standard", and it is the very quantity the A_2–M_w plot fits. A hand-entered R_g, R_h, or M_w is never overwritten by a later re-analysis.

The **default** source is the most-extrapolated the data support (a labeled choice). By case:

- **Zimm or Berry.** Between the two SLS extrapolations the Zimm value is the default, and the Berry value takes its place for each of R_g, M_w and A_2 where the Zimm one gives no value or is never the default (below) — past the qR_g limit of §12.4, for instance, which flags the Zimm R_g and M_w.
- **Extrapolated first.** While a quantity has any extrapolated value, its single-concentration values are listed but never the default, so a quantity whose every extrapolated value is one never chosen for you (*Never the default*, below) has no default until you pick one.
- **One concentration.** The Debye fit decides R_g and M_w. The Guinier value is listed beside the Debye value and is the default only where that concentration has no Debye value — so a single-concentration sample whose Debye R_g is flagged has no R_g default, even where the Guinier R_g passes its own check.
- **A tie.** Elsewhere, fit quality breaks a tie.
- **A flagged R_h.** It names its cause on its label, its row's badge carries the letter, and it can still be the default — a distribution peak flagged for an incomplete decay included, as a flagged cumulant is — except in the cases under *Never the default.* A Γ vs q^2 value that fails the linearity criteria names it on its label and can be the default too.
- **An arbitrary scale.** An M_w or A_2 on an arbitrary scale names its cause on its label — *uncalibrated, arbitrary scale*, *no usable dn/dc, arbitrary optical scale*, *no standard refractive index, arbitrary scale* — as does the label a Run writes onto the sample.
- **Never the default.** These are offered, each with its cause on its label, but never chosen for you:
  - a Zimm, Berry, Debye or Guinier R_g, M_w or A_2 whose fit failed a precondition other than its scale (past the qR_g limit, an ill-conditioned design, a non-positive intercept);
  - a size outside the range DLS can establish (§8.1.7);
  - a distribution peak whose maximum sits on an end of the R_h grid (§8.3.1);
  - every peak of a CONTIN fit whose α choice is flagged (§8.3.2);
  - a Mode Map family's R_h (§8.5).

  The label a Zimm or Berry **Run** writes onto the sample names the same cause after its scale words, and a family's label says when the family pooled a replicate average beside its own sources (§8.4). An uncalibrated run, a missing dn/dc or a blank standard refractive index never keeps a value from being the default.
- **Nothing to default to.** When no candidate can be the default, the re-selection of defaults clears a value the program stored for that quantity from a source it still lists, including one a Zimm/Berry **Run** wrote; the R_h row clears the same way when no DLS value is usable. A value you typed or picked is never cleared, and neither is one whose source is not listed, such as a form-factor value after you reopen a session and before you re-run its fit. To use a flagged value, pick it in the list.

A source you explicitly pick is firmer than the default, but not permanent:

- The re-selection of defaults never touches it.
- The two runs that write sample values replace it: a Zimm/Berry run replaces a picked R_g, M_w or A_2, and **Average derived results...** replaces a picked R_h. The run reports the replacement.
- Every other run offers its new fit in the dropdown instead of replacing your choice.
- A Settings change that clears the result a pick came from clears the pick with it, after you confirm; which setting clears what is §12.1.

To be asked before a run replaces a pick, tick **Ask before a Run replaces a picked value**, the **Runs** row at the end of the Settings tab (§12.9).

Each source dropdown groups its candidates by result type (extrapolated / distribution / averaged / single-condition) and hides the long tail of per-angle / per-concentration apparent values behind a **Show all single-condition results** checkbox; a value never chosen for you is always listed. The best-tier value stays selected either way. Four deserve a word:

- **The extrapolated R_h.** The Γ vs q^2 and D vs c values are your own runs on the DLS tab while those are current (§8.1.3). Otherwise the program fits the sample itself, one concentration (or angle) at a time, over the measurements the DLS tab ticks by default, which leave out a replicate average beside its own source runs (§8.4). A run of yours that pooled such an average, mixed concentrations in one Γ vs q^2 line or mixed angles in one D vs c line is offered, its label saying what it pooled, but never chosen for you.
- **Single-angle ladders.** A concentration series read at one angle offers its apparent M_w and A_2 under a *Single-angle ladder (apparent)* group — one entry per angle you have run, each naming its own θ. At one angle there is no $q \to 0$ extrapolation, so both values are apparent: the M_w is $M_w \cdot P(q)$ and the slope is $2A_2$ only where $P(q) \approx 1$. The program offers them but never selects one for you — which angle should represent the sample is your call, and a thermodynamic Zimm/Berry value always outranks them in the automatic default. The angle travels with your pick, into the dropdown's label and the **Source** column of the exported scaling CSV.
- **Distribution peaks.** If you have run a CONTIN, NNLS or lognormal distribution for a measurement, each resolved population appears as its own R_h source, labeled with its geometric-mean R_h (§8.3.3) and its percentage of the intensity, so you choose which population feeds ρ. The dominant peak ranks with a single-condition cumulant for the default; a minor peak ranks below every other source and is chosen for you only when no other value can be, and either one's label names a failed check. The percentage is the peak's intensity fraction (§8.3.3). Peaks are offered only from a distribution you have already computed.
- **The DLS tab's own cumulant fit.** While the Correlogram sub-tab's cumulant fit of a measurement is current, it is offered as that measurement's single-angle R_h, labeled *as fitted in the DLS tab* (§8.2.2); otherwise the dropdown offers the measurement re-fitted at the cumulant order set in Settings (§12.1.2).

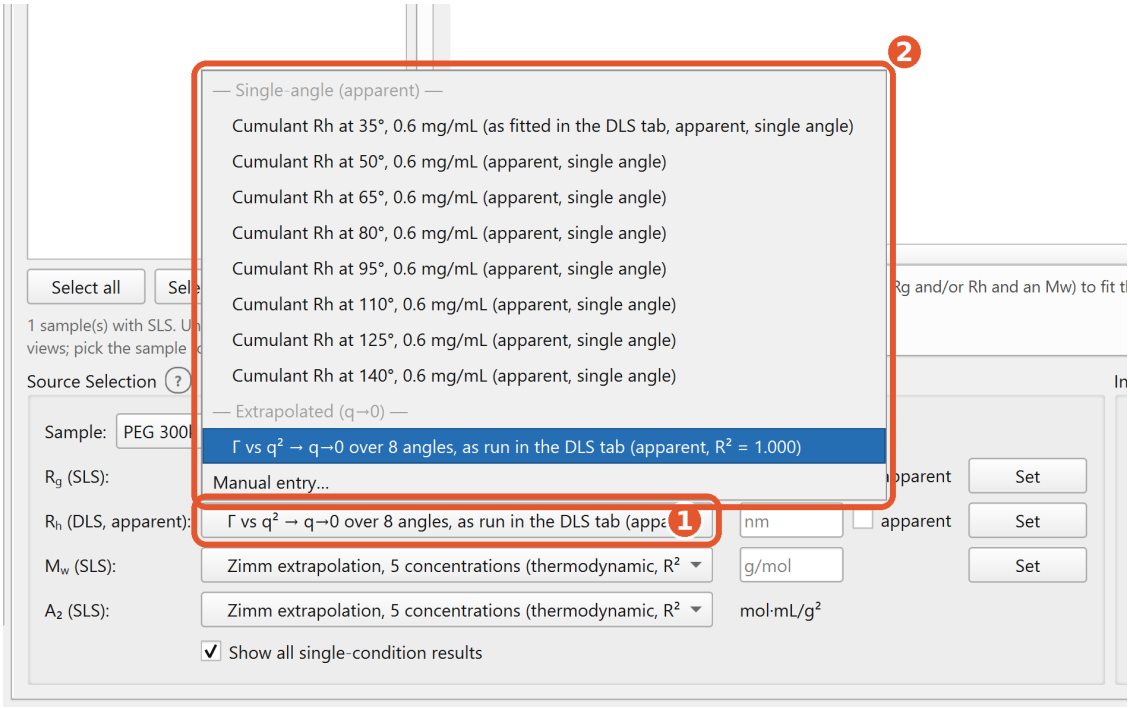


Figure 9: The R_h source dropdown (1) of the PEG 300k sample, open (2), with **Show all single-condition results** ticked: the candidates are grouped by result type — here the eight single-angle cumulant sizes under *Single-angle (apparent)*, and the Γ vs q^2 value under *Extrapolated (q→0)*, which is the selected best-tier default. A distribution peak, a replicate average, a form-factor size and a degenerate solution each appear under their own group when they exist.

10.1.3 Form-factor and degenerate sources

If you have run a **Form factor P(q)** fit on the SLS tab (§9.1.2), its R_g — and, when the run is calibrated, its M_w — also appears as a source, under its own *(model-dependent)* group, its label naming the model (*model-dependent (sphere P(q))*). A form-factor R_g assumes a particle shape, so it enters at the same tier as its slope-based sibling (a multi-concentration fit is thermodynamic, a single-concentration one apparent) but the program never auto-selects it over a slope-based value at the same tier — you pick it deliberately. The word travels into the table, whose R_g cell reads *(model-dependent)*, into the linear-reference caption (*includes model-dependent values*) and into the exported CSV (an **Rg source** column). A_2 is not offered from a form factor: at $q \to 0$ every shape converges to the same limit, so a form-factor A_2 would only duplicate the Zimm value. The form factors themselves are the **Theory-and-Equations-Guide** §4.8.

Ellipsoid and core–shell fits are often **degenerate**: more than one shape (prolate vs oblate, or different core/shell radius pairs) reproduces the same scattering curve, so the data alone cannot pick one. Each admissible solution is offered as its own pick under a *(model-dependent — degenerate)* group, labeled by its distinguishing shape parameters — an ellipsoid solution by prolate/oblate, axial ratio, and R_g; a core–shell solution by its core and outer radii. An ellipsoid label reads *axial ratio unresolved* where the fit could not identify the ratio, and *axial ratio imprecise* where it identified one whose relative SE is past the precision limit (the **Theory-and-Equations-Guide** §4.9.3, axial-ratio identifiability). Selecting one is guided manual entry: if you have independently resolved the shape (by electron microscopy, SAXS/SANS, or optical microscopy), you choose the

matching solution; the program records which one you picked and flags every downstream value (ρ, the scaling exponent, the g/h branching factors, and the exports) as resting on a user-selected degenerate solution. The program never picks a degenerate solution for you. When a fit's solutions agree on R_g (within about 10 %, or 1 nm) even though the shape is not pinned down, the dropdown additionally offers a single **size determined, shape ambiguous** value (the mean of the solutions' R_g), which *can* be selected automatically so you need not choose a shape. A contrast-matched core–shell with no defined R_g still offers none.

A form-factor source appears only while its fit is **current** — computed against the same mask, calibration, and settings the sample currently carries. If you change any of those after fitting (mask a point, edit the calibration, switch the **Regression SE estimator**), the form-factor option quietly disappears from the dropdown until you re-run the fit on the SLS tab (which also flags it stale there). A form-factor R_g you had already picked is kept across a reopened session (§7.4), with its value and its *model-dependent* word; the option itself is offered again only after you re-run the fit.

10.1.4 Branching factors *g* and *h*

Tag samples **Linear** (at least two, at different M_w) and **Branched** with the **Chain type** field on the Data tab (§6.1): the linear R_g–M_w and R_h–M_w laws are fit to the Linear samples and read at each Branched sample's M_w, and the table fills its g (from R_g) and h (from R_h) columns for every Branched row. The **Branching model** selector sets how g is read — a regular-star **arm number**, or a random tri-/tetra-functional **branching density n** (the **n (branch)** column) — and is model-dependent: the same g reads as a different n under a different model, and the **Interpretation** readout's n says *(model-dependent)*. In the table:

- a cell that cannot be computed reads "—"; hover it for why;
- selecting a row shows its readout in the **Interpretation** panel;
- a ± appears only from replicate branched measurements;
- **Export CSV...** writes the table.

What g, h and n are, and how to read them, is the **Theory-and-Equations-Guide** §5.3–5.4.

10.2 Scaling

The **Scaling** sub-tab shows two stacked plots over the included samples (one point per sample/fraction), using each sample's effective M_w (a hand-entered M_w wins) chosen in the same source panel. The lower plot is always A_2–M_w; the upper plot is a size scaling whose axis you swap between R_g–M_w (SLS) and R_h–M_w (DLS) with the **Show R_h–Mw / Show R_g–Mw** button. The **Interpretation** panel below the plots gives a plain-language reading of the fitted exponents. The axes follow **Settings ▶ Display Units** (§12.7). Two cautions the sub-tab surfaces:

- A plotted value can be on an **arbitrary scale** — an uncalibrated run or one with the standard's refractive index blank (§9.1.1), a missing $\mathrm{d}n/\mathrm{d}c$ (§6.5), or both — in which case a line under the plots names the cause and the samples whose value it is (§8.1.7). The line appears whenever such a point is drawn, including on a plot too sparse to fit a slope. Two things follow from it:
  - which coordinate is meaningless: on a size plot the value on an arbitrary scale is the M_w, so the horizontal position is the meaningless one and a trusted M_w is the fix; on the A_2 plot it may instead be the A_2 itself, where the vertical position is affected and supplying the missing input the line names is the fix;

  - whether the exponent survives: where every plotted value carries one common unknown factor, the fitted **exponent survives** — a common factor shifts the log–log intercept, not the slope — so it is the plotted values and the prefactor that are not absolute; where the factor differs between samples, as when the line names only some of them, it moves each point by its own amount and the slope with it, so the exponent is distorted as well.
- A_2 can be **negative** (below the θ-point), in which case that point cannot sit on the log A_2 axis and is excluded.

Curating which samples form the series is your responsibility — mixing solutes or solvents on one fit is not meaningful. **Export CSV...** writes the points and fits, with branched points in their own columns beside the fitted ones (§13).

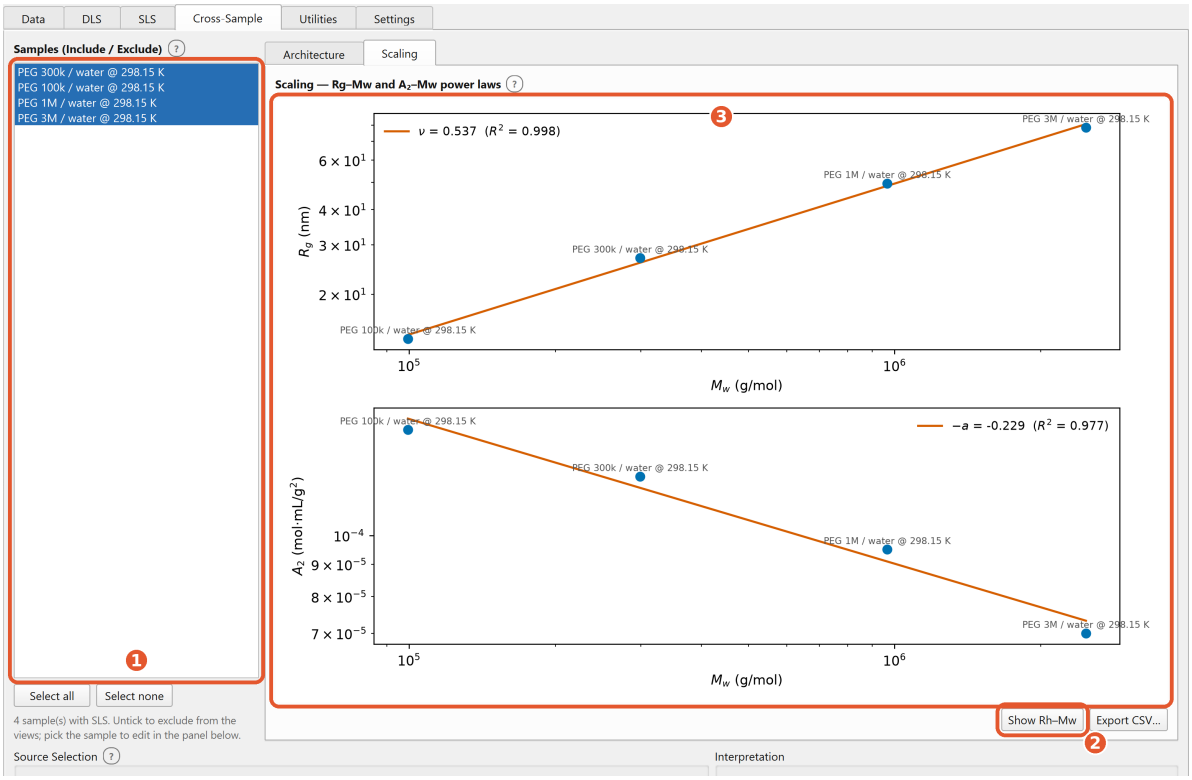


Figure 10: The Cross-Sample **Scaling** sub-tab over a PEG molecular-weight series. Each included sample (1) is one point; the upper plot fits R_g–M_w (swap to R_h–M_w with the button (2)) and the lower plot A_2–M_w, both on log–log axes (3). The fitted exponent is printed on each plot with its R^2 — here $\nu = 0.537$ on the size plot.

---

11. The Utilities Tab

The Utilities tab assesses data quality and generates synthetic test data. Its four sub-tabs are **Traces** (§11.1), **I·sin** θ (§11.2), **Synthetic Generator** (§11.3) and **Solvent Explorer** (§11.4). The diagnostics help identify slow-mode fluctuations, drift, and dust before committing to a full correlation-function or Zimm analysis.

The plots' anatomy, the axis-control bar and the layout are described once in §14; the export

buttons and files in §13.

11.1 Traces

Plots count-rate traces with overlays that flag what stands out, and a second panel with one of five diagnostics. Traces are loaded, attached and removed in the sidebar (§7.3), not here.

1. Tick the trace(s) to plot in the checklist (**Select all** / **Select none**).
2. Choose the **Scale** — **Absolute** or **Relative (baseline-normalized)**.
3. Tick the overlays you want (§11.1.2–11.1.4).
4. Pick a **Diagnostic** for the lower panel (§11.1.5); the summary line beneath the plot reports the numbers.

With several traces ticked, the summary line, the diagnostic panel and every **Export ...** button describe one trace — the highlighted one if it is among the ticked traces, otherwise the first ticked one — and the summary line starts with its name.

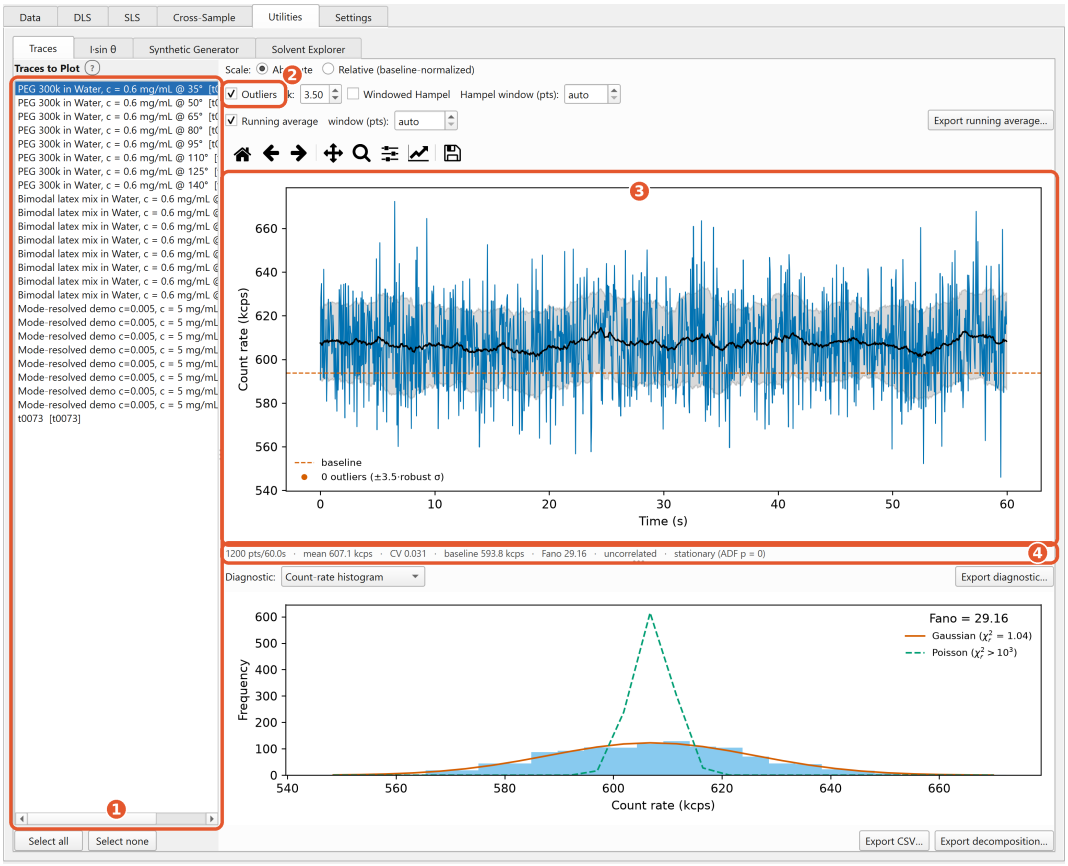


Figure 11: Utilities ▶ Traces. Tick the trace(s) to plot (1) and toggle the overlays (2). The plot of count rate against time (3) shows the robust baseline (dashed), the outlier overlay's count in the legend and the running average (solid) with its ±1 SD band; the summary line (4) reports the point count, mean, coefficient of variation, baseline, and the Fano factor. This trace flags no outlier though its Fano factor is far above 1 — the case §11.1.2 describes. The **Diagnostic** dropdown below it picks the second plot — here the count-rate histogram with its Gaussian and Poisson fits.

11.1.1 Summary statistics and baseline

The summary line under the plot reports the mean count rate and its coefficient of variation (CV) with the baseline. The baseline (the true scattering level) is estimated robustly, as the trace's 25th percentile, so upward dust spikes do not bias it, and drawn dashed. The summary line also reports an augmented Dickey–Fuller **stationarity** test's p-value; non-stationarity indicates drift or slow-mode behavior.

11.1.2 Outliers

Flags samples that stand out from the rest of the trace: a band centered on the trace's median and scaled by its own robust scatter, with a sample flagged when it falls outside $\pm k$ of that (**k**, default 3.5). The outlier count is printed in the plot legend beside the band's k. The band is global, so it flags departures from the baseline of any duration without saying how long they lasted, and it compares each sample against *this trace's* scatter, so a trace far noisier than photon statistics can still flag almost nothing — read the flag together with the **Fano factor** the summary line reports (§11.1.5). If more than half the samples are identical — a stalled or blocked detector, or a very coarsely quantized reading — the overlay reports **no usable scale** instead of flagging anything, an otherwise-obvious spike included. What the estimator is, and why a median and a robust scale, is the **Theory-and-Equations-Guide** §2.1.1.

11.1.3 Windowed Hampel

The same test against a moving baseline, which is what lets it say *when* something happened. Over a window centered on each sample it takes the median and the same robust scatter, draws that rolling median with its $\pm k$ band, and marks the samples that leave it. Its controls:

- **Windowed Hampel** turns the overlay on.
- **Hampel window (pts)** sets the window's width in points. Left on **auto**, it is sized from the trace length; the value used is shown in the summary line and written into the export. Widen it to catch longer events; narrow it to follow a faster-moving baseline, at the cost of more flags on a clean trace.
- **k** is the one shared with **Outliers** (§11.1.2).
- **Export decomposition...** writes time, count rate, rolling median, rolling scale, the flags and the stretches not judged as columns, with both overlays' settings in the header; the trace itself is exported unaltered.

The summary line adds an **event** count — consecutive flagged samples grouped into one event, with its duration — and how much of the trace was **not judged**, as percentages. A stretch not judged is a gap in the band: no full window fits there (the first and last half-window of every trace), or more than half the window shares one value; nothing is flagged inside a gap.

An event lasting up to about half the window can be flagged; anything longer is tracked *as* baseline, so a persistent offset — dust that settles and sits — appears as a **step in the rolling curve** with no flags at all, which is why the curve is drawn and not just the flags. The trade-off behind the window size is the **Theory-and-Equations-Guide** §2.1.2.

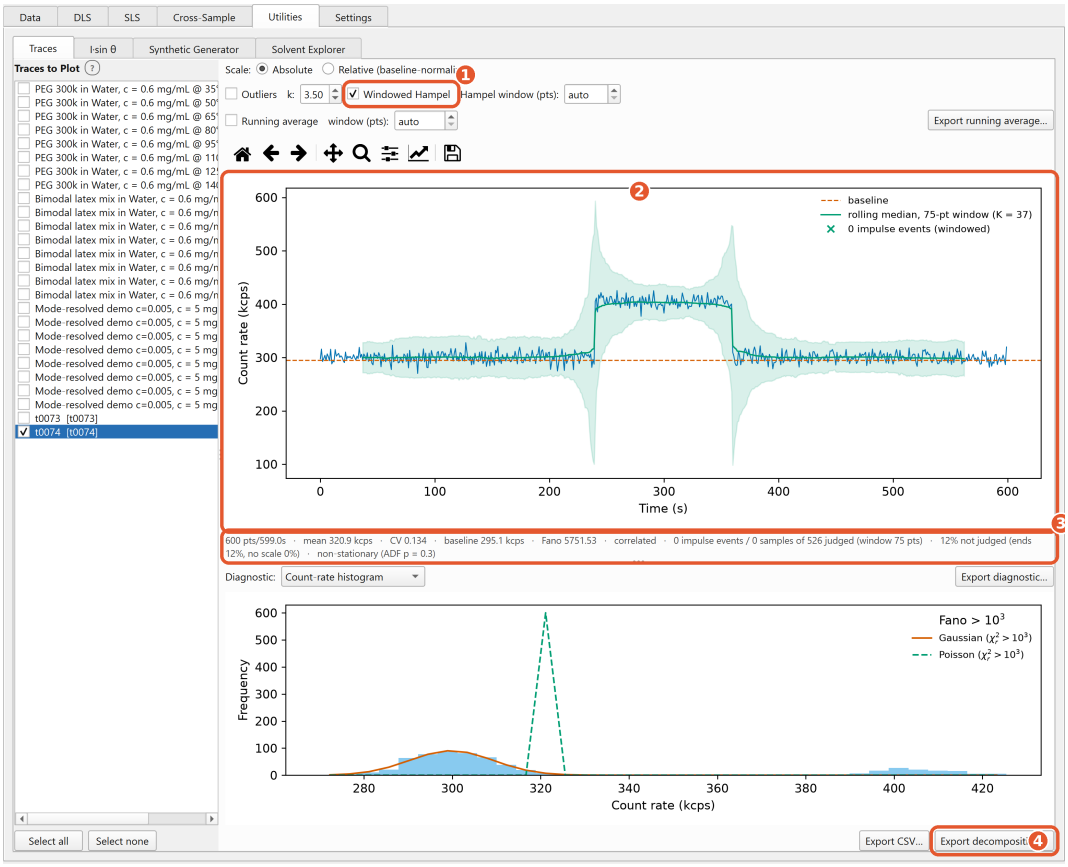


Figure 12: **Windowed Hampel** (1) over a trace with a long-residence offset. The rolling median and its $\pm k$ band (2) follow the step rather than flagging it, because the offset lasts longer than half the window; the band is absent over the first and last half-window, the stretch that was not judged. The summary line (3) reports the event count and the not-judged percentages; **Export decomposition...** (4) writes the rolling median, scale, flags and the stretches not judged as columns.

11.1.4 Running average

A sliding-window mean and standard deviation over **window (pts)** points (**auto**, about a twentieth of the trace), drawn solid with its ± 1 SD band; a wide window separates slow drift from shot noise. **Export running average...** writes the running mean and its within-window standard deviation at the window set there. **Export CSV...** writes the plotted trace as `(time, count rate)` columns.

11.1.5 The Diagnostic panel

The **Diagnostic** dropdown picks the lower plot; **Export diagnostic...** writes the plotted numbers. The histogram and block variance describe the trace's amplitudes; the Allan deviation says how long it may usefully be averaged; the spectrum and periodogram look for periodic contamination. Each is defined in the **Theory-and-Equations-Guide** §2.1.

- **Count-rate histogram.** Fitted with a Gaussian and a Poisson distribution (Freedman–Diaconis binning, Sturges fallback). The key diagnostic is the **Fano factor** (variance/mean in count space): a value near 1 is shot-noise limited (ideal Poisson), and a value above 1 signals excess fluctuations (slow modes, dust, drift). It is also printed in the summary line.

The reduced χ^2 of each fit and the coefficient of variation (against the shot-noise expectation) are reported beside it.

- **Block variance.** The standard error of the mean vs block size: flat for an uncorrelated trace; a rise-then-plateau is the signature of positive correlations (slow modes). A correlation flag reads the plateau against the small-block value (default threshold 1.5).
- **Allan deviation.** Answers "how long may I average?". While it falls with slope $-1/2$, averaging longer keeps paying; where it flattens or turns up, drift has taken over and averaging for longer than that buys nothing.
- **Power spectral density.** Finds periodic contamination — pump or stirrer vibration, temperature cycling, mains harmonics. The dashed line is where an ideal photon-counting detector would sit; broad power above it is excess noise, a narrow spike above it is something periodic. **segment (pts)** sets the points per averaged segment (longer segments resolve finer frequency detail but leave fewer to average; 0 = auto), and **window** the data window applied to each segment before transforming (Hann, or the two classical Welch windows).
- **Periodogram + false alarm.** Puts a number on a spectral spike: its line is the height noise alone would rarely reach, drawn for the **false alarm (%)** probability you set (1 % is the usual convention). Read it only when the trace is otherwise well behaved — the test assumes uncorrelated noise, so on a drifting trace it is marked **unreliable** rather than believed. A spectral line also makes the Allan curve jump about, so check the spectrum before trusting an averaging time.

11.2 I·sin θ

For an ideal, isotropic, dust-free scattering volume, I·sin θ is flat across angle. Curvature, asymmetry about 90°, or upturns at the extremes reveal alignment problems, stray light, or dust. Choose the sample from the **Sample** dropdown at the top of the sub-tab (only samples with a $c = 0$ solvent-blank curve appear, since the sub-tab plots the $c = 0$ intensities) and the **Scale**: **Absolute** (each curve at its own intensity, to compare the blanks a sample holds) or **Relative (normalized)** (each curve divided by its mean — for comparing solvents, e.g. toluene vs water, where absolute scale is meaningless). A note under the plot counts the curves and says how to read them. The sub-tab has no **Export CSV...**; save the plot with its toolbar's **Save** button.

> **Tip.** A flat I·sin θ is much harder to achieve with water than toluene. Overlaying normalized curves for both is an effective alignment/stray-light check.

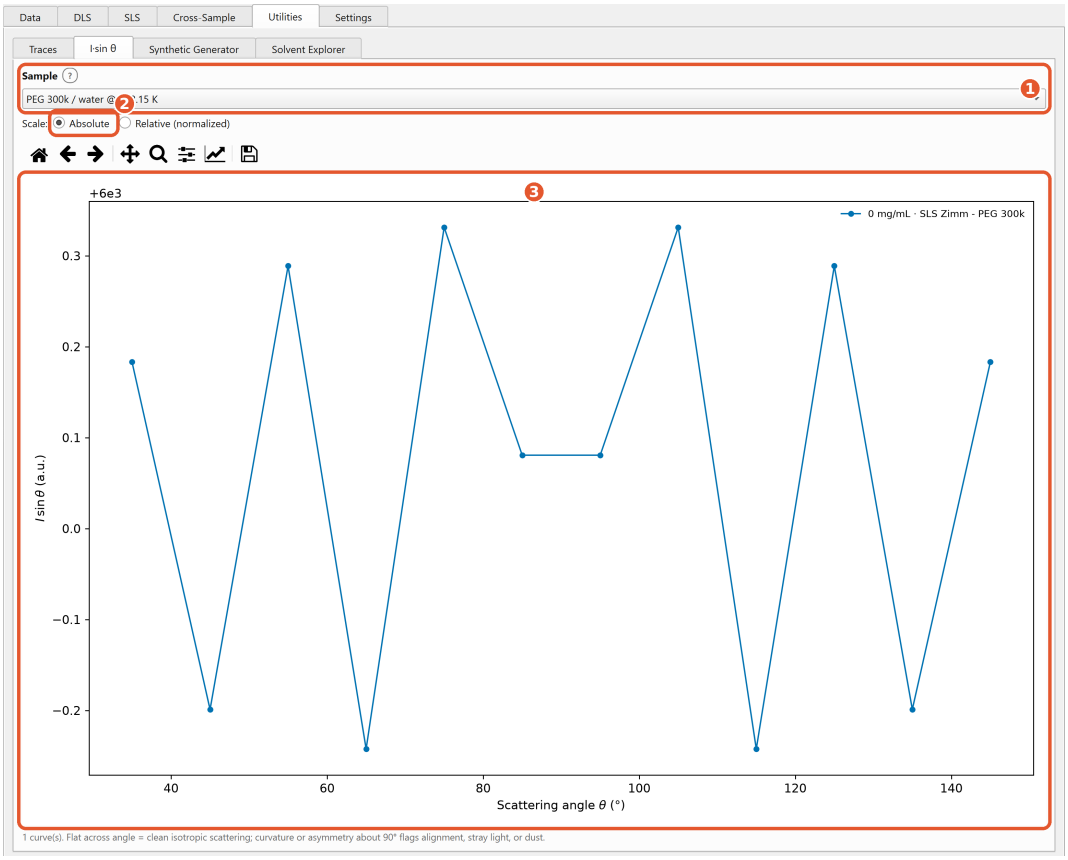


Figure 13: The **I·sin** θ sub-tab on the PEG 300k sample's $c = 0$ curve. Pick the sample (1) and the **Scale** (2); the plot (3) draws I·sin θ against angle for every $c = 0$ curve the sample holds. Read the axis with its offset: this curve is flat to about one part in 10^4 (the whole vertical span is about 0.57 on a level of 6000), which is what a clean, aligned blank looks like; a real misalignment shows as a trend across angle or an asymmetry about 90° that survives the offset.

11.3 Synthetic Generator

Generates artificial light-scattering data with known ground-truth parameters — to validate an analysis (recover a known answer) or produce test files. The forward model uses the same q, Stokes–Einstein, optical-constant, Rayleigh and Zimm relations the analysis tabs use, so loading the output and analyzing it with the fit its model matches recovers what you put in (§11.3.2).

1. Set the shared **Conditions**, and fill the input block(s) for the artifacts you want (§11.3.1).
2. Tick **Preview** and/or **Save** for each **artifact** — a kind of data set the generator makes — under **What to Generate** (§11.3.2).
3. Choose the **Output** folder, or tick **Add to workspace as sample**.
4. Click **Generate**.

Ticking **Preview** draws the raw generated data (the analysis figures come from loading the data into the analysis tabs) and never changes any state; **Save** writes loadable instrument files into the chosen folder; **Add to workspace as sample** injects a coherent sample under a parsed `name / solvent` identity, so DLS + SLS connect for ρ without touching disk. The sample takes one DLS artifact (the multi-angle set in preference to the two-mode set or a single correlogram), one SLS

artifact (the full Zimm set in preference to the others), any depolarized or mode-resolved set, and the trace.

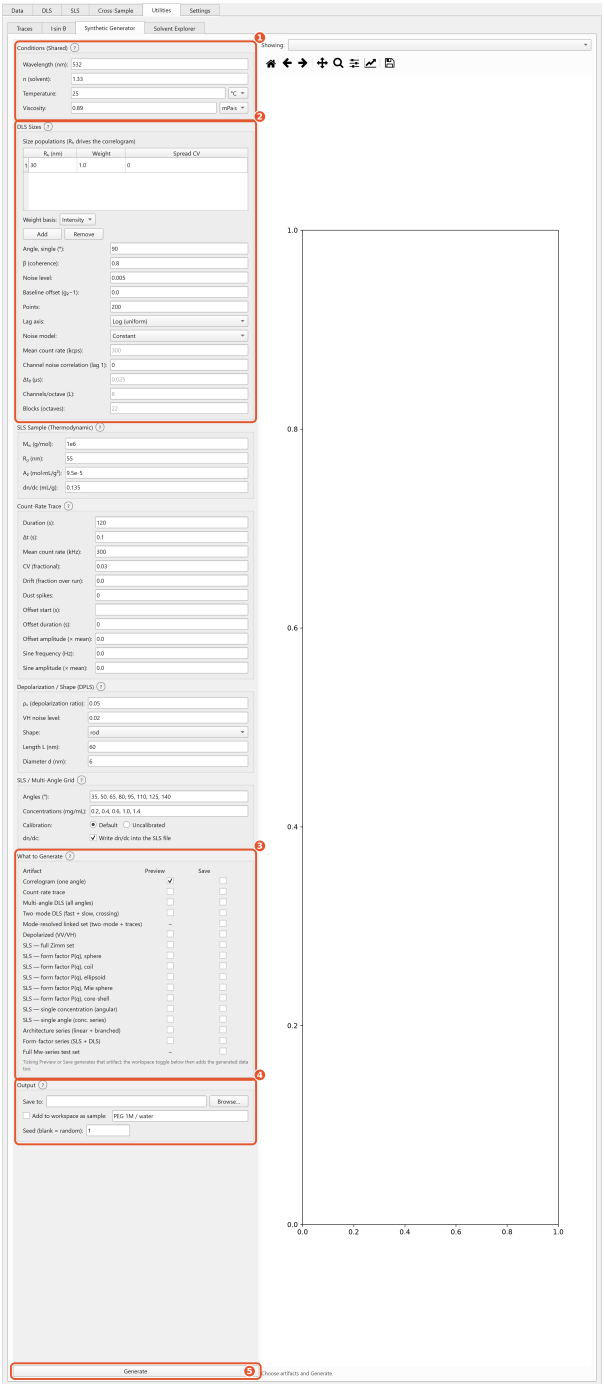


Figure 14: The **Synthetic Generator** panel: the shared **Conditions** (1) and the **DLS Sizes** block (2) with its populations table, noise model and lag axis; further down the SLS sample, count-rate trace, depolarization and grid blocks, then the **What to Generate** grid (3) with a **Preview** and a **Save** box per artifact, the **Output** row (4) with the folder, the workspace toggle and the seed, and **Generate** (5) pinned at the foot.

11.3.1 Inputs

DLS and *SLS* need different physical parameters, so they have separate input blocks that share the optical **Conditions (Shared)** — wavelength, n, temperature, viscosity — and the **SLS / Multi-Angle Grid** of angles and concentrations.

- **DLS Sizes** — a populations table (median R_h, relative scattering weight, lognormal spread; one row = monomodal, two = bimodal, …; **Add / Remove**) plus the coherence factor β, noise level, baseline offset and point count. The baseline offset adds a flat shift to the whole g_2-1 curve — the residual an imperfect correlator baseline leaves behind; leave it at 0 for a textbook correlogram, and raise it to see what a floating-baseline cumulant fit is for. It reaches the Correlogram, Multi-angle DLS and Depolarized (VV/VH) artifacts (the last on both its VV and its VH correlograms). The populations *are* the modality — there is no separate "bimodal" option, and the CONTIN/NNLS distribution is something you then recover by analyzing the generated correlogram, not a generated artifact.
- **Weight basis** (in **DLS Sizes**) — how the populations table's weights are read: **Intensity** (the default), the scattering amplitude, used directly; or **Number**, **Volume** or **Surface**, a composition whose intensity is forward-modeled with the chosen **Model** (Rayleigh point, RDG sphere or Mie sphere, the last two with a **Solute n**), so the generated correlogram's intensity distribution reweights back to it on the Distribution sub-tab (§8.3.3).
- **Lag axis** — *Log (uniform)* spaces the delay times evenly and uses the **Points** count; *Multi-τ* builds the staircase a real correlator produces, set by Δt_0, the channels per octave and the number of blocks (the **Theory-and-Equations-Guide** §7.1). Every instrument format of §3.1 carries a multi-τ axis (only a plain-text file can carry another grid), so it is the more realistic — and harder — test of a distribution fit. The choice applies to the Correlogram, Multi-angle DLS, Two-mode DLS and Mode-resolved linked set artifacts, which then ignore **Points**; every other artifact keeps a uniform grid.
- **Noise model** — *Constant* puts one standard deviation on every channel; *Per-channel (Biganzoli–Ferri)* varies it the way a correlator does — quietest through the decay, noisiest in the coarsest blocks — with **Noise level** setting the first channel's SD, the same per-channel variance the nonlinear cumulant weights its fit by; *Per-channel + shot noise (Biganzoli–Ferri, full)* adds the photon-counting terms a real count rate brings and takes the **Mean count rate** entered beside it, in kcps. All three models are the **Theory-and-Equations-Guide** §7.2.
- **Mean count rate** (beside the noise model). For a multi-angle set — Multi-angle DLS or Two-mode DLS — that value is the rate at 90°: each angle's rate follows the $1/\sin\theta$ volume law from it, its correlogram's noise is drawn at that rate, and its count-rate trace is built around the same number, so the rates the set commits and the traces it links agree. The Mode-resolved linked set draws at its own static intensities instead, so there the value only selects the model. A multi-angle set's per-channel noise is also drawn at the run duration its file states (60 s); a single correlogram states none.
- **SLS Sample (Thermodynamic)** — M_w, R_g, A_2, and $\mathrm{d}n/\mathrm{d}c$.
- **Count-Rate Trace** — the trace artifact's controls, in the table below.
- **Depolarization / Shape (DPLS)** (for the Depolarized (VV/VH) artifact) — a depolarization ratio ρ_v and a particle **Shape** (rod, ellipsoid or sphere) with its dimensions, which set the depolarized VV/VH channels over the shared angle grid. A sphere has one dimension, so its second field is grayed out.
- **SLS / Multi-Angle Grid** — the angles and concentrations, and two choices:
  - **Calibration** — *Default* writes calibration metadata into a saved SLS file (and, when

Add to workspace as sample is ticked, prefills the session calibration so M_w is absolute); *Uncalibrated* omits it, so the analysis is flagged arbitrary-scale (§6.5).

– **Write dn/dc into the SLS file** — untick it to produce a file that withholds the increment (the intensities are still generated with it): the program then analyzes it on the arbitrary optical scale, as it would a real file with no $\mathrm{d}n/\mathrm{d}c$ (§6.5).

Count-Rate Trace control	What it sets
Duration (s), **Δt (s)**	the trace's length and its sampling interval
Mean count rate (kHz)	its mean
CV (fractional)	a fixed relative scatter, not shot noise — what the Fano factor measures
Drift (fraction over run)	a drift over the run
Dust spikes	a number of single-sample multiplicative hits, the impulses both outlier overlays are built to find
Offset start (s), **Offset duration (s)**, **Offset amplitude (× mean)**	a step that persists for its duration, the case §11.1.3 describes; an empty start adds none
Sine frequency (Hz), **Sine amplitude (× mean)**	a periodic component of known frequency, the ground truth for the spectrum and the periodogram; above half the sampling rate it aliases, as on a real instrument

11.3.2 Artifacts

Artifact	What it tests	Saved as
Correlogram (one angle)	cumulant / single / double / KWW / CONTIN / NNLS	generic two-column CSV
Count-rate trace	trace diagnostics (outliers, windowed Hampel, drift, Fano, ADF)	two-column CSV
Multi-angle DLS (all angles)	Γ vs q^2	multi-angle `.ASC` (+ a trace per angle)
Two-mode DLS (fast + slow, crossing)	the Mode Map: two modes whose $\Gamma(q^2)$ lines cross	multi-angle `.ASC` (+ a trace per angle)
Mode-resolved linked set (two-mode + traces)	the SLS Mode-Resolved sub-tab: a two-mode DLS set with the traces a Derived Intensity series needs	a folder (save only)
Depolarized (VV/VH)	static Cabannes (δ^2, f) + dynamic DDLS (D_r, shape)	two ALV `.ASC` (VV + VH)
SLS — full Zimm set	Zimm / Berry (thermodynamic)	multi-concentration SLS `.csv`
SLS — form factor P(q), sphere · coil · ellipsoid · Mie sphere · core-shell	the five **P(q) model** fits (large qR_g curvature)	multi-concentration SLS `.csv` each
SLS — single concentration (angular)	Debye / Guinier (apparent)	SLS `.csv`

Artifact	What it tests	Saved as
SLS — single angle (conc. series)	calibration-free A_2 / single-angle ladder (apparent)	SLS `.csv`
Architecture series (linear + branched)	Cross-Sample branching g / h / arm-number n	a linear + branched folder
Form-factor series (SLS + DLS)	Cross-Sample ρ shape verdict, degenerate-solution selection, model-dependent $\nu \approx 1/3$ scaling, apparent / uncalibrated	an SLS + paired-DLS folder
Full M_w-series test set	everything (a homologous series)	a Clean + a Messy folder (save only)

The ellipsoid and core-shell form-factor sets also exercise the Cross-Sample tab's degenerate solutions (§10.1.3).

The SLS full Zimm set and the SLS sets of the Full M_w-series test set follow the Zimm form exactly, so a Zimm fit recovers the R_g you entered at any qR_g, while a Berry fit reads it low once qR_g passes about 1. The two largest samples of the M_w-series set lie past the default qR_g limit (§12.4), where the Zimm R_g is flagged and the Cross-Sample default takes the Berry value (§10.1.2); with the sources left at their defaults, the **Scaling** sub-tab plots their smaller Berry R_g and fits an exponent ν below the one the set was built with. Each of the set's two folders holds a `parameters.txt` with the values to enter and the true values.

The **Depolarized** artifact bundles both depolarized-scattering inputs: a static VV/VH intensity pair (its ρ_v, δ^2, and Cabannes factor are shown — copy the two intensities into the SLS **Anisotropy correction** check, §9.1.3), and a dynamic VV + VH multi-angle set that, injected, is already tagged VV/VH so the DLS tab's DDLS sub-tab pairs it and recovers D_r and the shape. The **Architecture series** is a batch: injecting it adds a whole linear reference ladder plus branched samples, each already tagged **Linear** / **Branched**, so the Cross-Sample **Architecture** sub-tab reads g / h / n out of the box.

11.4 Solvent Explorer

A standalone look-up and visual calculator over the whole solvent library. It is global — it needs no loaded sample, ignores the sidebar selection, and never writes into a measurement. Choose a solvent, temperature, and wavelength to read n and η with their uncertainty and a **confidence-tier badge** reading **primary** or **estimate** — the same two tiers as the Data tab's **Measured properties** and **Estimated properties** sections (§6.3; all library solvents appear here, grouped the same way). Two plots, side by side on one shared figure, show at a glance the range of conditions the library can serve:

- η **and** n **vs temperature** (twin axes — viscosity on the left, on its own log-capable scale; refractive index on the right), and
- n **vs wavelength**,

each drawn only across the range the library covers, with a shaded **confidence band** (the viscosity's can be too narrow to see on its log scale) and a marker at your chosen condition. The band

is a conservative bound at each condition, including an allowance for how much independent laboratories' published measurements typically disagree — narrowest near the conditions the sources measure best, growing toward the edges of the validity range, and flat where a source states no temperature shape (the **Theory-and-Equations-Guide** §1.1.2). Like every library uncertainty it is displayed, never folded into a reported ±. The solvent the Explorer opens on is **Settings ▶ Solvent Library ▶ Default solvent** (§12.5).

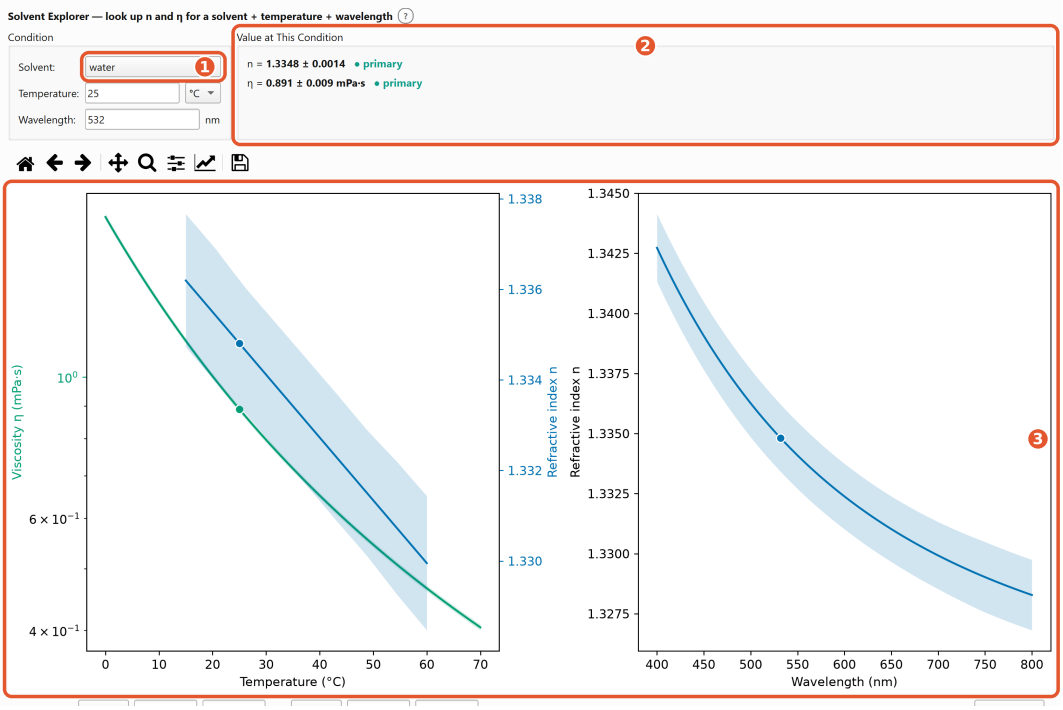


Figure 15: The Solvent Explorer (Utilities ▶ Solvent Explorer): pick a solvent (1); read n and η at the chosen temperature and wavelength with their uncertainty and a confidence-tier badge (2). The plots (3) draw η and n vs temperature and n vs wavelength, only across the range the library covers; the n curves carry a visible shaded confidence band, while η's is too narrow to see at this scale.

---

12. The Settings Tab

Global defaults and appearance, in the groups the tab shows, top to bottom. A setting that affects a number **seeds** the per-measurement control on the tab where it takes effect, and a setting that every fit reads says below what it clears when changed. Nothing here is applied until you press **Update** at the foot of the tab (§12.10).

12.1 DLS Analysis Defaults

12.1.1 Cumulant method

Nonlinear (Frisken's moment expansion, the default — fits $g_2{-}1$ directly with a floating baseline) or *Linear* (Koppel's classic log-polynomial fit); how the two differ is the **Theory-and-Equations-Guide** §3.2.3 (the Linear fit) and §3.2.5 (the Nonlinear fit). It applies to every cumulant-based

step: the Correlogram sub-tab's cumulant, Γ vs q^2, D vs c, DDLS, replicate averaging and the Cross-Sample size candidates. The results table shows which method produced each result (and its fitted baseline).

Switching it asks first — the dialog says how many results, and how many picked values, will go, and **Cancel** changes nothing — and then:

- clears the cumulant-based results, every Γ vs q^2 and D vs c result among them, a reopened session's Summary rows included, whose fits are not kept with the session (§7.4);
- clears a sample R_h taken from a cleared result, one you picked yourself on the Cross-Sample tab included, since the fit it named does not exist under the new method — so ρ is blank until you re-run or re-pick;
- keeps the distributions, SLS, and any hand-entered R_h.

12.1.2 Cumulant order

1 to 4. It is the order every fit *without an order box of its own* runs at — Γ vs q^2, D vs c, DDLS, replicate averaging and the single-angle cumulant R_h the Cross-Sample tab re-fits — and it seeds the Correlogram sub-tab's own **Cumulant order** box, whose per-measurement value wins (§8.2.2). Changing it clears, after you confirm, those results and a Cross-Sample pick taken from one of them, but never a Correlogram fit that ran at its own box.

12.1.3 Cumulant residual weighting

- *Sourced variance (default)* — each method's own weight, the Biganzoli–Ferri per-channel variance. The export records it as `biganzoli-full` when a committed count rate entered it and `biganzoli-classical` when none did; the synthetic generator's *Per-channel (Biganzoli–Ferri)* noise models plant the same variance (§11.3.1).
- *Unweighted* — plain least squares on the method's own residual, for comparing with a fit made without a noise model, or for seeing how much the noise model moves a size and PDI; it is not a more accurate answer. The fit records `uniform`, the correlogram, Γ vs q^2 and D vs c exports name it (§13), and the DLS results titles read *unweighted* (§8.1.7).

The choice applies to every cumulant fit the program shows or uses, but not to the cumulant each exponential or lognormal fit starts from. A fit whose default weight cannot be formed from the measurement's inputs falls back to the unweighted form. That fit is the one choosing *Unweighted* would give, so no flag is set; the fit records the fallback and why the weight could not be formed, as a plain line under the sub-tab's results (§8.1.7), in the Summary cell's ⓘ (§8.8) and in the export's Comments. Switching the choice clears existing cumulant-based results after you confirm, exactly as for the method. What each weighting is, and why an opted-into choice travels with the number, is the **Theory-and-Equations-Guide** §3.2.11 and §3.2.12 (*The other two weightings, and the fallback*; *Which weight is the default*).

12.1.4 Skip initial lag channels

Drops the first N channels (the shortest lags) from *every* DLS fit — set it to your correlator's afterpulsing count when a fit is dragged to a falsely small size by a noisy first channel (default 0; the skip and the delay-window minimum compose, the later of the two winning). Because every DLS fit reads it, changing it asks first and then clears **every DLS result**:

- every per-measurement fit and its Summary row;
- the Γ vs q^2, D vs c and DDLS results, and every replicate average;

- the live Mode Map, with the mode-family fits and the amplitude report read off it;
- the sample R_h values taken from them, a Cross-Sample pick of every DLS kind included.

It keeps the recorded mode sets and their results, SLS and any hand-entered value. The mode families you picked on the Mode Map are kept as your assignments, but the new setting moves the modes, so a re-run map can orphan their picks (§8.5).

12.1.5 Cumulant fit cutoff

Sets where the *linear* cumulant's fitted range stops: it keeps the run of channels around the correlogram's peak that stays above this fraction of the peak, and drops the rest. The default is 0.1; ISO 22412 prescribes 0.01, and the **Theory-and-Equations-Guide** §3.2.3 states the program's cutoff as a benchmarked deviation from the standard and says which of its companion clauses are implemented. Under the *nonlinear* method, which fits every live channel in the window, it still sets the range the reported RMS error is measured over and the starting guess the fit departs from. Lowering it can make a measurement look *less* polydisperse than it is — the extra channels are the noisiest, the fitted curve flattens, and the reported PDI falls, which can clear the polydispersity warning. Changing it clears existing cumulant-based results after you confirm, exactly as for the method (§12.1.1), and the value a fit ran at travels into its export as the *Fit cutoff* column (§8.2.3) and into the Summary export (§8.8).

12.2 DLS Distribution Defaults

The starting values of the distribution fits (CONTIN, NNLS, lognormal):

- the R_h grid (min / max / pts), for every method;
- the **Residual weighting** — *Sourced variance (default)* or ${g_1}^2$ *(older form)* (§8.3.2);
- CONTIN's alone: the **CONTIN α selection** (robust GCV, the L-curve corner or the F-test), the **CONTIN α (min / max)** range, the **GCV robustness γ** and the **F-test probability to reject**.

They seed the Distribution sub-tab's (§8.3) and the Mode Map's (§8.5) own fields, which you can still change per measurement there; replicate averaging of a distribution method (§7.1.1) has no fields of its own and runs on these values unless its prompt copies the Distribution sub-tab's. Changing a value here does not touch a result already computed. An α end that is not a number above zero is refused where you type it, and an α minimum above the maximum is refused at **Update**, with the reason. How CONTIN chooses α is in the **Theory-and-Equations-Guide** §3.4.5.

12.3 File Loading

- **Brookhaven .dat baseline.** *Calculated* (the default — formed from the acquisition's totals, so it uses the whole measurement) or *Measured* (the average of the baseline channels the acquiring software set aside; it matches what Particle Explorer reports, but depends on those channels lying past the decay). Applies only when the file itself does not name a baseline (§5.2). The two typically differ by a few tenths of a percent; that makes no difference to the default fits, which float or estimate a baseline of their own, and shifts the size by a few percent for the linear cumulant method, which assumes none. How a baseline is determined at all is the **Theory-and-Equations-Guide** §3.1.2.

12.4 SLS Defaults

- **Standard geometry.** The scattering geometry of the calibration standard — *VU* (a vertical laser with no analyzer), *VV* (polarized) or *VH* (depolarized). It seeds the **Calibrant (toluene) geometry** field of the SLS Calibration panel (§9.1.1), where the per-sample value wins. The geometry-aware Rayleigh ratio is the **Theory-and-Equations-Guide** §1.6.2.
- **Guinier qRg validity limit.** The largest qR_g at which a size read from the small-qR_g slope counts as valid; the default is 1.0. Despite its name it governs every fit that reads one: above it the Zimm, Debye, Guinier and Amplitude Guinier R_g are flagged, the Zimm, Debye and Guinier M_w with them, and a Berry fit's R_g and M_w are labeled *past the Zimm limit* rather than flagged (§9.1.4). Each of these fits shows the qR_g(max) it is compared with. Why one limit serves them all, and where the expansion holds, is the **Theory-and-Equations-Guide** §4.6 (the Guinier plot).

12.5 Solvent Library

- **Default solvent.** The solvent the Solvent Explorer (§11.4) starts on. A convenience default only — it never sets a value in any analysis.

12.6 Uncertainty

The **Regression SE estimator** is the covariance estimator behind every reported regression ± — the SLS Zimm/Berry, Debye, Guinier and calibration-free A_2 fits, the DLS Γ vs q^2 and k_D, and the exponents of the scaling laws (§10.2). *HC3 (robust)*, the default, is a heteroscedasticity-consistent SE that never under-reports when point precision varies across angles or concentrations; use it unless you have a specific reason not to. *Classical OLS* is the textbook SE, offered for like-for-like comparison with classical software, a published table or a spreadsheet; it can under-report on a short, high-leverage concentration ladder, which is why it is not the default. The choice is recorded on each result and written into the export (an OLS ± is labeled "± from classical OLS, not HC3", §13); point estimates are unaffected. Switching clears existing ±-bearing results — and a Cross-Sample pick taken from one, and a form-factor source's currency (§10.1.3) — after you confirm; the Γ vs q^2, D vs c and DDLS sub-tabs keep drawing their fit, marked out of date, until you run it again (§8.1.3). The two estimators are compared in the **Theory-and-Equations-Guide** §6.1 (HC3 standard errors).

12.7 Display Units

One unit per plotted quantity — **Delay time** τ, **Count rate**, **Scattering q^2**, **Diffusion coeff. D**, **Decay rate** Γ, **Concentration**, **Radius (R_h / R_g)** and **Molecular weight M_w** — for the plot axes and for the Cross-Sample table and its hand-entry fields (§10.1). The **Scattering q^2** choice (nm^{-2} or m^{-2}) applies to every q^2 axis — the DLS Γ vs q^2 and DDLS plots and the SLS plots (excess Rayleigh ratio, Debye, Guinier, form factors, and the Zimm grid, whose printed spacing constant k follows the same unit). The **Concentration** choice governs the D vs c and calibration-free A_2 axes; legend concentration labels always read mg/mL. Switching a unit redraws the axes and re-labels the tables — no fitted or reported number changes with the unit choice. The Data tab's per-row **Unit** column (§6.1) is a separate, per-field choice.

12.8 Result Formatting

- **No-uncertainty precision (sig. figs).** The significant figures shown for every result that has no honest uncertainty — a single-correlogram fit, an NNLS/CONTIN peak, and every derived number that carries no ± (Γ, PDI, R^2, qR_g, …). Default 3; lower it for a more conservative look, at the cost of the trailing digits of everything without a ±. A value that *has* a ± is rounded to the place its ± supports, and this setting does not change that, with one exception: in the DDLS **Shape Models** table (§8.7) and the R_g and R_h cells of the Cross-Sample **Architecture** sub-tab's table (§10.1.1), a ± so large that it would round the value to zero leaves the value at this precision, shown beside its ± at two significant figures. The program never fabricates a ± to show more digits. The **Theory-and-Equations-Guide** §6 states the rounding rule.

12.9 Appearance

- **Theme.** *System* (follows your OS light/dark setting) by default; *Light* or *Dark* overrides the OS so the application stays in that mode regardless of system preferences.
- **Plot palette.** The series colors for every plot: *okabe_ito* (the default, a colorblind-safe palette), *tab10* or *grayscale*. Series also vary by marker and line style, so they stay distinct in grayscale and for colorblind readers. It applies to plots drawn afterwards.
- **Plot theme. Match plot to app theme (dark)**: when on and the app theme is dark, the on-screen plots use a dark background too. Saved and exported images always stay white for clean, print-parity figures — this affects only what you see on screen.
- **UI density.** *Compact*, *Comfortable* or *Large* — the application-wide text size and spacing. *Large* aids readability; *Compact* fits more on screen.
- **Tooltips. Show tooltips on hover** — the passive hover tooltips throughout the program. The **?** help buttons still work on click when this is off, and so does a click on a result's notes line or lettered value (§8.1.7).
- **Session. Reopen last session on startup** — the program's one form of persistence (§7.4): when on, the current workspace is kept when you close the program and reopened the next time you launch it. Off by default.
- **Runs. Ask before a Run replaces a picked value** — a value you explicitly picked on the Cross-Sample tab (§10.1.2) is replaced by the two runs that write sample values, a Zimm/Berry run and replicate averaging, and the run always reports the replacement afterwards. Off (the default), the run proceeds; on, a confirm names the picked value(s) first, and **Cancel** keeps the pick and runs nothing. Hand-entered values are never replaced by any run, with or without this setting.

12.10 Update and Restore defaults

The two buttons pinned under the form. **Update** applies every change on the tab — the confirms above (for the cumulant inputs, the skipped channels and the SE estimator) appear at this point, and **Cancel** on any of them reverts that setting. **Restore defaults** returns every setting to its factory value and applies it, with the same confirms where a fit would be cleared. Settings persist across restarts.

---

13. Exporting Results

Every analysis result can be written to a CSV that OriginLab imports directly. On any analysis sub-tab, click **Export CSV...**, accept or edit the proposed filename in the save dialog, and the file lands in the folder of your last export (your Documents folder on first use); the plot toolbars' save button shares that folder and naming, so a plot image lands next to its CSV with a matching name. The open-file dialogs remember the folder you last loaded data from, separately.

13.1 Where the Buttons Are

Each button writes the result currently on screen, and is enabled once a result has been computed:

Sub-tab	What its **Export CSV...** writes
DLS ▶ Correlogram (§8.2)	the fit of each ticked measurement under the **Method** shown, with the order, method, weighting and cutoff recorded (§8.2.3)
DLS ▶ Distribution (§8.3)	each ticked measurement·method's distribution and peaks; **Export α-sweep...** the CONTIN sweep
DLS ▶ Γ vs q^2 / D vs c (§8.4, §8.6)	the points table and the fit, its last columns the cumulant order and the recorded weighting
DLS ▶ Mode Map (§8.5)	the per-point map with each point's family, α, selector and γ, every family's fit, and the amplitude report
DLS ▶ DDLS (§8.7)	the per-angle Γ_{VV}/Γ_{VH} and D_r, the combined $D_t/D_r/R_{h,t}/\tau_{rot}$, and the rod, sphere and ellipsoid shape dimensions when available
DLS ▶ Summary (§8.8)	one row per result or peak from both tables, workspace-wide, in one set of columns (sample, measurement, method, R_h and its ±, PDI, …)
SLS ▶ Analysis (§9.1)	the last method you ran — Zimm/Berry, Debye, Guinier, the single-angle ladder, calibration-free A_2, the form factor, or the excess Rayleigh series
SLS ▶ Mode-Resolved (§9.3)	a mode's per-mode analysis, its partition recorded in the Comments
Cross-Sample ▶ Architecture (§10.1)	the table — ρ, the sources, the branching factors
Cross-Sample ▶ Scaling (§10.2)	the R_g–M_w (or R_h–M_w) and A_2–M_w points and fits
Utilities ▶ Traces (§11.1)	the trace; **Export decomposition...** the Windowed Hampel columns; **Export running average...** the running mean and its standard deviation; **Export diagnostic...** the lower panel's numbers
The sidebar (§7.3)	right-click a trace ▶ export it as `(time, count rate)` columns

The DDLS file flags each shape dimension "MODEL … not measured" in its Comments cell, and its rod and ellipsoid ± cells also state their Monte-Carlo census: how many of the draws a shape reproduced, and the seed. The Scaling file puts branched points in their own columns beside the fitted ones and, when any samples were dropped before the fit, adds an *Excluded* column counting them.

When a sub-tab holds several curves at once (an NNLS+CONTIN distribution overlay, or both size and A_2 scaling plots), the export writes one file per curve; if any of those files already exist, the program lists them and asks before overwriting.

13.2 The File

- **Label rows.** Each file carries the four Origin label rows — **Long Name, Units, Comments, Parameters** — so axis labels and units populate on import.
- **Layout.** Tabular outputs (correlogram fits, distributions, Rayleigh ratios) are written column by column with units; scalar results (M_w, R_g, Γ, …) as single-value labeled columns.
- **Zimm datasets** use a wide layout: one $Kc/\Delta R_\theta$ column per concentration, the concentration in the Parameters row. Its q^2 column is the union of the concentrations' angles, so where they do not share one grid each ordinate still sits on the row of the q^2 it was measured at, and a blank cell there means no value was written — that concentration has no point at that angle, or its ordinate is not a usable number (a masked or dropped point).
- **Missing values and flags.** Missing values are empty cells. A flag column that was evaluated (*PDI valid*, *Is diffusive*, *Guinier valid*, *Is apparent*, …) reads `yes` or `no`; blank there means the flag has no value at all (for example *Single-exponential valid* when no rod length was given).
- **Caveats.** A result's notes — its failed preconditions and how it was obtained (§8.1.7) — are written as clauses, without the guide section, into the **Comments** cell of the columns each is about, each clause once.
- **Derived series.** A fit over several derived series (§9.2) names each one's record in the Comments; in a Zimm or Berry export, a concentration's $Kc/\Delta R$ column names that concentration's record and the two intercept columns name every one.
- **Uncalibrated results.** An uncalibrated result writes "uncalibrated, arbitrary scale" into the **Comments** cell of the affected columns only — their SE columns included — never changing the row count, so the Origin import stays intact.
- **Standard errors.** Every SE column's Comments cell says what kind of ± it holds ("± statistical; excludes calibration & dn/dc"; for the calibration-free $2A_2M_w$ product, that its ± is statistical and that the product cancels both scales) and, when Settings selects *Classical OLS* as the **Regression SE estimator** (§12.6), names it ("± from classical OLS, not HC3").
- **The residual weight.** A distribution export carries three columns recording what its residual weight was and what it read — *Residual weighting*, *Sample time provenance* and *Run duration form*. The correlogram-fit export records the cumulant's residual weighting in its own column, and the Γ vs q^2 and D vs c exports end with a column for the weighting their per-point fits ran under (§8.4). The Summary export records it for each per-measurement row (§8.8). The DDLS result, a replicate average, the Summary's sample-level rows and a sample's R_h do not record which weighting produced them, so a number there fitted off the default is not marked as such.

13.3 The Name

The save dialog proposes a filename in one of three forms. An export of a single measurement is named for its sample, the analysis and the method, then the measurement's own name and its angle — `PVP-water_correlogram_cumulant_batch 1-3_90deg.csv`. An export that belongs to a whole sample names the sample and the analysis — `PVP-water_debye.csv`. An export that belongs to no single sample keeps a plain name, e.g. `dls_summary.csv`. The sample part comes from the committed solute and solvent; the temperature is not part of a file name, so two samples that differ only in temperature propose the same name — edit it in the dialog. Characters that are not allowed in filenames are replaced with `-`.

When one export writes several files, the dialog shows the part they all share, and each file's own ending is added to whatever you type: type `tuesday-batch` and you get `tuesday-batch_1_90deg.csv`

through `tuesday-batch_6_90deg.csv`. A measurement exported on its own and the same measurement exported in a batch therefore land on the same name. In the rare case where two files of one batch would end up with the same name, the program appends each measurement's identifier (the `m0001`-style **ID** its sidebar tooltip shows) so neither is overwritten.

An export that comes from exactly one measurement also carries a **Source file** column holding that measurement's file name, so a single CSV taken out of a batch still says which measurement it came from. An export built from several measurements names its sources in the Comments cells instead.

---

14. Visualization

Every plot shows what was computed: correlogram fits with residuals, size distributions, Γ vs q^2 diagnostics, Debye plots, and the full Zimm/Berry grid. Distributions from different methods can be overlaid (e.g. NNLS and CONTIN on one axis), making their trade-off visible at a glance. The plot conventions every DLS sub-tab shares — color for the measurement, line style for the method, the on-screen key against the saved figure's full legend — are §8.1's.

Data-quality and reliability **flags are interface overlays**, not part of the figure: a saved plot image is always clean. On screen, where a result is apparent or uncalibrated, the interface marks it next to the plot, so the on-screen chart is never read without its caveat.

The plots of the DLS **Distribution**, Γ **vs** $\mathbf{q^2}$, **Mode Map**, **D vs c** and **DDLS** sub-tabs and of the three SLS sub-tabs carry an **axis-control bar** beneath them (the Solvent Explorer, §11.4, carries one per axis): per-axis linear/log toggles, explicit min/max boxes, and an autoscale button. Axis units follow **Settings ▶ Display Units** (§12.7); the theme, palette and density of the whole window are **Settings ▶ Appearance** (§12.9).

Within the analysis tabs, the stacked plots and their control columns are resizable — drag the **splitter grips** (the visible three-dot handles) between them, including the fit/residual divider, which carries a persistent grip line. Every control column has the same shape: the measurements to tick, then the settings with **Run** directly under them, then the results. **Export CSV...**, the status line and any ⚠ warning stay at the foot of the column, so they remain in view when the settings scroll. The sidebar on the left is resizable (drag the divider) and long sample names show in full on hover; the window itself can be freely resized (tab contents scroll when the window is small).

See the **Theory-and-Equations-Guide** for the equations and derivations behind every plotted quantity.